

1 Quantum and Classical Statistics

1.1 A very short summary of quantum mechanics

In many cases we will assume that the Hamiltonian has the form

$$H = \frac{p^2}{2m} + V(x). \quad (1.1)$$

This definition holds both for classical and quantum mechanics, although we typically put a hat on operators in quantum mechanics. x and p are vectors of f dimensions, where in many situations f is 3 times the number of atoms and $V(x)$ is a scalar field of these variables.

Eigenstates are defined by the time-independent Schrödinger equation:

$$\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle \quad (1.2)$$

but remember that wavefunctions do not need to be eigenstates (but can be defined as linear combinations of eigenstates).

Consider a system (e.g. a single molecule) described by the state $|\psi\rangle$. An expectation value of an operator, $\hat{A}$, is given by

$$\langle \hat{A} \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle. \quad (1.3)$$

If $\hat{A} = A(\hat{x})$, this can be evaluated in the position representation by inserting a complete set of position states $1 = \int |x\rangle \langle x| dx$ to give

$$\langle \hat{A} \rangle_\psi = \int \langle \psi | \hat{A} | x \rangle \langle x | \psi \rangle dx \quad (1.4a)$$

$$= \int \langle \psi | x \rangle A(x) \langle x | \psi \rangle dx \quad (1.4b)$$

$$= \int \psi(x)^* A(x) \psi(x) dx, \quad (1.4c)$$

which is the formula you may be more familiar with. For the specific example of finding the probability of finding the particle at a point $x = a$, use $\hat{A} = \delta(\hat{x} - a)$ to give

$$\langle \delta(\hat{x}_a) \rangle_\psi = \int \psi(x)^* \delta(x - a) \psi(x) dx \quad (1.5a)$$

$$= |\psi(a)|^2. \quad (1.5b)$$

1.2 Quantum statistical ensembles

If we do a measurement on a large number of molecules, the observable will be an average over all the molecules, leading to

$$\langle \hat{A} \rangle = \sum_n P_n \langle \psi_n | \hat{A} | \psi_n \rangle, \quad (1.6)$$

where P_n is the probability that a molecule chosen at random is in state ψ_n and we sum over all possible states $|\psi_n\rangle$. The ensemble of molecules is thus defined by the probabilities.

We define the density operator

$$\hat{\rho} = \sum_n P_n |\psi_n\rangle\langle\psi_n|, \quad (1.7)$$

and note that we can obtain the same ensemble average using

$$\langle\hat{A}\rangle_\rho = \text{Tr} [\hat{\rho} \hat{A}], \quad (1.8)$$

where the quantum trace is defined analogously to the matrix trace as a sum over diagonal elements

$$\text{Tr} [\hat{O}] = \sum_n \langle\psi_n|\hat{O}|\psi_n\rangle. \quad (1.9)$$

Using these definitions, it is easy to show [make sure you can do this]

$$\text{Tr} [\hat{\rho}] = \sum_{n,m} P_m \langle\psi_n|\psi_m\rangle \langle\psi_m|\psi_n\rangle = \sum_n P_n = 1, \quad (1.10)$$

where we have used the fact that the wavefunctions are orthonormal and that the probabilities sum to 1. This is the normalization condition for the density operator.

To find the population of a particular state, use the projection operator $\hat{A} = |\psi_m\rangle\langle\psi_m|$. This gives

$$\langle\hat{A}\rangle_\rho = \text{Tr} [\hat{\rho} |\psi_m\rangle\langle\psi_m|] = \langle\psi_m|\hat{\rho}|\psi_m\rangle = P_m. \quad (1.11a)$$

Also note the properties of the trace. It is invariant to cyclic permutation

$$\text{Tr}[ABC] = \text{Tr}[CAB] = \text{Tr}[BCA]$$

and to unitary transformation ($S^\dagger = S^{-1}$):

$$\text{Tr}[S^\dagger AS] = \text{Tr}[A].$$

Because of these properties, we can find even more general definitions of the density operator which are unitary transforms of the form given above. The expression can be rewritten in an arbitrary basis set, $\{|\phi_i\rangle\}$ by using the completeness of the states, $1 = \sum_i |\phi_i\rangle\langle\phi_i|$. Then the expectation value can be then written as

$$\langle\hat{A}\rangle_\rho = \sum_n P_n \langle\psi_n|\hat{A}|\psi_n\rangle \quad (1.12a)$$

$$= \sum_{n,i,j} P_n \langle\psi_n|\phi_j\rangle \langle\phi_j|\hat{A}|\phi_i\rangle \langle\phi_i|\psi_n\rangle \quad (1.12b)$$

$$= \sum_{n,i,j} \langle\phi_i|\psi_n\rangle P_n \langle\psi_n|\phi_j\rangle A_{ji} \quad (1.12c)$$

$$= \sum_{i,j} \langle\phi_i|\hat{\rho}|\phi_j\rangle A_{ji} \quad (1.12d)$$

$$= \sum_{i,j} \rho_{ij} A_{ji} = \text{Tr} [\hat{\rho} \hat{A}]. \quad (1.12e)$$

In the final line we have defined the density matrix with elements $\rho_{ij} = \langle \phi_i | \hat{\rho} | \phi_j \rangle$ and note that the sums are equivalent to a standard matrix product and trace. Note that whereas the wavefunction $|\psi\rangle$ behaves like a vector, the density operator $\hat{\rho}$ behaves like a matrix.

If all molecules are in the same state then one of the probabilities will be 1 and all others 0, leading to

$$\hat{\rho} = |\psi\rangle\langle\psi|.$$

This is called a pure state, as opposed to a mixed state. To determine the purity, compute

$$\text{Tr}[\hat{\rho}^2],$$

which is clearly equal to 1 only for a pure state.

1.3 Classical statistical ensembles

In quantum mechanics the state of the system is completely determined by the wavefunction. In classical mechanics one specifies the positions and momenta of all particles. The classical equivalent of the density operator is the probability density function $\rho(x, p)$. This should be normalized over all phase space such that

$$(2\pi\hbar)^{-f} \iint \rho(x, p) \, dx \, dp = 1. \quad (1.13)$$

The average of an observable $A \equiv A(x, p)$ in a particular distribution ρ is

$$\boxed{\langle A \rangle_{\rho}^{\text{cl}} = (2\pi\hbar)^{-f} \iint \rho(x, p) A(x, p) \, dx \, dp,} \quad (1.14)$$

where f are the number of degrees of freedom (typically 3 times the number of atoms).

One thus sees that the classical analogue of the trace is an integral over phase space. The prefactor is required to keep the units consistent and its value related to Planck's constant is determined to ensure quantum-classical correspondence, as we will show later.

1.4 Wigner functions

The Wigner distribution function is defined as

$$\rho_W(x, p) = \int \langle x + \frac{1}{2}\xi | \hat{\rho} | x - \frac{1}{2}\xi \rangle e^{-ip\cdot\xi/\hbar} d\xi. \quad (1.15)$$

Note the normalization such that $(2\pi\hbar)^{-f} \iint \rho_W(x, p) \, dx \, dp = 1$.

One can further show that [check that you can do this: Hint, use property of delta function $(2\pi\hbar)^{-f} \int e^{\pm ip\cdot x/\hbar} dp = \delta(x)$]

$$\langle \hat{A} \rangle_{\rho} = \text{Tr} [\hat{\rho} \hat{A}] = (2\pi\hbar)^{-f} \iint \rho_W(x, p) A_W(x, p) \, dx \, dp, \quad (1.16)$$

where $A_W(x, p)$ is defined in terms of $\hat{A}$ in the same way as equation (1.15). It is easy to show that if it is an operator of position only, then this is equivalent to simply removing the hats! In this way, we can compute quantum-mechanical ensemble averages using a classical-type method.

For example, the ground-state harmonic oscillator wavefunction is

$$\psi_0(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2},$$

where $\alpha = m\omega/\hbar$, and the first excited state is

$$\psi_1(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \sqrt{2\alpha} x e^{-\alpha x^2/2}.$$

In either case, we use $\hat{\rho} = |\psi_n\rangle\langle\psi_n|$ to get

$$\rho_W(x, p) = 2e^{-\frac{p^2}{m\omega\hbar} - \frac{m\omega x^2}{\hbar}}$$

for the ground state and

$$\rho_W(x, p) = 2 \left(\frac{2p^2}{m\omega\hbar} + \frac{2m\omega x^2}{\hbar} - 1 \right) e^{-\frac{p^2}{m\omega\hbar} - \frac{m\omega x^2}{\hbar}}$$

for the excited state.

1.5 Thermal equilibrium and partition functions

Statistical properties are average values of an equilibrium distribution and are therefore not time dependent. The partition function is one of the most important quantities as other thermodynamic properties can be derived from it. Throughout we use $\beta = (k_B T)^{-1}$.

1.5.1 Thermal density matrix

The density operator which describes thermal equilibrium is the Boltzmann operator

$$\hat{\rho}_{\text{eq}} = Q^{-1} e^{-\beta \hat{H}}, \quad (1.17)$$

where this is the exponential of a matrix [not the same as a matrix of exponentials!] defined by

$$e^{\hat{A}} = \sum_{k=0}^{\infty} \frac{1}{k!} \hat{A}^k. \quad (1.18)$$

We require $\text{Tr}[\hat{\rho}] = 1$ which implies that the normalization factor is

$$Q = \text{Tr} \left[e^{-\beta \hat{H}} \right]. \quad (1.19)$$

This is the most general expression for the quantum partition function.

We can evaluate the trace in the basis of eigenstates of the Hamiltonian, $|\psi_n\rangle$, to give

$$\begin{aligned} Q &= \sum_n \langle \psi_n | e^{-\beta \hat{H}} | \psi_n \rangle \\ &= \sum_n \langle \psi_n | e^{-\beta E_n} | \psi_n \rangle \\ &= \sum_n e^{-\beta E_n}. \end{aligned}$$

Here we used the Taylor series expression for the matrix exponential, equation (1.18), as well as the eigenvalue relationship, $\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle$.

1.5.2 Classical density

The probability density in classical mechanics is $\rho(x, p)$, for instance $\rho_{\text{eq}}(x, p) = e^{-\beta H(x, p)} / Q_{\text{cl}}$ is the Boltzmann distribution.

In many cases one cannot actually compute the normalization factor Q_{cl} , but can nonetheless obtain equilibrium averages from thermostatted molecular dynamics simulations or Metropolis Monte Carlo (see further reading on Molecular Simulation):

$$\langle A \rangle_{\text{eq}}^{\text{cl}} = \frac{\iint e^{-\beta H(x, p)} A(x, p) \, dx \, dp}{\iint e^{-\beta H(x, p)} \, dx \, dp}. \quad (1.20)$$

This expression should be interpreted as taking an average of $A(x, p)$ from an unnormalized distribution $e^{-\beta H}$.

1.5.3 Partition functions

The quantum partition function can be defined as

$$Q = \sum_n e^{-\beta E_n}, \quad (1.21)$$

where E_n are the eigenvalues of the Hamiltonian. This formula is therefore only useful if the eigenvalues are known or can be computed.

The classical partition function can be defined as

$$Q_{\text{cl}} = (2\pi\hbar)^{-f} \iint e^{-\beta H(x, p)} \, dx \, dp, \quad (1.22)$$

where f is the number of degrees of freedom. The momentum integral is a Gaussian so can be performed explicitly to give

$$Q_{\text{cl}} = \left(\frac{m}{2\pi\beta\hbar^2} \right)^{f/2} \int e^{-\beta V(x)} \, dx. \quad (1.23)$$

1.5.4 Example: Free particle

The free particle potential is given by $V(x) = 0$. Therefore the classical partition function for a free particle in one degree of freedom is

$$Q_{\text{cl}} = \lim_{L \rightarrow \infty} \left(\frac{m}{2\pi\beta\hbar^2} \right)^{1/2} \int_{-L/2}^{L/2} dx \quad (1.24a)$$

$$= \lim_{L \rightarrow \infty} \left(\frac{m}{2\pi\beta\hbar^2} \right)^{1/2} L \quad (1.24b)$$

$$Q_{\text{cl}}/L = \left(\frac{m}{2\pi\beta\hbar^2} \right)^{1/2}. \quad (1.24c)$$

Therefore for a free particle, it only makes sense to talk about the partition-function per unit length. For an atom or molecule in three-dimensional space, the translational partition function is $(m/2\pi\beta\hbar^2)^{3/2}$ per unit volume.

For the quantum version, we will use more sophisticated techniques introduced later in the course. However, note now that the result is exactly equivalent to the classical version in this case. For an alternative derivation, use the states of a particle in a box of length L and then take the limit $L \rightarrow \infty$.

1.5.5 Example: Harmonic Oscillator

The one-dimensional harmonic oscillator potential is given by $V(x) = \frac{1}{2}m\omega^2x^2$. Therefore the classical partition function is

$$Q_{\text{cl}} = \frac{1}{\beta\hbar\omega} \quad \text{for a 1D harmonic oscillator.} \quad (1.25)$$

The quantum energy levels are $E_n = (n + \frac{1}{2})\hbar\omega$ for $n \in \{0, 1, 2, \dots\}$ so that

$$Q = \sum_{n=0}^{\infty} e^{-\beta(n+\frac{1}{2})\hbar\omega} \quad (1.26a)$$

$$= e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n \quad (\text{N.B. geometric series}) \quad (1.26b)$$

$$= \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} \quad (1.26c)$$

$$= [2 \sinh \beta\hbar\omega/2]^{-1}. \quad (1.26d)$$

1.5.6 Example: rotational partition function

The moment of inertia for a *linear* system is $I = \sum_i m_i r_i^2$ where r_i is the distance from the centre of mass to the atom i . The rotational constant is then given by $B = \hbar^2/2I$.

For a linear system, the energy levels are $E_J = BJ(J+1)$ with degeneracy $2J+1$. The quantum partition function is thus given by

$$Q = \sum_{J=0}^{\infty} (2J+1) e^{-\beta BJ(J+1)} \quad \text{for a linear rotor.} \quad (1.27)$$

The classical partition function can be found by approximating this sum by an integral to give

$$Q_{\text{cl}} = \frac{1}{\beta B} \quad \text{for a linear rotor.} \quad (1.28)$$

For a nonlinear rotor, the energy levels are more complicated. However, the classical partition function is known to be

$$Q_{\text{cl}} = \sqrt{\frac{\pi}{\beta^3 ABC}} \quad \text{for a nonlinear rotor,} \quad (1.29)$$

where A , B and C are the rotational constants defined as B above for the three principal moments of inertia.

1.5.7 Example: average energy

The average energy of a quantum system in thermal equilibrium is

$$\langle E \rangle = \text{Tr} [\hat{\rho}_{\text{eq}} \hat{H}] \quad (1.30a)$$

$$= Q^{-1} \sum_n E_n e^{-\beta E_n}, \quad (1.30b)$$

where E_n are the eigenvalues of the Hamiltonian.

For a classical system,

$$\langle E \rangle^{\text{cl}} = (2\pi\hbar)^{-f} \iint Q^{-1} e^{-\beta H(x,p)} H(x,p) dx dp \quad (1.31)$$

In both cases, the average energy can be defined directly from the partition function:

$$\langle E \rangle = -\frac{d}{d\beta} \ln Q. \quad (1.32)$$

1.6 Transition-state theory

Eyring TST is normally defined as

$$k_{\text{TST}}(T) = \frac{k_B T}{h} \frac{Q^\ddagger(T)}{Q_r(T)} e^{-\beta V^\ddagger}, \quad (1.33)$$

where $Q_r(T)$ and $Q^\ddagger(T)$ are the partition functions of the reactants and transition state (per unit length/volume etc. as appropriate) and $V^\ddagger$ is the barrier height. One typically separates the vibrations and rotations using the rigid-rotor approximation and uses quantum harmonic-oscillator partition functions for the vibrational modes and classical partition functions for the rotations. One degree of freedom of the transition state corresponds to the reaction coordinate and is ignored in this theory. We will show where this formula comes from later in the course.

1.7 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Basic Quantum mechanics and bra-ket notation

- Dirac, Quantum Mechanics (Oxford, 1948) [beautiful mathematical development but with few examples]
- Griffiths & Schroeter, Introduction to Quantum Mechanics (Cambridge), chapter 3

Statistical ensembles

- Tuckerman, Statistical Mechanics (Oxford), chapters 4.1-4.5 and 10

Molecular Simulations

- Tuckerman, Statistical Mechanics (Oxford)
- Frenkel and Smit, Understanding Molecular Simulations (Academic Press)

Wigner functions

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapter 5
- Schleich, Quantum Optics in Phase Space (Wiley), chapter 3

Partition functions

- Atkins, Physical Chemistry (Oxford), chapter 15
- McQuarrie, Statistical Mechanics

Transition-state theory

- Atkins, Physical Chemistry (Oxford), section 21C

2 Quantum and Classical Dynamics

2.1 Classical dynamics

2.1.1 Hamiltonian formulation

- Dynamics of the particles are described as trajectories in phase space with coordinates q_k and momenta p_k .
- State of system is a point in $6N$ -dimensional phase space ($3N$ coordinates and $3N$ momenta).

Hamiltonian formulation: introduced by William Rowan Hamilton (1805-1865), an Irish physicist who also introduced Hamilton's principle, Hamiltonian mechanics, Hamilton-Jacobi equation, quaternions, Nabla symbol, tensors etc.

Based on a Hamiltonian, H , the equations of motion are

$$\frac{dq_k}{dt} \equiv \dot{q}_k = \frac{\partial H}{\partial p_k} \quad (2.1a)$$

$$\frac{dp_k}{dt} \equiv \dot{p}_k = -\frac{\partial H}{\partial q_k}. \quad (2.1b)$$

For the Hamiltonian

$$H = \sum_k \frac{p_k^2}{2m_k} + V(q_1, q_2, \dots), \quad (2.2)$$

we can recover Newton's equation of motion:

$$\dot{q}_k = \frac{p_k}{m_k} \quad (2.3a)$$

$$\dot{p}_k = -\frac{dV}{dq_k} \equiv F_k. \quad (2.3b)$$

Differentiating the first line again and comparing the two shows that $\dot{p}_k = F_k = m_k \ddot{q}_k$, which is Newton's second law.

Therefore the time-evolution of a dynamical variable $A \equiv A(\vec{p}, \vec{q})$ is given by

$$\frac{dA}{dt} = \sum_k \frac{\partial A}{\partial q_k} \dot{q}_k + \frac{\partial A}{\partial p_k} \dot{p}_k \quad (2.4a)$$

$$= \sum_k \frac{\partial A}{\partial q_k} \frac{\partial H}{\partial p_k} - \frac{\partial A}{\partial p_k} \frac{\partial H}{\partial q_k} \quad (2.4b)$$

$$= \{A, H\} = -\{H, A\}, \quad (2.4c)$$

where the curly brackets are known as Poisson brackets defined as above. We can easily show that energy is conserved using the operator H in this formula to give $\frac{dH}{dt} = 0$.

2.1.2 Classical dynamics of harmonic oscillator

For a harmonic oscillator, $V(q) = \frac{1}{2}kq^2$, the equations of motion are solved by

$$q(t) = q_0 \cos \omega t + \frac{p_0}{m\omega} \sin \omega t \quad (2.5a)$$

$$p(t) = p_0 \cos \omega t - m\omega q_0 \sin \omega t, \quad (2.5b)$$

where $\omega = \sqrt{k/m}$.

2.2 Quantum dynamics

2.2.1 Propagator

The time-dependent Schrödinger equation (TDSE) is

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle. \quad (2.6)$$

This is a linear equation so the general solution of the TDSE is:

$$|\Psi(t)\rangle = \hat{U}(t, t_0) |\Psi(t_0)\rangle. \quad (2.7)$$

The time-evolution operator is obtained by solving the equation:

$$i\hbar \frac{\partial \hat{U}(t, t_0)}{\partial t} = \hat{H} \hat{U}(t, t_0). \quad (2.8)$$

The special (but common) case of a time-independent $\hat{H}$ leads to

$$\hat{U}(t, t_0) = \exp\left(-\frac{i\hat{H}(t-t_0)}{\hbar}\right), \quad (2.9)$$

where the phase factor has been included to ensure that $\hat{U}(t_0, t_0) = 1$.

Thus if $|\Psi(t_0)\rangle$ is a solution of the time-independent Schrödinger equation (TISE),

$$\hat{H} |\Psi(t_0)\rangle = E |\Psi(t_0)\rangle$$

i.e. is an eigenstate, then

$$\hat{U}(t-t_0) |\Psi(t_0)\rangle = e^{-iE(t-t_0)/\hbar} |\Psi(t_0)\rangle \quad (2.10)$$

2.2.2 Expectation values

What we have used so far is called the “**Schrödinger picture**”.

$$\begin{aligned} \text{e.g: } \langle A(t) \rangle &= \langle \Psi(t) | \hat{A} | \Psi(t) \rangle \\ &= \langle e^{-\frac{i\hat{H}t}{\hbar}} \Psi(0) | \hat{A} | e^{-\frac{i\hat{H}t}{\hbar}} \Psi(0) \rangle. \end{aligned} \quad (2.11)$$

Properties:

1. Operators are stationary (time independent)
2. State vector Ψ and expectation values evolve in time.

We now turn to **The “Heisenberg picture”**. The expectation value can equivalently be written

$$\langle A(t) \rangle = \langle \Psi(0) | e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} | \Psi(0) \rangle \quad (2.12a)$$

$$= \langle \Psi(0) | \hat{A}(t) | \Psi(0) \rangle \quad (2.12b)$$

where

$$\hat{A}(t) = e^{i\frac{\hat{H}t}{\hbar}} \hat{A} e^{-i\frac{\hat{H}t}{\hbar}}. \quad (2.13)$$

Properties:

1. State vector Ψ is stationary.
2. Operators are time dependent.

For the special case of time-independent Hamiltonians:

$$\frac{d}{dt} \hat{A} = \frac{i}{\hbar} \left(\hat{H} e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} - e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} \hat{H} \right) + e^{i\hat{H}t/\hbar} \frac{\partial \hat{A}}{\partial t} e^{-i\hat{H}t/\hbar} \quad (2.14a)$$

$$= \frac{i}{\hbar} [\hat{H}, \hat{A}] + \frac{\partial \hat{A}}{\partial t} \quad (2.14b)$$

The final term is 0 if the operator isn't explicitly time-dependent.

For position and momentum operators $\hat{q}$ and $\hat{p}$ and a Hamiltonian $\hat{H} = \hat{p}^2/2m + V(\hat{q})$,

$$[\hat{H}, \hat{p}] = [V(\hat{q}), \hat{p}] = i\hbar \frac{\partial V}{\partial \hat{q}} \quad (2.15a)$$

$$[\hat{H}, \hat{q}] = [\hat{p}^2/2m, \hat{q}] = -i\hbar \hat{p}/m \quad (2.15b)$$

and therefore

$$\frac{d\hat{p}}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{p}] = - \left(\frac{\partial V}{\partial \hat{q}} \right) \quad (2.16a)$$

$$\frac{d\hat{q}}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{q}] = \hat{p}/m \quad (2.16b)$$

This looks remarkably similar to the classical equations of motion.

In fact, it means that the average position/momentum follow a trajectory

$$\frac{d}{dt} \langle \hat{p}(t) \rangle = - \langle \nabla V(\hat{q}(t)) \rangle \frac{d}{dt} \langle \hat{q}(t) \rangle \quad == \quad \langle \hat{p}(t) \rangle / m \quad (2.17a)$$

which is known as the Ehrenfest theorem. In special cases only, $\langle \nabla V(\hat{q}) \rangle = \nabla V(\langle \hat{q} \rangle)$, e.g. for the harmonic oscillator where the gradient is a linear function of the position, and the average position/momentum really do exactly follow the ordinary classical trajectory. For this reason, the dynamics of the quantum harmonic oscillator has many similarities to the dynamics of the classical harmonic oscillator. However, in general, the average of the wavepacket only approximately follows the classical motion and can deviate significantly after a long propagation time.

2.2.3 Differences between mixed states and superpositions of states

A linear combination (or superposition) of eigenstates such as

$$\Psi(x) = c_1\psi_1(x) + c_2\psi_2(x) \quad (2.18)$$

depends on time as

$$\Psi(x, t) = c_1\psi_1(x, t) + c_2\psi_2(x, t) \quad (2.19a)$$

$$= c_1 e^{-iE_1 t/\hbar} \psi_1(x) + c_2 e^{-iE_2 t/\hbar} \psi_2(x). \quad (2.19b)$$

Thus the probability density oscillates as

$$|\Psi(x, t)|^2 = |c_1|^2 |\psi_1(x)|^2 + |c_2|^2 |\psi_2(x)|^2 + 2 \operatorname{Re}[c_1^* c_2 \psi_1(x)^* \psi_2(x)] \cos(E_2 - E_1)t/\hbar. \quad (2.20)$$

The dynamics of mixed states is simple as they are independent. For the density operator,

$$\hat{\rho} = P_1 |\psi_1\rangle\langle\psi_1| + P_2 |\psi_2\rangle\langle\psi_2|, \quad (2.21)$$

there is no time dependence,

$$\hat{\rho}(t) = e^{-iE_1 t/\hbar} P_1 e^{iE_1 t/\hbar} |\psi_1\rangle\langle\psi_1| + e^{-iE_2 t/\hbar} P_2 e^{iE_2 t/\hbar} |\psi_2\rangle\langle\psi_2| = \hat{\rho}.$$

This is true only for density matrices which are diagonal in the eigenstate basis, and the interference terms appear as a consequence of off-diagonal elements. The density matrix

$$\hat{\rho} = c_1 c_1^* |\psi_1\rangle\langle\psi_1| + c_1 c_2^* |\psi_1\rangle\langle\psi_2| + c_2 c_1^* |\psi_2\rangle\langle\psi_1| + c_2 c_2^* |\psi_2\rangle\langle\psi_2| \quad (2.22)$$

has the time dependence

$$\hat{\rho}(t) = |c_1|^2 |\psi_1\rangle\langle\psi_1| + e^{-i(E_1 - E_2)t/\hbar} c_1 c_2^* |\psi_1\rangle\langle\psi_2| + e^{-i(E_2 - E_1)t/\hbar} c_2 c_1^* |\psi_2\rangle\langle\psi_1| + |c_2|^2 |\psi_2\rangle\langle\psi_2|. \quad (2.23)$$

The off-diagonal elements are thus termed “coherence” terms.

2.2.4 Tunnelling in ammonia

We first consider states in the localized left or right potential minimum ϕ_ℓ , ϕ_r . The lowest lying stationary states have energies E_0 and E_1 . The corresponding wave functions are

$$\psi_0 \simeq \frac{1}{\sqrt{2}}(\phi_\ell + \phi_r) \quad (2.24a)$$

$$\psi_1 \simeq \frac{1}{\sqrt{2}}(\phi_\ell - \phi_r). \quad (2.24b)$$

We assume that at $t = 0$ the wave packet is localized in the left well.

$$\Psi(t = 0) = \frac{1}{\sqrt{2}}(\psi_0 + \psi_1) = \phi_\ell \quad (2.25)$$

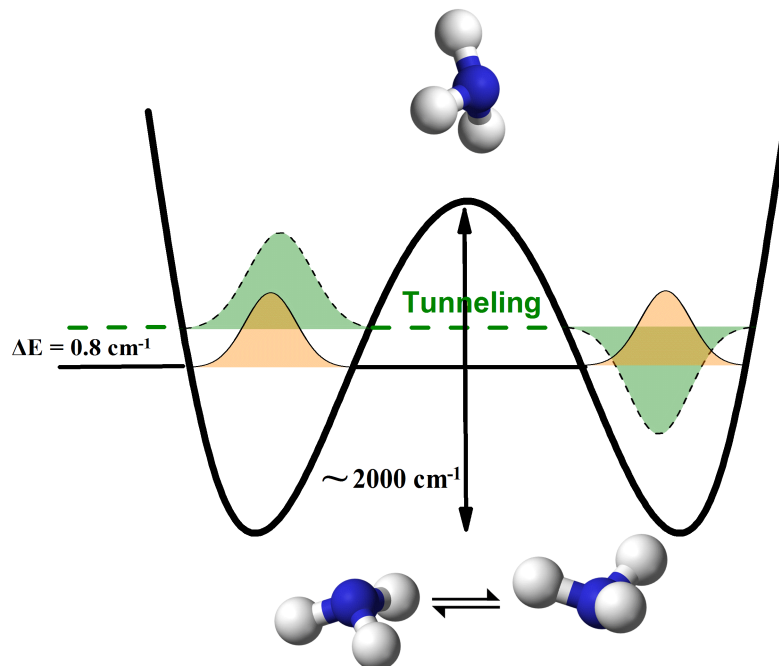


Figure 1: Tunnelling and ammonia inversion

This is not an eigenstate (stationary solution) but a superposition state or wave packet.

$$\Psi(t) = \frac{1}{\sqrt{2}} e^{\frac{-iE_0 t}{\hbar}} (\psi_0 + \psi_1 e^{\frac{-i(E_1 - E_0)t}{\hbar}}). \quad (2.26)$$

For $t_1 = \frac{h}{2(E_1 - E_0)}$ the wave function becomes

$$\Psi(t_1) = \frac{1}{\sqrt{2}} (\psi_0 - \psi_1) = \phi_r \quad (2.27)$$

which gives the oscillation period $T = \frac{h}{\Delta E}$ where $\Delta E = E_1 - E_0$ is the tunnelling splitting. This relation is a general result for 2-state wave packet.

The wavefunction returns to its initial state, called a revival.

2.3 Observables

2.3.1 Time averages

The probability density in classical mechanics is $\rho(\vec{p}, \vec{q})$, for instance $\rho_{\text{eq}} = e^{-\beta H(\vec{p}, \vec{q})}/Q$ is the Boltzmann distribution. The average of an observable $A \equiv A(\vec{p}, \vec{q})$ in a particular distribution ρ is

$$\langle A(t) \rangle_{\rho} = (2\pi\hbar)^{-f} \iint \rho(\vec{p}_0, \vec{q}_0) A(\vec{p}_t, \vec{q}_t) d\vec{p}_0 d\vec{q}_0. \quad (2.28)$$

For the special case of the equilibrium distribution, which due to conservation of energy, is time independent,

$$\langle A(t) \rangle_{\text{eq}} = (2\pi\hbar)^{-f} \iint A(\vec{p}_t, \vec{q}_t) \rho_{\text{eq}}(\vec{p}_t, \vec{q}_t) d\vec{p}_t d\vec{q}_t \quad (2.29)$$

Liouville's theorem states that $d\vec{p}_0 d\vec{q}_0 = d\vec{p}_t d\vec{q}_t$ such that

$$\langle A(t) \rangle_{\text{eq}} = (2\pi\hbar)^{-f} \iint A(\vec{p}_t, \vec{q}_t) \rho_{\text{eq}}(\vec{p}_t, \vec{q}_t) d\vec{p}_t d\vec{q}_t \quad (2.30a)$$

$$= (2\pi\hbar)^{-f} \iint A(\vec{p}, \vec{q}) \rho_{\text{eq}}(\vec{p}, \vec{q}) d\vec{p} d\vec{q} \quad (2.30b)$$

$$= \langle A(0) \rangle_{\text{eq}} \equiv \langle A \rangle_{\text{eq}} \quad (2.30c)$$

where we simply rename the integration variables to prove equivalence with the average value of A at time 0.

So although properties such as bond lengths are oscillating, the ensemble averaged bond length remains constant when in equilibrium.

With quantum mechanics, similar formulae exist

$$\langle \hat{A}(t) \rangle_{\rho} = \text{Tr}[\hat{\rho} \hat{A}(t)] \quad (2.31a)$$

$$= \text{Tr} \left[\hat{\rho} e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} \right] \quad (2.31b)$$

$$= \text{Tr} \left[e^{-i\hat{H}t/\hbar} \hat{\rho} e^{i\hat{H}t/\hbar} \hat{A} \right] \quad (2.31c)$$

$$= \text{Tr} \left[\hat{\rho}(t) \hat{A} \right] \quad (2.31d)$$

where

$$\hat{\rho}(t) = e^{-i\hat{H}t/\hbar} \hat{\rho} e^{i\hat{H}t/\hbar}, \quad (2.32)$$

which implies $\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}]$ (compare with equation (2.13) equation (2.14b) and note minus sign difference). For the equilibrium case,

$$\langle \hat{A}(t) \rangle_{\text{eq}} = \text{Tr} \left[\hat{\rho}_{\text{eq}}(t) \hat{A} \right] \quad (2.33a)$$

$$= \text{Tr} \left[\hat{\rho}_{\text{eq}} \hat{A} \right] \equiv \langle \hat{A} \rangle_{\text{eq}} \quad (2.33b)$$

as the density operator is time independent, i.e. commutes with propagators.

2.3.2 Correlation functions

In order to gain information about the dynamical processes, we define time-correlation functions.

$$\langle A(0)B(t) \rangle = (2\pi\hbar)^{-f} \iint \rho(\vec{p}_0, \vec{q}_0) A(\vec{p}_0, \vec{q}_0) B(\vec{p}_t, \vec{q}_t) d\vec{p}_0 d\vec{q}_0 \quad (2.34)$$

After a long time the correlation between two measurements decays, i.e. the system decorrelates and $\langle A(0)B(t) \rangle \rightarrow \langle A \rangle \langle B \rangle$ as $t \rightarrow \infty$. This is a consequence of chaotic dynamics.

Quantum case

$$\langle \hat{A}(0)\hat{B}(t) \rangle = \text{Tr} \left[\hat{\rho} \hat{A} e^{i\hat{H}t/\hbar} \hat{B} e^{-i\hat{H}t/\hbar} \right] \quad (2.35)$$

2.3.3 Example: harmonic oscillator

To compute classical position autocorrelation functions for harmonic oscillator, use $A = B = x$.

$$\langle x(0)x(t) \rangle = \frac{1}{2\pi\hbar} \iint \rho_{\text{eq}}(x_0, p_0) x_0 x_t dx_0 dp_0 \quad (2.36a)$$

$$= \frac{1}{2\pi\hbar Q} \iint e^{-\beta H(x_0, p_0)} x_0 x_t dx_0 dp_0 \quad (2.36b)$$

$$= \frac{\beta\hbar\omega}{2\pi\hbar} \iint e^{-\beta \left(\frac{p_0^2}{2m} + \frac{1}{2} m\omega^2 x_0^2 \right)} x_0 \left[x_0 \cos \omega t + \frac{p_0}{m\omega} \sin \omega t \right] dx_0 dp_0 \quad (2.36c)$$

$$= \frac{\beta\omega}{2\pi} \sqrt{\frac{2\pi m}{\beta}} \int e^{-\frac{1}{2}\beta m\omega^2 x_0^2} x_0^2 \cos \omega t dx_0 \quad (2.36d)$$

$$= \frac{\beta\omega}{2\pi} \sqrt{\frac{2\pi m}{\beta}} \sqrt{\frac{2\pi}{(\beta m\omega^2)^3}} \cos \omega t \quad (2.36e)$$

$$= \frac{1}{\beta m\omega^2} \cos \omega t \quad (2.36f)$$

2.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Fundamentals of Quantum mechanics

- Dirac, Quantum Mechanics (Oxford, 1948)

Time-dependent Quantum mechanics

- Tannor, Introduction to quantum mechanics: a time-dependent perspective

Time-dependent ensembles

- Tuckerman, Statistical Mechanics (Oxford)
- Zwanzig, Nonequilibrium statistical mechanics (Oxford)

Properties of correlation functions

- Nitzan, Chemical Dynamics in Condensed Phases (Oxford) Sec. 6.3

Molecular Dynamics

- Tuckerman, Statistical Mechanics (Oxford)
- Frenkel and Smit, Understanding Molecular Simulations (Academic Press)

3 Quantum dynamics of molecules

3.1 Born-Oppenheimer approximation

The Born-Oppenheimer approximation has been introduced in previous lectures. Briefly, we write the full molecular Hamiltonian in Cartesian coordinates:

$$\begin{aligned}\hat{H} &= \sum_i -\frac{\nabla_{e,i}^2}{2} + \sum_i -\frac{\nabla_{N,i}^2}{2M_i} + \left(\sum_{j>i} \frac{1}{|\vec{r}_i - \vec{r}_j|} + \sum_{j>i} \frac{Z_i Z_j}{|\vec{R}_i - \vec{R}_j|} - \sum_{i,j} \frac{Z_j}{|\vec{r}_i - \vec{R}_j|} \right) \\ &\equiv \hat{T}_e + \hat{T}_N + \hat{V}_e + \hat{V}_N + \hat{V}_{eN}\end{aligned}\quad (3.1)$$

in atomic units.

- $\{\vec{r}, \nabla_e\}$ refers to electrons.
- $\{\vec{R}, \nabla_N\}$ refers to nuclei with masses M_i .

The TISE in the full coordinate space of nuclei and electrons is

$$\hat{H}(\vec{r}, \vec{R})\psi(\vec{r}, \vec{R}) = E\psi(\vec{r}, \vec{R}). \quad (3.2)$$

Before tackling the solution of this equation, let's consider a simpler problem of solving just the electronic part with nuclei at a fixed configuration $\vec{R}$.

$$\hat{H}_e\varphi(\vec{r}; \vec{R}) = V(\vec{R})\varphi(\vec{r}; \vec{R}), \quad (3.3)$$

where $\hat{H}_e = \hat{T}_e + \hat{V}_e + \hat{V}_{eN} + \hat{V}_N$.

We then assume that the full molecular wave function (adiabatic wave function) can be written as a product of this electronic wave function and a nuclear wave function:

$$\psi(\vec{r}, \vec{R}) = \varphi(\vec{r}; \vec{R})\chi(\vec{R}). \quad (3.4)$$

The reasoning for this assumption is that the electron mass is much smaller than the masses of the nuclei and thus that they should relax on a fast time-scale such that the nuclear dynamics effectively only feel the electrons in a mean-field sense. However, as we shall see this argument sometimes catastrophically breaks down.

Substituting this ansatz into full TISE equation (3.2) we obtain

$$\begin{aligned}\hat{H}(\vec{r}, \vec{R})[\varphi(\vec{r}; \vec{R})\chi(\vec{R})] &= V(\vec{R})\varphi(\vec{r}; \vec{R})\chi(\vec{R}) \\ &\quad - \sum_i \frac{1}{2M_i} \left[\varphi(\vec{r}; \vec{R})\nabla_i^2\chi(\vec{R}) + 2\nabla_i\varphi(\vec{r}; \vec{R})\nabla_i\chi(\vec{R}) + \nabla_i^2\varphi(\vec{r}; \vec{R})\chi(\vec{R}) \right] \quad (3.5)\end{aligned}$$

$$= E\varphi(\vec{r}; \vec{R})\chi(\vec{R}), \quad (3.6)$$

where ∇_i is used to represent $\nabla_{N,i}$.

The last two terms are proportional to m_e/M_N to some power and are therefore several orders of magnitude smaller than the other terms. Neglecting them yields

$$\hat{H}_{B.O.}(\vec{R})\chi(\vec{R}) = [\hat{T}_N + V(\vec{R})]\chi(\vec{R}) = E\chi(\vec{R}), \quad (3.7)$$

which is a TISE for nuclei under an adiabatic Hamiltonian whose effective potential is $V(\vec{R})$; this is the potential-energy surface. The set of approximations leading to equation (3.7) are known as the Born-Oppenheimer approximation.

Often classical mechanics is used to approximate the nuclear dynamics, and one simply uses the classical version of the Born-Oppenheimer Hamiltonian, $\sum_i P_i^2/2M_i + V(\vec{R})$.

3.2 Harmonic approximation

If the potential-energy surface exhibits a minimum at $\vec{R}_0$, and the dynamics is known to oscillate with small amplitudes around it, it is a commonly used approximation to replace the PES by a harmonic oscillator. This is typically done by truncating the Taylor series at second order

$$V(\vec{R}) = V(\vec{R}_0) + \frac{1}{2}(\vec{R} - \vec{R}_0)^T \cdot \nabla^2 V(\vec{R}_0) \cdot (\vec{R} - \vec{R}_0) + \dots, \quad (3.8)$$

where there is no first order term as $\nabla V(\vec{R}_0) = 0$ by definition. The Hamiltonian is therefore

$$H(\vec{P}, \vec{R}) = \sum_i \frac{P_i^2}{2M_i} + V_0 + \frac{1}{2}(\vec{R} - \vec{R}_0)^T \mathbf{K}(\vec{R} - \vec{R}_0), \quad (3.9)$$

where $V_0 = V(\vec{R}_0)$ and the Hessian is $\mathbf{K} = \nabla^2 V(\vec{R}_0)$.

Next we take the normal-mode transform

$$\vec{p} = \mathbf{U}^T \mathbf{M}^{-1/2} \vec{P} \quad (3.10)$$

$$\vec{q} = \mathbf{U}^T \mathbf{M}^{1/2} (\vec{R} - \vec{R}_0), \quad (3.11)$$

where $\mathbf{M}$ is a diagonal matrix with elements M_i along the diagonal and the orthogonal matrix $\mathbf{U}$ has columns which are the eigenvectors of the mass-weighted Hessian such that $\mathbf{U}^T \mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2} \mathbf{U}$ is diagonal with values λ_k along the diagonal.

One thus has a set of independent harmonic oscillators:

$$H(\vec{q}, \vec{p}) = V_0 + \sum_k \frac{1}{2}(p_k^2 + \lambda_k q_k^2). \quad (3.12)$$

3.3 Adiabatic and diabatic representation

The description of molecular dynamics in electronically excited states almost systematically requires a description beyond the Born-Oppenheimer approximation. Such descriptions can be made either in the (unique) adiabatic representation or in one (of infinitely many) diabatic representations. The ensemble of electronic states of any of these representations forms a complete basis.

3.3.1 Adiabatic representation

It is possible to go beyond the Born-Oppenheimer approximation by writing a formally exact solution*

$$\psi(\vec{r}, \vec{R}) = \sum_n \varphi_n(\vec{r}; \vec{R}) \chi_n(\vec{R}). \quad (3.13)$$

*we neglect here the continuum which is, in principle, required

where $\varphi_n(\vec{r}; \vec{R})$ are the set of all electronic states solving

$$\hat{H}_e \varphi_n(\vec{r}; \vec{R}) = V_n(\vec{R}) \varphi_n(\vec{r}; \vec{R}) \quad (3.14)$$

Substituting this ansatz equation (3.13) into equation (3.2), multiplying from the left with $\varphi_m(\vec{r}; \vec{R})$ and integrating over the electronic coordinates yields a set of coupled equations:

$$\sum_n \langle \varphi_m | \hat{H} | \varphi_n \rangle \chi_n(\vec{R}) = \sum_n \langle \varphi_m | \hat{T}_N | \varphi_n \rangle \chi_n(\vec{R}) + \sum_n \langle \varphi_m | \varphi_n \rangle V_n(\vec{R}) \chi_n(\vec{R}) \quad (3.15)$$

$$= [\hat{T}_N + V_m(\vec{R})] \chi_m(\vec{R}) - \sum_n \hat{\Lambda}_{mn} \chi_n(\vec{R}) \quad (3.16)$$

$$= E \chi_m(\vec{R}) \quad (3.17)$$

where $\hat{\Lambda}_{mn}$ are the nonadiabatic couplings defined as

$$\hat{\Lambda}_{mn} = \delta_{mn} \hat{T}_N - \langle \varphi_m | \hat{T}_N | \varphi_n \rangle. \quad (3.18)$$

For example, taking only the lowest two levels, the equation can be written in matrix notation as

$$\begin{pmatrix} \hat{T}_N + V_1(\vec{R}) - \hat{\Lambda}_{00} & -\hat{\Lambda}_{01} \\ -\hat{\Lambda}_{10} & \hat{T}_N + V_2(\vec{R}) - \hat{\Lambda}_{11} \end{pmatrix} \begin{pmatrix} \chi_1(\vec{R}) \\ \chi_2(\vec{R}) \end{pmatrix} = E \begin{pmatrix} \chi_1(\vec{R}) \\ \chi_2(\vec{R}) \end{pmatrix} \quad (3.19)$$

Inserting the position-representation of the nuclear kinetic energy operator gives

$$\hat{\Lambda}_{mn} = \frac{1}{2M} [2\hat{F}_{mn} \cdot \nabla + \hat{G}_{mn}] \quad (3.20)$$

$$\hat{F}_{mn} = \langle \varphi_m | \nabla | \varphi_n \rangle \quad (3.21)$$

$$\hat{G}_{mn} = \langle \varphi_m | \nabla^2 | \varphi_n \rangle \quad (3.22)$$

with Dirac brackets referring only to integration over electronic coordinates. Note that these are operators as they include ∇ , which is a symbol for the multi-dimensional derivative with respect to mass-weighted nuclear coordinates, and to ease the notation, we have assumed that the coordinates have been mass-weighted such that M is the same for all degrees of freedom; $\hat{F}_{mn}$ is a vector with a number of components equal to the number of nuclear degrees of freedom; $\hat{G}_{mn}$ is a scalar. Since the electronic wave functions are normalized at all nuclear geometries, $\langle \varphi_m | \varphi_n \rangle = \delta_{mn}$, it follows that $\hat{F}_{mn}$ is a anti-Hermitian matrix, *i.e.*

$$\hat{F}_{mn} = -\hat{F}_{nm}^*. \quad (3.23)$$

Since electronic wave functions can always be chosen real-valued, it means that diagonal effects of $\hat{F}_{mn}$ vanish: $\hat{F}_{nn}(\vec{R}) = 0$.

As a first approximation, we could keep only the diagonal terms in equation (3.15). We would then obtain the Born–Oppenheimer Schrödinger equation for the nuclei with an additional contribution from $\hat{G}_{nn}$ which is usually small but non-vanishing

$$\left[\hat{T}_N + V_n(\vec{R}) - \frac{\hat{G}_{nn}(\vec{R})}{2M} \right] \chi_n(\vec{R}) = E \chi_n(\vec{R}) \quad (3.24)$$

This is known as the Born-Huang approximation.

In order to find out if we can do this, we should consider the magnitude of the off-diagonal components. Thinking of the nonadiabatic coupling operators as matrices, one can show that $\mathbf{G} = (\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot \mathbf{F}$ and therefore $\hat{T}_N - \Lambda = -\frac{1}{2M}(\nabla + \mathbf{F})^2$. Using this, one does not have to worry about the nonadiabatic scalar couplings G_{nm} anymore.

The question is therefore whether $\hat{F}_{mn}$ is large or small. To find out, we apply first ∇ and then $\langle \varphi_m |$ (where $m \neq n$) to the electronic Schrödinger equation

$$\nabla H_e |\varphi_n\rangle = \nabla V_n |\varphi_n\rangle \quad (3.25)$$

$$(\nabla H_e) |\varphi_n\rangle + H_e |\nabla \varphi_n\rangle = (\nabla V_n) |\varphi_n\rangle + V_n |\nabla \varphi_n\rangle \quad (3.26)$$

$$\langle \varphi_m | (\nabla H_e) |\varphi_n\rangle + \langle \varphi_m | H_e |\nabla \varphi_n\rangle = (\nabla V_n) \langle \varphi_m | \varphi_n\rangle + V_n \langle \varphi_m | \nabla \varphi_n\rangle \quad (3.27)$$

$$\langle \varphi_m | (\nabla H_e) |\varphi_n\rangle + V_m \langle \varphi_m | \nabla \varphi_n\rangle = (\nabla V_n) \times 0 + V_n \langle \varphi_m | \nabla \varphi_n\rangle \quad (3.28)$$

and therefore

$$\hat{F}_{mn} = \frac{\langle \varphi_m | (\nabla H_e) |\varphi_n\rangle}{V_n - V_m}. \quad (3.29)$$

This will become large whenever the states n and m become close in energy, e.g. at avoided crossings. Near these configurations, the Born-Oppenheimer approximation breaks down and it may be possible for the wave function to flow from one state into another. At conical intersections where $V_n = V_m$, the expression blows up. It is therefore necessary to find a different representation which is more convenient.

3.3.2 Diabatic representation

The alternative to the adiabatic basis is a diabatic basis which is chosen such that all electronic matrix elements of the derivatives $\frac{\partial}{\partial R}$ and $\frac{\partial^2}{\partial R^2}$ are as small as possible. There are infinitely many ways of choosing a diabatic basis. The simplest version consists in choosing the electronic basis functions to be independent of the nuclear geometry $\varphi(\vec{r}; \vec{R}) \rightarrow \varphi(\vec{r}; \vec{R}_0)$. This is called the “crude adiabatic” basis in which

$$\psi(\vec{r}, \vec{R}) = \sum_n \varphi_n(\vec{r}; \vec{R}_0) \chi_n^{(0)}(\vec{R}). \quad (3.30)$$

The basis functions $\varphi_n(\vec{r}; \vec{R}_0)$ are solutions of the electronic Schrödinger equation at geometry $\vec{R}_0$:

$$\hat{H}_e(\vec{r}; \vec{R}_0) \varphi_n(\vec{r}; \vec{R}_0) = [\hat{T}_e + V(\vec{r}; \vec{R}_0)] \varphi_n(\vec{r}; \vec{R}_0) \quad (3.31)$$

$$= V_n(\vec{R}_0) \varphi_n(\vec{r}; \vec{R}_0) \quad (3.32)$$

where $V(\vec{r}; \vec{R}_0)$ includes $\hat{V}_e$, $\hat{V}_N$, and $\hat{V}_{eN}$. Substituting equation (3.30) into equation (3.2), multiplying from the left with $\varphi_m(\vec{r}; \vec{R}_0)$ and integrating over the electronic degrees of freedom we obtain

$$\hat{H} \psi(\vec{r}, \vec{R}) = [\hat{T}_N + \hat{H}_e(\vec{r}; \vec{R})] \psi(\vec{r}, \vec{R}) \quad (3.33)$$

$$= [\hat{T}_N + \hat{H}_e(\vec{r}; \vec{R}_0) + V(\vec{r}; \vec{R}) - V(\vec{r}; \vec{R}_0)] \psi(\vec{r}, \vec{R}) \quad (3.34)$$

$$\langle \varphi_m | \hat{H} | \psi \rangle = \sum_n \langle \varphi_m | \hat{T}_N + V_n(\vec{R}_0) + V(\vec{r}; \vec{R}) - V(\vec{r}; \vec{R}_0) | \varphi_n \rangle \chi_n^{(0)}(\vec{R}) \quad (3.35)$$

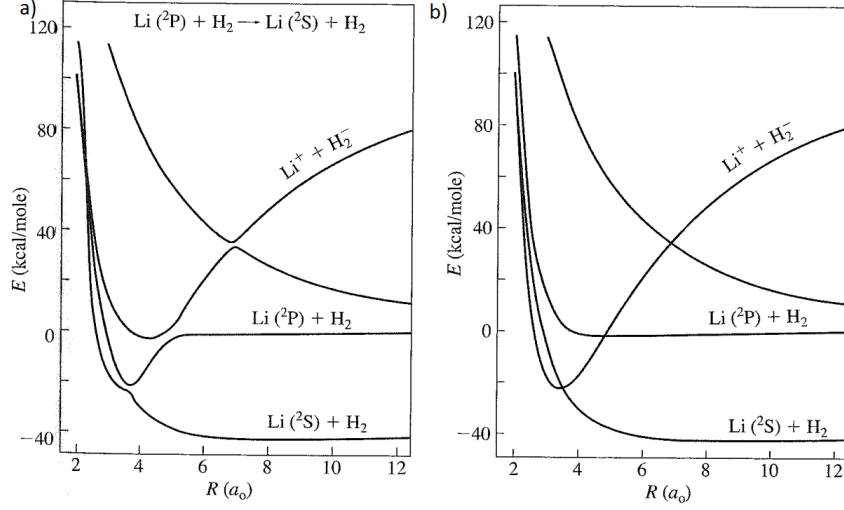


Figure 2: Slices of the four lowest potential energy surfaces of a particular symmetry of LiH_2 as a function of the Li-H_2 separation. The orientation of the H_2 internuclear axis is 60° relative to the Li-H_2 direction, and the H_2 internuclear distance is $1.4 \text{ \AA}$ (adiabatic representation). Note that the adiabatic potentials, being of the same symmetry, obey a “non-crossing” rule. At the avoided crossings, the physical characters of the adiabatic electronic states generally change, for example, from ionic to covalent. (b) Diabatic representation, potentials have continuous physical character. (taken from Introduction to quantum mechanics, David J. Tannor)

and therefore

$$\sum_n \{\hat{T}_N \delta_{nm} + U_{mn}(\vec{R})\} \chi_n^{(0)} = E \chi_m^{(0)}, \quad (3.36)$$

where $U_{mn}(\vec{R}) = \langle \varphi_m(\vec{R}_0) | V_n(\vec{R}_0) + \hat{V}(\vec{r}, \vec{R}) - \hat{V}(\vec{r}, \vec{R}_0) | \varphi_n(\vec{R}_0) \rangle$ is the coupling between the states. As $\hat{H}$ must be Hermitian, so is the diabatic coupling matrix, $U_{mn} = U_{nm}^*$. (Again, brackets imply integration over the electronic coordinates only).

3.3.3 Converting between representations

The adiabatic representation is complicated due to the $\hat{\Lambda}_{nm}$ operators which involve derivatives. In certain cases it is possible to transform the states into a diabatic representation where the derivative couplings disappear. This is not always possible, but sometimes the transformation makes them small enough that they can be neglected.

Let us consider the simplest case where there are only two relevant adiabatic states and suggest the transformation

$$\begin{pmatrix} \tilde{\varphi}_1(\vec{r}; \vec{R}) \\ \tilde{\varphi}_2(\vec{r}; \vec{R}) \end{pmatrix} = \mathbf{S}(\vec{R}) \begin{pmatrix} \varphi_1(\vec{r}; \vec{R}) \\ \varphi_2(\vec{r}; \vec{R}) \end{pmatrix} \quad \mathbf{S}(\vec{R}) = \begin{pmatrix} \cos \gamma(\vec{R}) & \sin \gamma(\vec{R}) \\ -\sin \gamma(\vec{R}) & \cos \gamma(\vec{R}) \end{pmatrix} \quad (3.37)$$

where we will choose function $\gamma(\vec{R})$ later.

Consider the $\hat{F}$ operator in the new basis:

$$\tilde{F}_{12}(\vec{R}) = \langle \tilde{\varphi}_1 | \nabla \tilde{\varphi}_2 \rangle \quad (3.38)$$

$$= \sum_{nm} \langle \varphi_n | S_{1n} (\nabla S_{2m}) | \varphi_m \rangle + \sum_{nm} \langle \varphi_n | S_{1n} S_{2m} | \nabla \varphi_m \rangle \quad (3.39)$$

$$= -\nabla \gamma(\vec{R}) + F_{12}(\vec{R}), \quad (3.40)$$

where the final expression follows from differentiating the transformation matrix and using trigonometric identities.

If we choose $\gamma(\vec{R})$ such that equation (3.40) is equal to 0, then we will have removed the derivative coupling terms. This is easy to do for a one-dimensional system by choosing $\gamma(R) = \int_0^R F_{12}(R') dR'$.

Therefore in the diabatic basis, our Schrödinger equation becomes

$$\begin{pmatrix} \hat{T}_N + U_{11}(\vec{R}) & U_{12}(\vec{R}) \\ U_{21}(\vec{R}) & \hat{T}_N + U_{22}(\vec{R}) \end{pmatrix} \begin{pmatrix} \tilde{\chi}_1(\vec{R}) \\ \tilde{\chi}_2(\vec{R}) \end{pmatrix} = E \begin{pmatrix} \tilde{\chi}_1(\vec{R}) \\ \tilde{\chi}_2(\vec{R}) \end{pmatrix} \quad (3.41)$$

with the diabatic potential-energy matrix

$$\mathbf{U}(\vec{R}) = \mathbf{S}(\vec{R})^T \begin{pmatrix} V_1(\vec{R}) & 0 \\ 0 & V_2(\vec{R}) \end{pmatrix} \mathbf{S}(\vec{R}) \quad (3.42)$$

Note the simplicity of equation (3.41) compared to equation (3.19). The transformation from the adiabatic into the diabatic basis has also transformed derivative couplings $(\frac{d}{dR})_{mn}$ into coordinate couplings $(\hat{U}(R))_{mn}$. Diagonal elements of U are diabatic surfaces and off-diagonal elements are non-adiabatic couplings which can cause non-adiabatic transitions. In practice one looks for a conventional “diabatization scheme” that minimizes the coupling. The complete elimination of couplings is only possible for **diatomics**. The situation for **polyatomics** is generally more complicated and some nonadiabatic couplings remain.

Converting from the diabatic representation to the adiabatic representation is however much easier. The $\mathbf{S}$ matrix is defined uniquely as the one which diagonalizes the $\mathbf{U}$ matrix, i.e. rearrange equation (3.42).

3.4 Adiabatic dynamics: Quantum adiabatic theorem

We consider a quantum system with discrete level structure and a Hamiltonian given by:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t). \quad (3.43)$$

The matrix representation of $\hat{H}$ in the basis of eigenstates of $\hat{H}_0$ is:

$$H(t) = \begin{pmatrix} E_a & V_{ab}(t) & \dots \\ V_{ba}(t) & E_b & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

We seek a solution of the corresponding TDSE, which can be written as

$$\psi(t) = \begin{pmatrix} \psi_a(t) \\ \psi_b(t) \\ \vdots \end{pmatrix}.$$

One can define a unitary transformation that diagonalizes the instantaneous Hamiltonian $H(t)$:

$$\begin{aligned} S^\dagger(t)H(t)S(t) &= D(t) \\ S^\dagger(t)\psi(t) &= \tilde{\psi}(t), \end{aligned} \quad (3.44)$$

where $D(t)$ is a diagonal matrix and $S^\dagger(t) = S^{-1}(t)$.

The TDSE can be written as

$$i\hbar \frac{\partial}{\partial t} \left(S(t)\tilde{\psi}(t) \right) = i\hbar \left(S(t) \frac{\partial \tilde{\psi}(t)}{\partial t} + \frac{\partial S(t)}{\partial t} \tilde{\psi}(t) \right) = H(t)S(t)\tilde{\psi}(t). \quad (3.45)$$

Pre-multiplying both sides by $S^\dagger(t)$ and rearranging gives

$$i\hbar \frac{\partial}{\partial t} \tilde{\psi}(t) = D(t)\tilde{\psi}(t) - i\hbar S^\dagger(t) \frac{\partial S(t)}{\partial t} \tilde{\psi}(t). \quad (3.46)$$

N.B. $S^\dagger(t)S(t) = I$ so $0 = \frac{\partial}{\partial t} S^\dagger(t)S(t) = \frac{\partial S^\dagger(t)}{\partial t} S(t) + S^\dagger(t) \frac{\partial S(t)}{\partial t}$ so the last line can be alternatively written as

$$i\hbar \frac{\partial}{\partial t} \tilde{\psi}(t) = D(t)\tilde{\psi}(t) + i\hbar \frac{\partial S^\dagger(t)}{\partial t} S(t)\tilde{\psi}(t).$$

If $H(t)$ is slowly varying, so will be $S(t)$, and in this case $\frac{\partial S(t)}{\partial t}$ may be neglected. This is called the *adiabatic approximation*.

Adiabatic theorem: If $\psi(0)$ is an eigenfunction of $H(0)$ and $H(t)$ is a slowly-varying function of time, then $\psi(t)$ will evolve in such a way as to remain an eigenfunction of $H(t)$ for all time, in the limit of infinitely-slowly-varying $H(t)$. (see A.Messiah Quantum Mechanics, North Holland Publishing Company, 1967 for the detailed derivation).

The Born–Oppenheimer approximation is an example of an adiabatic approximation, although in this case, the off-diagonal couplings depend implicitly on time via the nuclear positions. Thus if the nuclei move slowly enough, the couplings can be neglected.

3.5 Intersections between adiabatic potential-energy surfaces

Particularly interesting is the case of **conical intersections**. We examine the intersection of adiabatic surfaces. In diatomics, adiabatic curves belonging to electronic states of the same symmetry generally do not cross but exhibit avoided crossings. The proof of this non-crossing rule is simple in a diabatic basis:

$$\mathbf{U} = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}.$$

where $U_{21} = U_{12}^*$ so that the matrix is Hermitian. The off-diagonal elements are zero, unless the states 1 and 2 have the same electronic symmetry. Diagonalization gives the adiabatic potentials

$$V_{\pm} = \frac{U_{11} + U_{22}}{2} \pm \frac{1}{2} \sqrt{(U_{11} - U_{22})^2 + 4|U_{12}|^2}. \quad (3.47)$$

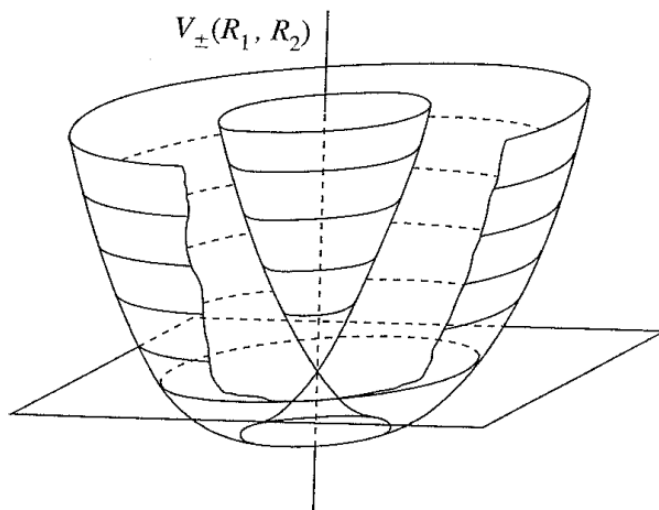


Figure 3: Potential energy functions $V_{\pm}$ as a function of two non-symmetric nuclear coordinates showing the geometry of a conical intersection. (taken from Introduction to quantum mechanics, David J. Tannor)

The energy gap between the curves is

$$\Delta V = V_+ - V_- = \sqrt{(U_{11} - U_{22})^2 + 4|U_{12}|^2} \quad (3.48)$$

Curves only cross if $U_{11}(R) = U_{22}(R)$ and $U_{12}(R) = 0$ which represent two conditions to be met for one variable R . There is in general no reason for both these conditions to be met simultaneously in a one-dimensional system, leading to the non-crossing rule for diatomics. In two dimensions there is generally a single (or a discrete set of) points where the two surfaces touch leading to so-called conical intersections. In $n > 2$ dimensions, the intersection space is normally a manifold of dimension $n-2$ resulting from the two constraints given above e.g. triatomics with $3N - 6 = 3$ degrees of freedom have a one-dimensional “seam” of conical intersections.

Consider the simple case of

$$\mathbf{U} = \begin{pmatrix} ax & by \\ by & -ax \end{pmatrix} \quad (3.49)$$

and convert to polar coordinates $ax = r \cos \theta$ and $by = r \sin \theta$:

$$\mathbf{U} = \begin{pmatrix} r \cos \theta & r \sin \theta \\ r \sin \theta & -r \cos \theta \end{pmatrix}. \quad (3.50)$$

This diabatic potential-energy matrix has eigenvalues $\pm r$ and the lower (Born–Oppenheimer) state has the eigenvector $(c_1, c_2) = (\sin \frac{\theta}{2}, -\cos \frac{\theta}{2})$. The total electronic wavefunction, $\psi = c_1 \phi_1 + c_2 \phi_2$, thus changes sign when θ goes from 0 to 2π . Therefore in order for the total molecular wavefunction to be single valued everywhere, the nuclear wavefunction must also change sign. In this way, the

existence of a conical intersection can have strong effects on spectroscopy and dynamics even when the dynamics occurs at energies far below the intersection and otherwise proceeds purely on the ground-state Born–Oppenheimer PES. This known as the *geometric-phase effect*.

3.6 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Adiabatic and diabatic representations

- W. Domcke and D. R. Yarkony and H. Köppel, Conical Intersections: Electronic Structure, Dynamics and Spectroscopy [chapter 1 has a good description of nonadiabatic coupling]
- Tannor, Introduction to quantum mechanics: a time-dependent perspective [chapter 12]
- May and Kühn, Charge and Energy Transfer Dynamics in Molecular Systems (Wiley, 2011)

Geometric-phase effect

- Baer, Beyond Born–Oppenheimer (Wiley, 2006) [goes deep into maths, well beyond this course, but maybe you’ll find it interesting in the future]

4 Light–matter interaction and spectra

4.1 Time-dependent Hamiltonians

The TDSE is:

$$i\hbar \frac{\partial \Psi(t)}{\partial t} = \hat{H}(t)\Psi(t). \quad (4.1)$$

In this section we introduce time-dependent perturbation theory. We assume that the Hamiltonian can be written as a time-independent and a (possibly) time-dependent term, $\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t)$. It is used to develop a series of successive approximations to the evolving wave functions. Time-dependent perturbation theory is:

- Useful if dynamics under $\hat{H}_0$ are known
- Only recommended if $\hat{H}_1 \ll \hat{H}_0$
- derived either using standard perturbation theory or more directly from the interaction picture. We use the latter derivation here.

The propagator associated with $\hat{H}_0$ is given by

$$i\hbar \frac{\partial}{\partial t} \hat{U}_0(t, t_0) = \hat{H}_0 \hat{U}_0(t, t_0). \quad (4.2)$$

As before, the solution to this is $\hat{U}_0(t, t_0) = e^{-i\hat{H}_0(t-t_0)/\hbar}$.

The full propagator obeys the equation

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H}(t)\hat{U}(t, t_0). \quad (4.3)$$

If $\hat{H}(t)$ is time-dependent, the solution to this equation cannot be written down in a simple way. Nonetheless we can define the interaction-picture propagator as

$$\hat{U}_I(t, t_0) = \hat{U}_0(t, t_0)^\dagger \hat{U}(t, t_0). \quad (4.4)$$

Substituting $\hat{U} = \hat{U}_0 \hat{U}_I$ into the TDSE gives

$$i\hbar \frac{\partial}{\partial t} \hat{U}_I(t, t_0) = \hat{H}_I(t) \hat{U}_I(t, t_0) \quad (4.5)$$

$$\text{where} \quad \hat{H}_I(t) = e^{\frac{i\hat{H}_0(t-t_0)}{\hbar}} \hat{H}_1(t) e^{-\frac{i\hat{H}_0(t-t_0)}{\hbar}}.$$

Exercise: derive this formula.

Equation (4.5) is solved by

$$\hat{U}_I(t, t_0) = 1 + \sum_{n=1}^{\infty} \hat{U}_I^{(n)}(t, t_0) \quad (4.6)$$

where

$$\hat{U}_I^{(n)}(t, t_0) = \frac{1}{(i\hbar)^n} \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau_{n-1} \cdots \int_{t_0}^{\tau_2} d\tau_1 \hat{H}_I(\tau_n) \hat{H}_I(\tau_{n-1}) \cdots \hat{H}_I(\tau_1) \quad (4.7)$$

with ordering $t > \tau_n > \tau_{n-1} > \dots > \tau_1 > t_0$.

Exercise: Show that this is a valid solution.

Therefore

$$\hat{U}(t, t_0) = \hat{U}_0(t, t_0) + \sum_{n=1}^{\infty} \hat{U}^{(n)}(t, t_0) \quad (4.8)$$

where

$$\begin{aligned} \hat{U}^{(n)}(t, t_0) = & \frac{1}{(i\hbar)^n} \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau_{n-1} \dots \int_{t_0}^{\tau_2} d\tau_1 \hat{U}_0(t, \tau_n) \hat{H}_1(\tau_n) \hat{U}_0(\tau_n, \tau_{n-1}) \\ & \times \hat{H}_1(\tau_{n-1}) \dots \hat{U}_0(\tau_2, \tau_1) \hat{H}_1(\tau_1) \hat{U}_0(\tau_1, t_0). \end{aligned} \quad (4.9)$$

Substituting $\hat{U}_0(t, t') = e^{\frac{-i\hat{H}_0(t-t')}{\hbar}}$ gives

$$\begin{aligned} \psi^{(1)}(t) &= \frac{1}{i\hbar} \int_{t_0}^t dt' e^{\frac{-i\hat{H}_0(t-t')}{\hbar}} \hat{H}_1(t') e^{\frac{-i\hat{H}_0(t'-t_0)}{\hbar}} \psi(t_0) \\ \psi^{(2)}(t) &= \frac{1}{(i\hbar)^2} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' e^{\frac{-i\hat{H}_0(t-t')}{\hbar}} \hat{H}_1(t') e^{\frac{-i\hat{H}_0(t'-t'')}{\hbar}} \hat{H}_1(t'') e^{\frac{-i\hat{H}_0(t''-t_0)}{\hbar}} \psi(t_0). \end{aligned} \quad (4.10)$$

4.2 Interaction of light with matter

Here we make a couple of assumptions.

Dipole approximation: valid if wavelength of light is much greater than size of molecule

Optical domain: assume that $\hat{V}_{\text{el}}$ dominates over $\hat{V}_{\text{mag}}$.

We therefore introduce the light-matter interaction:

$$\hat{V}_{\text{el}}(t) = -\vec{\mu}_{\text{el}} \cdot \vec{\mathcal{E}}(t) \quad (4.11)$$

where $\vec{\mathcal{E}}(t)$ is the instantaneous electric field of the light. In other courses on EPR and NMR, you may have seen a similar expression for the magnetic interaction:

$$\hat{V}_{\text{mag}}(t) = -\vec{\mu}_{\text{mag}} \cdot \vec{B}(t). \quad (4.12)$$

In this course, we will only consider electric-field interactions and thus drop the subscript “el” in the following.

The (electric) dipole moment vector is defined in terms of the positions of all charges (including electrons and nuclei) in the molecule

$$\vec{\mu} = \sum_i Z_i \vec{R}_i - \sum_j \vec{r}_j. \quad (4.13)$$

With a continuous-wave monochromatic laser, one has the electric field

$$\vec{\mathcal{E}}(t) = \vec{\mathcal{E}}_0 \cos(\omega t) \quad (4.14)$$

and in the opposite limit of a short pulse at time t_0 :

$$\vec{\mathcal{E}}(t) \propto \delta(t - t_0). \quad (4.15)$$

In reality the fields are typically intermediate between these two limits.

4.3 Dipole correlation functions and infra-red spectra

Here, we will assume that the Born–Oppenheimer approximation holds and add a continuous-wave light field. Our total wavefunctions can be written as a product of an electronic part, $|\phi\rangle$, and a time-dependent nuclear part $|\chi(t)\rangle$. We integrate out all electronic variables to get the Born–Oppenheimer Hamiltonian $\hat{H}_0$ and a perturbing Hamiltonian

$$\hat{H}_1(t) = -\hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t), \quad (4.16)$$

where the dipole moment operator, which depends on nuclear positions only, is

$$\hat{\mu}(\vec{R}) = \langle \phi | \vec{\mu} | \phi \rangle_{\vec{r}} \quad (4.17)$$

and here the bracket indicates integration over electronic variables only.

We choose our initial wavefunction, $|\chi(t_0)\rangle$, to be a vibrational eigenstate on the ground electronic state and we wish to calculate how much population of this state is lost to another vibrational state. This will be proportional to the amount of light absorbed, which is what is measured in an IR spectrum.

The first-order perturbed nuclear wavefunction at time t is

$$|\chi^{(1)}(t)\rangle = \frac{-1}{i\hbar} \int_{t_0}^t dt' e^{-i\hat{H}_0(t-t')/\hbar} \hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t') e^{-i\hat{H}_0(t'-t_0)/\hbar} |\chi(t_0)\rangle \quad (4.18)$$

Integrating over the absolute-value squared of the first-order perturbed wavefunction gives

$$\begin{aligned} \langle \chi^{(1)}(t) | \chi^{(1)}(t) \rangle &= \frac{1}{\hbar^2} \int_{t_0}^t dt'' \int_{t_0}^t dt' \langle \chi(t_0) | e^{i\hat{H}_0(t''-t_0)/\hbar} \hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t'') e^{i\hat{H}_0(t-t'')/\hbar} \\ &\quad \times e^{-i\hat{H}_0(t-t')/\hbar} \hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t') e^{-i\hat{H}_0(t'-t_0)/\hbar} | \chi(t_0) \rangle \end{aligned} \quad (4.19)$$

where here the bracket indicates integration over nuclear variables. We can now take the limits that $t_0 \rightarrow -\infty$ and $t \rightarrow \infty$ and remove phase terms which anyway cancel to give the absorption spectrum as

$$I(\omega) \propto \langle \chi^{(1)}(\infty) | \chi^{(1)}(\infty) \rangle \quad (4.20)$$

$$\propto \frac{1}{\hbar^2} \iint_{-\infty}^{\infty} dt'' dt' \langle \chi(-\infty) | \hat{\mu} e^{i\hat{H}_0(t'-t'')/\hbar} \hat{\mu} e^{-i\hat{H}_0(t'-t'')/\hbar} | \chi(-\infty) \rangle \cos(\omega t'') \cos(\omega t') \quad (4.21)$$

Bearing in mind that

$$\cos(\omega t') \cos(\omega t'') = \frac{1}{2} [\cos \omega(t' - t'') + \cos \omega(t' + t'')] \quad (4.22)$$

we see that only the first term contributes, as the second averages out to 0. Therefore, changing variables again using $t = t' - t''$, we get

$$\langle \chi^{(1)}(\infty) | \chi^{(1)}(\infty) \rangle \propto \int_{-\infty}^{\infty} dt \langle \chi(-\infty) | \hat{\mu} e^{i\hat{H}_0 t/\hbar} \hat{\mu} e^{-i\hat{H}_0 t/\hbar} | \chi(-\infty) \rangle \cos(\omega t) \quad (4.23)$$

$$\propto \int_{-\infty}^{\infty} dt \langle \chi(-\infty) | \hat{\mu} \cdot \hat{\mu}(t) | \chi(-\infty) \rangle \cos(\omega t) \quad (4.24)$$

This is easily extended to treat an ensemble of initial states described by the density matrix $\hat{\rho}$ and gives the formula for the absorption as the Fourier (cosine) transform

$$I(\omega) \propto \int_{-\infty}^{\infty} C_{\mu\mu}(t) \cos(\omega t) dt \quad (4.25)$$

where the dipole-dipole correlation function is

$$C_{\mu\mu}(t) = \text{Tr}[\hat{\rho} \hat{\mu} \hat{\mu}(t)]. \quad (4.26)$$

In the simplest cases, we can expand the dipole operator as a Taylor series around the minimum-energy geometry, $\vec{R}_0$, as

$$\hat{\mu}(\vec{R}) = \mu(\vec{R}_0) + \nabla \mu(\vec{R}_0) \cdot (\vec{R} - \vec{R}_0) + \dots \quad (4.27)$$

Truncating this series after the first term gives a simple approximation. Then in equilibrium we have

$$C_{\mu\mu}(t) \approx [\nabla \mu(\vec{R}_0)]^2 \text{Tr}[\hat{\rho}_{\text{eq}}(\vec{R} - \vec{R}_0) \cdot (\vec{R}(t) - \vec{R}_0)] \quad (4.28)$$

because all other terms are identically zero. This therefore becomes the position-position correlation function. Note that you already computed this correlation function for the harmonic oscillator in the second lecture. It was proportional $\cos(\omega_0 t)$, where ω_0 is the frequency of the oscillator. By Fourier-transforming it, you can find the infra-red spectrum of the harmonic oscillator, which gives two delta functions, one at $\omega = \omega_0$ and the other at $\omega = -\omega_0$ corresponding to absorption and emission of light.

We can check that this approach based on correlation functions gives us the same spectrum as the time-independent approach that you are more familiar with. To do this, we evaluate the trace in equation (5.55) in the basis of (vibrational) eigenstates to get

$$C_{\mu\mu}(t) = \sum_{nm} P_n \langle n | \hat{\mu} | m \rangle \langle m | \hat{\mu}(t) | n \rangle \quad (4.29)$$

$$= \sum_{nm} P_n \mu_{nm} e^{iE_m t/\hbar} \mu_{mn} e^{-iE_n t/\hbar} \quad (4.30)$$

where $\mu_{nm} = \langle n | \hat{\mu} | m \rangle$ is the transition dipole moment. Taking the Fourier transform gives the spectrum

$$I(\omega) \propto \int_{-\infty}^{\infty} \sum_{nm} P_n |\mu_{nm}|^2 \left[e^{i(\omega_{nm} - \omega)t} + e^{i(\omega_{nm} + \omega)t} \right] dt \quad (4.31)$$

$$\propto \sum_{nm} P_n |\mu_{nm}|^2 [\delta(\omega_{nm} - \omega) + \delta(\omega_{nm} + \omega)] \quad (4.32)$$

where $\omega_{nm} = (E_m - E_n)/\hbar$. This is a stick spectrum of lines appearing at frequencies corresponding to the gaps between energy levels weighted by the Boltzman distribution of the initial state and the transition dipole moment squared. The first delta function describes absorption whereas the second describes stimulated emission.

4.4 Rabi oscillations in two-level system

In this section we allow the field strength to be strong, and thus cannot employ the perturbation approach.

We consider a two-level system described by a field-free Hamiltonian $\hat{H}_{\text{mol}}$ and a time-dependent interaction $\hat{V}(t)$ with an electromagnetic field:

$$\mathbf{H} = \begin{pmatrix} \hbar\omega_a & -\mu\epsilon \cos(\omega t) \\ -\mu\epsilon \cos(\omega t) & \hbar\omega_b \end{pmatrix}, \quad (4.33)$$

where $\mu\epsilon = \langle \psi_a | \hat{\mu} | \psi_b \rangle E_0$.

We use the interaction picture to write the wave function as

$$\psi(t) = a(t)e^{-i\omega_a t}\psi_a + b(t)e^{-i\omega_b t}\psi_b. \quad (4.34)$$

And therefore the Schrödinger equation is

$$\dot{a}(t) = \frac{i\mu\epsilon}{2\hbar} \left[e^{i(\omega-\omega_{ab})t} + e^{-i(\omega+\omega_{ab})t} \right] b(t) \quad (4.35a)$$

$$\dot{b}(t) = \frac{i\mu\epsilon}{2\hbar} \left[e^{-i(\omega-\omega_{ab})t} + e^{i(\omega+\omega_{ab})t} \right] a(t), \quad (4.35b)$$

where $\omega_{ab} = \omega_b - \omega_a$.

Only approximate solutions of this system of coupled differential equations are known analytically. We employ the “rotating wave approximation” which is valid if the frequency of the light, ω , is close to ω_{ab} , the energy gap. If the detuning $\Delta\omega = \omega - \omega_{ab}$ is small, i.e. $\Delta\omega \ll \omega_{ab}$ then one of the terms will oscillate very fast, and the other very slowly. We can ignore the fast oscillations as they will cancel out in the integration

Therefore within this approximation, we find

$$\dot{a}(t) = \frac{i\mu\epsilon}{2\hbar} e^{i\Delta\omega t} b(t) \quad (4.36a)$$

$$\dot{b}(t) = \frac{i\mu\epsilon}{2\hbar} e^{-i\Delta\omega t} a(t). \quad (4.36b)$$

These equations, along with the initial conditions $|a(0)| = 1$ and $b(0) = 0$, are solved by

$$a(t) = e^{i\Delta\omega t/2} \left(\cos(\Omega t/2) - i \frac{\Delta\omega}{\Omega} \sin(\Omega t/2) \right) \quad (4.37a)$$

$$b(t) = e^{-i\Delta\omega t/2} \frac{\mu\epsilon}{2\hbar\Omega} 2i \sin(\Omega t/2). \quad (4.37b)$$

Therefore for a state starting in a , the population of b goes like

$$|b(t)|^2 = \left(\frac{\mu\epsilon}{\hbar\Omega} \right)^2 \sin^2(\Omega t/2), \quad (4.38)$$

where $\Omega = \sqrt{\Delta\omega^2 + \mu^2\epsilon^2/\hbar^2}$.

- Note that the population dynamics are independent of detuning, $\Delta\omega$, at short times.
- At resonance, $\Delta\omega = 0$, it is possible to put all the population into the excited state if the laser pulse is turned off at $t = \pi/\Omega$.

4.5 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Time-dependent perturbation theory

- Tuckerman, Statistical Mechanics (Oxford), chapter 14
- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapter 9

Spectra and dipole correlation functions

- Heller, The Semiclassical Way to Dynamics and Spectroscopy, chapter 17
- May and Kühn, Charge and Energy Transfer Dynamics in Molecular Systems (Wiley, 2011)

Rabi oscillations

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapter 15
- Grynberg, Aspect and Fabre, Introduction to quantum optics (Cambridge, 2010)

5 Wavepacket dynamics

5.1 Basis functions for a free particle

A free particle has a continuum of states which are eigenfunctions of the Hamiltonian,

$$\hat{H}_0 = \frac{\hat{p}^2}{2m}. \quad (5.1)$$

The eigenstates of this are also the eigenstates of $\hat{p}$

$$\hat{p} |p\rangle = p |p\rangle \quad (5.2)$$

$$\hat{H}_0 |p\rangle = E_p |p\rangle, \quad (5.3)$$

where $E_p = p^2/2m$.

The position-representation of the eigenstate is

$$\langle x|p\rangle = \phi_p(x) \quad (5.4)$$

and is found by solving

$$\hat{p}\phi_p(x) = -i\hbar \frac{\partial}{\partial x} \phi_p(x) = p\phi_p(x) \quad (5.5)$$

to give

$$\boxed{\langle x|p\rangle = \phi_p(x) = (2\pi\hbar)^{-\frac{1}{2}} e^{+ipx/\hbar}.} \quad (5.6)$$

The normalization term has been chosen such that

$$\langle p'|p\rangle = \int_{-\infty}^{\infty} \phi_{p'}(x)^* \phi_p(x) dx \quad (5.7)$$

$$= (2\pi\hbar)^{-1} \int_{-\infty}^{\infty} e^{-i(p'-p)x/\hbar} dx \quad (5.8)$$

$$= \delta(p' - p) \quad (5.9)$$

This is a subtly different normalization scheme from what is used for discrete states where $\langle n|n\rangle = 1$. The momentum eigenstates form a complete basis set such that any one-dimensional wave function (not just those describing free particles) can be written as

$$|\psi\rangle = \int_{-\infty}^{\infty} |p\rangle \langle p|\psi\rangle dp. \quad (5.10)$$

In the same way there are position states

$$\boxed{\langle p|x\rangle = \langle x|p\rangle^* = (2\pi\hbar)^{-\frac{1}{2}} e^{-ipx/\hbar}} \quad (5.11)$$

which are complete such that

$$|\psi\rangle = \int_{-\infty}^{\infty} |x\rangle \langle x|\psi\rangle dx \quad (5.12)$$

and normalized such that

$$\langle x' | x \rangle = \delta(x' - x). \quad (5.13)$$

They are thus like a set of infinitely small pixels which together with the weights $\langle x | \psi \rangle = \psi(x)$ can define a state $|\psi\rangle$.

So the evolution of a free-particle wavepacket is

$$|\psi(t)\rangle = \hat{U}_0(t, t_0) |\psi(t_0)\rangle \quad (5.14)$$

$$= e^{-i\hat{H}_0(t-t_0)/\hbar} |\psi(t_0)\rangle \quad (5.15)$$

$$= e^{-\frac{i\hat{p}^2(t-t_0)}{2m\hbar}} |\psi(t_0)\rangle \quad (5.16)$$

$$= \int e^{-\frac{i\hat{p}^2(t-t_0)}{2m\hbar}} |p\rangle \langle p | \psi(t_0)\rangle dp \quad (5.17)$$

$$= \int |p\rangle e^{-\frac{ip^2(t-t_0)}{2m\hbar}} \langle p | \psi(t_0)\rangle dp. \quad (5.18)$$

We use the word “wavepacket” to emphasize that the wavefunction is a linear combination of eigenstates and therefore has interesting dynamics. In many cases the wavepacket is localized in space at all times.

If we know the position representation of the wavepacket, it is possible to compute the momentum-representation using a Fourier transform:

$$\langle p | \psi(t_0)\rangle = \int \langle p | x \rangle \langle x | \psi(t_0)\rangle dx \quad (5.19)$$

$$= (2\pi\hbar)^{-\frac{1}{2}} \int e^{-ipx/\hbar} \langle x | \psi(t_0)\rangle dx \quad (5.20)$$

Likewise we can convert back using an inverse Fourier transform

$$\langle x | \psi(t_0)\rangle = \int \langle x | p \rangle \langle p | \psi(t_0)\rangle dp \quad (5.21)$$

$$= (2\pi\hbar)^{-\frac{1}{2}} \int e^{ipx/\hbar} \langle p | \psi(t_0)\rangle dp. \quad (5.22)$$

For example, if the initial wavepacket is a Gaussian $\langle x | \psi(t_0)\rangle = \mathcal{N}^{-1} e^{-\alpha(x-x_0)^2/2 + ip_0(x-x_0)/\hbar}$, where x_0 is the centre of the wavepacket and p_0 is the expectation value of the momentum, then the momentum representation is

$$\langle p | \psi(t_0)\rangle = \frac{\mathcal{N}^{-1}}{(2\pi\hbar)^{\frac{1}{2}}} \int e^{-ipx/\hbar} e^{-\alpha(x-x_0)^2/2 + ip_0(x-x_0)/\hbar} dx \quad (5.23)$$

$$= \frac{\mathcal{N}^{-1}}{(2\pi\hbar)^{\frac{1}{2}}} \int e^{-ipx_0/\hbar} e^{-\alpha(x-x_0)^2/2 - i(p-p_0)(x-x_0)/\hbar} dx \quad (5.24)$$

$$= \frac{\mathcal{N}^{-1}}{(2\pi\hbar)^{\frac{1}{2}}} \sqrt{\frac{2\pi}{\alpha}} e^{-ipx_0/\hbar} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} \quad (5.25)$$

$$= \frac{\mathcal{N}^{-1}}{(\hbar\alpha)^{\frac{1}{2}}} e^{-ipx_0/\hbar} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} \quad (5.26)$$

which is another Gaussian (with an unimportant phase factor) centred around p_0 , hence confirming that this parameter can be interpreted as the initial momentum.

Therefore the wavepacket at time t is

$$\langle x|\psi(t)\rangle = \int \langle x|p\rangle e^{-\frac{ip^2(t-t_0)}{2m\hbar}} \langle p|\psi(t_0)\rangle dp \quad (5.27)$$

$$= \frac{\mathcal{N}^{-1}}{\sqrt{2\pi\alpha\hbar^2}} \int e^{ipx/\hbar} e^{-\frac{ip^2(t-t_0)}{2m\hbar}} e^{-ipx_0/\hbar} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} dp \quad (5.28)$$

$$= \frac{\mathcal{N}^{-1}}{\sqrt{2\pi\alpha\hbar^2}} \int e^{ip(x-x_0)/\hbar} e^{-\frac{ip^2(t-t_0)}{2m\hbar}} e^{-\frac{(p-p_0)^2}{2\alpha\hbar^2}} dp \quad (5.29)$$

$$= \mathcal{N}_t^{-1} e^{-\alpha_t(x-x_t)^2/2 + ip_0(x-x_t)/\hbar + ic_t} \quad (5.30)$$

where

$$x_t = x_0 + \frac{p_0}{m}t \quad (5.31)$$

$$\alpha_t = \alpha/(1 + i\hbar\alpha t/m) \quad (5.32)$$

$$c_t = \frac{p_0^2}{2m}t + \frac{i}{2}\hbar \log(1 + i\hbar\alpha t/m) \quad (5.33)$$

So it remains a Gaussian centred around x_t (which moves like a classical particle), with an ever increasing width.*

5.2 Wavepacket dynamics

In previous sections we have propagated wavefunctions by expanding them as a linear combination of eigenstates. In many cases, one does not want to do this, as solving for the eigenstates can be a difficult problem. It will not be possible to solve the problem analytically in general. Nonetheless, it is possible to evolve the wavefunction numerically using the propagator in an appropriate basis.

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle \quad (5.34)$$

First use the property of the propagator:

$$\hat{U}(t, t_0) = \hat{U}(t, t_N) \hat{U}(t_N, t_{N-1}) \cdots \hat{U}(t_2, t_1) \hat{U}(t_1, t_0). \quad (5.35)$$

This is just like saying “travelling forward by a hour is the same as travelling forward by a minute 60 times.”

The Hamiltonian is a sum of kinetic and potential operators, which do not commute with each other.

$$\hat{U}(t_{n+1}, t_n) = e^{-i\hat{H}\epsilon/\hbar} \quad (5.36a)$$

$$= e^{-i(\hat{T}+\hat{V})\epsilon/\hbar} \quad (5.36b)$$

$$\approx e^{-i\hat{V}\epsilon/2\hbar} e^{-i\hat{T}\epsilon/\hbar} e^{-i\hat{V}\epsilon/2\hbar} + \mathcal{O}(\epsilon^3) \quad (5.36c)$$

$$\approx e^{-i\hat{T}\epsilon/\hbar} e^{-i\hat{V}\epsilon/\hbar} + \mathcal{O}(\epsilon^2) \quad (5.36d)$$

*You can derive these formulas for yourself using mathematica. Also see the discussion in Tannor, chapter 3.

where we have given two alternative approximations, one more accurate than the other.

Exercise: take Taylor expansions (in ϵ) of all three expressions [equation (5.36b)–(5.36c)] and confirm that the errors are of order ϵ^2 and ϵ^3 .*

It is easy to apply the potential-energy operator in the position representation because $\hat{V}|x\rangle = V(x)|x\rangle$ and $\langle x|\hat{V} = \langle x|V(x)$ and likewise it is easy to apply the kinetic-energy operator in the momentum representation because $\hat{T}|p\rangle = \frac{p^2}{2m}|p\rangle$ and $\langle p|\hat{T} = \langle p|\frac{p^2}{2m}$. Therefore in order to propagate by the full Hamiltonian, we alternate between $\hat{V}$ and $\hat{T}$ steps and use Fourier transforms [equation (5.19)] to change representation each time. This is called the *split operator* approach.

First we consider the potential-energy term

$$\langle x|\psi^a(t_n)\rangle = \langle x|e^{-i\hat{V}\epsilon/2\hbar}|\psi(t_n)\rangle \quad (5.37)$$

$$= e^{-iV(x)\epsilon/2\hbar} \langle x|\psi(t_n)\rangle \quad (5.38)$$

which simply multiplies the wavefunction by the phase term. Next, we carry out a Fourier transform [equation (5.19)] to obtain $\langle p|\psi^a(t_n)\rangle$. Then we apply the kinetic-energy term

$$\langle p|\psi^b(t_n)\rangle = \langle p|e^{-i\hat{T}\epsilon/\hbar}|\psi^a(t_n)\rangle \quad (5.39)$$

$$= e^{-ip^2\epsilon/2m\hbar} \langle p|\psi^a(t_n)\rangle \quad (5.40)$$

This is inverse Fourier transformed [equation (5.21)] to obtain $\langle x|\psi^b(t_n)\rangle$. Finally, operation by the second potential-energy term gives

$$\langle x|\psi(t_{n+1})\rangle = \langle x|e^{-i\hat{V}\epsilon/2\hbar}|\psi^b(t_n)\rangle \quad (5.41)$$

$$= e^{-iV(x)\epsilon/2\hbar} \langle x|\psi^b(t_n)\rangle \quad (5.42)$$

You should convince yourself that these steps are equivalent to

$$\langle x|\psi(t_{n+1})\rangle = \langle x|\hat{U}(t_{n+1}, t_n)|\psi(t_n)\rangle \quad (5.43)$$

$$\approx \langle x|e^{-i\hat{V}\epsilon/2\hbar}e^{-i\hat{T}\epsilon/\hbar}e^{-i\hat{V}\epsilon/2\hbar}|\psi(t_n)\rangle \quad (5.44)$$

$$= \iint \langle x|e^{-i\hat{V}\epsilon/2\hbar}e^{-i\hat{T}\epsilon/\hbar}|p\rangle \langle p|e^{-i\hat{V}\epsilon/2\hbar}|x'\rangle \langle x'|\psi(t_n)\rangle dx' dp \quad (5.45)$$

$$= \iint e^{-iV(x)\epsilon/2\hbar} \langle x|p\rangle e^{-ip^2\epsilon/2m\hbar} \langle p|x'\rangle e^{-iV(x')\epsilon/2\hbar} \langle x'|\psi(t_n)\rangle dx' dp \quad (5.46)$$

Ultimately, all of these formulas can be easily derived simply from the completeness relations.

5.3 Electronic absorption spectrum

If we have two uncoupled electronic states, the interaction with the light field may cause transitions between the two. For this, we can use the Born–Oppenheimer expansion in terms of the ground and excited states, $|\phi_g\rangle$ and $|\phi_e\rangle$.

The dipole operator of interest is

$$\hat{\mu}_{eg}(\vec{R}) = \langle \phi_e | \vec{\mu} | \phi_g \rangle_{\vec{r}} \quad (5.47)$$

*Solutions are given in Tannor, Exercises 11.1 and 11.2.

and when applied to the initial wavefunction $|\phi_g\rangle |\chi_g(t_0)\rangle$, leads to the perturbed wavefunction on the electronically excited state:

$$|\chi_e^{(1)}(t)\rangle = \frac{-1}{i\hbar} \int_{t_0}^t e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} \cdot \vec{E}_0 \cos(\omega t') e^{-i\hat{H}_g(t'-t_0)/\hbar} |\chi_g(t_0)\rangle dt' \quad (5.48)$$

Let us assume that the initial wavefunction $|\chi_g(t_0)\rangle$ is an eigenstate of $\hat{H}_g$ with eigenvalue $\hbar\omega_g$. Then,

$$|\chi_e^{(1)}(t)\rangle = \frac{-\vec{E}_0}{i\hbar} \cdot \int_{t_0}^t e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt' \quad (5.49)$$

We can interpret this as taking the initial wavepacket $\chi_g(t_0)$ and multiplying it by the $\vec{R}$ -dependent function μ_{eg} . This modified wavepacket is then placed in the excited electronic state, and because it is typically no longer at the equilibrium position, starts to move (see Figure 4).

Following a similar approach to that of the last lecture, we can show that the absorption spectrum can be obtained as the Fourier transform of the dipole-dipole correlation function

$$I(\omega) \propto \langle \chi_e^{(1)}(t) | \chi_e^{(1)}(t) \rangle \quad (5.50)$$

$$\propto \iint_{t_0}^t e^{+i\omega_g(t''-t_0)} \cos(\omega t'') \langle \chi_g(t_0) | \hat{\mu}_{eg} e^{+i\hat{H}_e(t-t'')/\hbar} e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{ge} |\chi_g(t_0)\rangle \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt' dt'' \quad (5.51)$$

$$= \frac{1}{2} \iint_{t_0}^t \langle \chi_g(t_0) | \hat{\mu}_{ge} e^{-i\hat{H}_e(t''-t')/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle e^{i\omega_g(t''-t')} [\cos \omega(t' + t'') + \cos \omega(t'' - t')] dt' dt'' \quad (5.52)$$

where $\hat{\mu}_{ge} = \hat{\mu}_{eg}^\dagger = \langle \phi_g | \vec{\mu} | \phi_e \rangle_{\vec{r}}$. Neglecting the term with $\cos \omega(t' + t'')$, taking the limits $t_0 \rightarrow -\infty$ and $t \rightarrow \infty$, and renaming $t' - t'' = t$, we obtain

$$I(\omega) \propto \int_{-\infty}^{\infty} \langle \chi_g(-\infty) | \hat{\mu}_{ge} e^{-i\hat{H}_e t/\hbar} \hat{\mu}_{eg} | \chi_g(-\infty) \rangle e^{i\omega_g t} \cos(\omega t) dt \quad (5.53)$$

For an equilibrium ensemble, we write

$$I(\omega) \propto \int_{-\infty}^{\infty} C_{\mu\mu}(t) \cos(\omega t) dt \quad (5.54)$$

where

$$C_{\mu\mu}(t) = \text{Tr}[\hat{\rho} \hat{\mu}_{ge} e^{-i\hat{H}_e(t''-t')/\hbar} \hat{\mu}_{eg} e^{i\hat{H}_g(t''-t')}] \quad (5.55)$$

This is the same as above if $\hat{\rho} = |\chi_g\rangle\langle\chi_g|$ but also generalizes the result for an equilibrium ensemble.

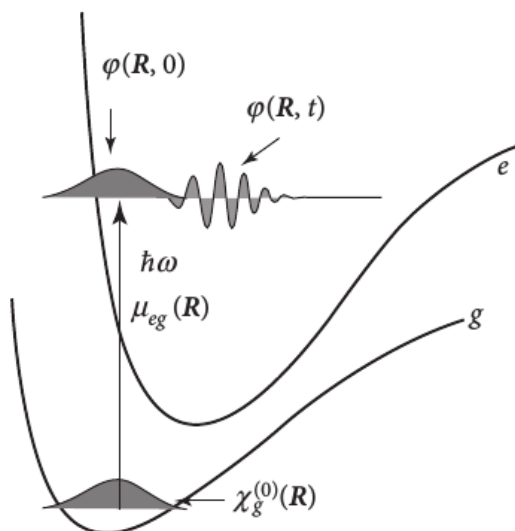


Figure 4: The initial wavepacket on the ground state is $\chi_g^{(0)}(\mathbf{R})$ and this is modified by the dipole operator to give $\phi(\mathbf{R}, 0) = \hat{\mu}\chi_g^{(0)}(\mathbf{R})$ which is now placed on the excited state. This evolves to $\phi(\mathbf{R}, t) = e^{-i\hat{H}_e t/\hbar}\phi(\mathbf{R}, 0)$. From Heller, *The Semiclassical Way to Dynamics and Spectroscopy*.

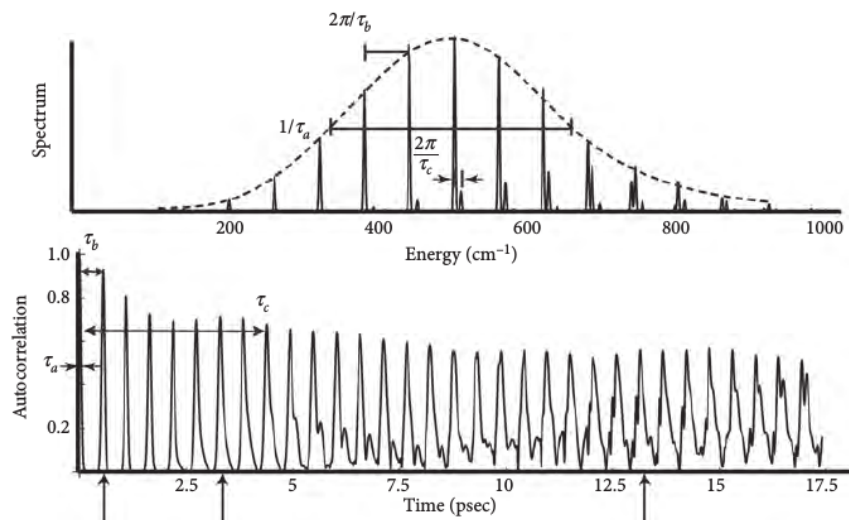


Figure 5: The spectrum and correlation function corresponding to electronic absorption. From Heller, *The Semiclassical Way to Dynamics and Spectroscopy*.

5.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Bras and kets

- Feynman Lectures Volume III chapter 5, <https://www.feynmanlectures.caltech.edu>

Wavepacket propagation

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapters 2, 3 and 11.7.1

Electronic absorption spectroscopy

- Heller, The Semiclassical Way to Dynamics and Spectroscopy, chapter 17
- Tannor, Introduction to quantum mechanics: a time-dependent perspective, section 14.1

Hopefully you may be able to read them on the internet.

6 Intramolecular electronic transitions

In this lecture, we will look more carefully at how molecules change electronic states, either because the transition is induced by a light field or because of nonadiabatic coupling. First we want to understand under what circumstances transitions may take place. We will look at the problem from a time-independent and time-dependent perspective.

6.1 Fermi's Golden-rule

The so-called Fermi golden rule goes back to Wentzel and Dirac (~ 1927) but was named after Fermi's lecture notes (~ 1950).

Consider a state which can decay by (for example) a nonadiabatic transition into a continuum or quasi-continuum. This is the physical situation which occurs in predissociation. Now the Hamiltonian matrix is

$$\mathbf{H} = \begin{pmatrix} E_1 & \Delta_{12} & \Delta_{13} & \cdots \\ \Delta_{12} & E_2 & \Delta_{23} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (6.1)$$

If we start in state m , $b_m = 1$ but $b_n = 0$ for all $n \neq m$. We will use first-order perturbation theory and thus make the approximation that Δ_{nm} is small such that higher-order perturbation terms are not important. This gives

$$b_n(t) = \frac{1}{i\hbar} \int_0^t e^{-iE_n(t-t')/\hbar} \Delta_{nm} e^{-iE_m t'/\hbar} dt' \quad (6.2)$$

$$= \frac{\Delta_{nm}}{i\hbar} e^{-iE_n t/\hbar} \int_0^t e^{i\Omega_{nm} t'} dt' \quad (6.3)$$

$$= \frac{\Delta_{nm}}{i\hbar} e^{-iE_n t/\hbar} \left[\frac{e^{i\Omega_{nm} t'}}{i\Omega_{nm}} \right]_0^t \quad (6.4)$$

$$= \frac{\Delta_{nm}}{\hbar} e^{-iE_n t/\hbar} \frac{e^{i\Omega_{nm} t} - 1}{-\Omega_{nm}}, \quad (6.5)$$

where $\Omega_{nm} = (E_n - E_m)/\hbar$.

Therefore the population of state n at time t is

$$P_{m \rightarrow n}(t) = |b_n(t)|^2 = \frac{\Delta_{nm}^2}{\hbar^2} \frac{4 \sin^2(\Omega_{nm} t/2)}{\Omega_{nm}^2}. \quad (6.6)$$

At short times,

$$\lim_{t \rightarrow 0} P_{m \rightarrow n}(t) = \frac{\Delta_{nm}^2}{\hbar^2} t^2 \quad (6.7)$$

the population increases quadratically with t . This is the same as the behaviour we saw in tunnelling in the ammonia molecule. It is a sign of *coherent* dynamics.

If we are interested not in the rate of a particular transition but of any transition away from m into any state n , we can compute the time-dependent population lost from state m as

$$P_{m \rightarrow ?} = \sum_n P_{m \rightarrow n}(t) \quad (6.8)$$

This function is particularly interesting to study in the long-time limit. In this limit, $\frac{4 \sin^2(\Omega_{nm}t/2)}{\Omega_{nm}^2}$ tends to a delta function $2\pi t \delta(\Omega_{nm}) = 2\pi \hbar t \delta(E_n - E_m)$ to give

$$\lim_{t \rightarrow \infty} P_{m \rightarrow ?}(t) = \frac{2\pi}{\hbar} t \sum_n \Delta_{nm}^2 \delta(E_n - E_m). \quad (6.9)$$

Therefore, in the long-time limit, the population increases linearly with time. Thus a transition is only possible if the final state has the same energy as the initial state. Energy is required to be conserved in such transitions. It is only valid if the first-order perturbation theory is appropriate. This would not be the case if you had a two-state system as then, in the long-time limit, one would need to take higher-order terms into account in order to obtain the oscillatory dynamics we saw previously. However, if there are a huge number of possible final states, each one only increases its population very slightly, and thus the first-order perturbation is valid and hence taken together the population transition occurs linearly in time. This is a sign of *incoherent* dynamics. You can easily see for yourself how the dynamics changes completely in these two regimes using the simple Bixon-Jortner model and starting in a reactant state which is coupled to a large number of product states uncoupled to each other. Plotting the population of the reactant state against time will show incoherent dynamics if the density of product states is high enough and coherent dynamics if the density is low enough. In intermediate regimes, it gets complicated and there is no simple phenomenological description.

We can define the rate constant for transitions out of m :

$$k_m = \lim_{t \rightarrow \infty} P_{m \rightarrow ?}(t)/t \quad (6.10)$$

$$= \sum_n \frac{2\pi}{\hbar} \Delta_{nm}^2 \delta(E_n - E_m). \quad (6.11)$$

To find the rate in thermal equilibrium, we simply sum over the rates from initial states m weighted by the Boltzmann distribution

$$k(T) = \sum_m e^{-\beta E_m} \sum_n \frac{2\pi}{\hbar} \Delta_{nm}^2 \delta(E_n - E_m). \quad (6.12)$$

6.2 Floquet theory

In the previous section, we saw that transitions only occur if the energy of the initial and final states match. At first sight this does not appear to be the case for electronic absorption spectroscopy, in which a ground-to-excited state transition occurs. Floquet theory provides a useful way to think about the interaction between molecules and light, which will help us understand how these transitions occur within the Golden-rule picture.

If the Hamiltonian is time-dependent and *periodic*, there is a special solution of the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H}(t) |\Psi(t)\rangle. \quad (6.13)$$

We insert the Ansatz $|\Psi(t)\rangle = e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle$ to obtain

$$E_\lambda e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle + i\hbar e^{-iE_\lambda t/\hbar} \frac{\partial}{\partial t} |\Phi_\lambda(t)\rangle = \hat{H}(t) e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle \quad (6.14)$$

which can be rearranged to give

$$\hat{H}(t) |\Phi_\lambda(t)\rangle - i\hbar \frac{\partial}{\partial t} |\Phi_\lambda(t)\rangle = E_\lambda |\Phi_\lambda(t)\rangle. \quad (6.15)$$

This defines the Floquet eigenvalues and eigenstates. The most general solution of the TDSE is thus $|\Psi(t)\rangle = \sum_\lambda a_\lambda e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle$.

The Floquet theorem states that if the Hamiltonian is periodic, then so are the functions $|\Phi_\lambda(t)\rangle$ which solve the previous equation. Therefore we can expand the Hamiltonian and wavefunction as a Fourier series:

$$\hat{H}(t) = \sum_{n=-\infty}^{\infty} e^{in\omega t} \hat{H}^{(n)} \quad (6.16)$$

$$|\Phi_\lambda(t)\rangle = \sum_{n=-\infty}^{\infty} e^{in\omega t} |\phi_\lambda^{(n)}\rangle \quad (6.17)$$

where the period is $2\pi/\omega$.

$$\sum_{n'} e^{in'\omega t} \hat{H}^{(n')} \sum_n e^{in\omega t} |\phi_\lambda^{(n)}\rangle - i\hbar \sum_n in\omega e^{in\omega t} |\phi_\lambda^{(n)}\rangle = E_\lambda \sum_n e^{in\omega t} |\phi_\lambda^{(n)}\rangle \quad (6.18)$$

multiply both sides by $e^{-im\omega t}$ and average time over one period

$$\sum_{n,n'} e^{i(n'+n-m)\omega t} \hat{H}^{(n')} |\phi_\lambda^{(n)}\rangle - i\hbar \sum_n in\omega e^{i(n-m)\omega t} |\phi_\lambda^{(n)}\rangle = E_\lambda \sum_n e^{i(n-m)\omega t} |\phi_\lambda^{(n)}\rangle \quad (6.19)$$

$$\sum_{n,n'} \delta_{n',m-n} \hat{H}^{(n')} |\phi_\lambda^{(n)}\rangle - i\hbar \sum_n in\omega \delta_{nm} |\phi_\lambda^{(n)}\rangle = E_\lambda \sum_n \delta_{nm} |\phi_\lambda^{(n)}\rangle \quad (6.20)$$

$$\sum_n \hat{H}^{(m-n)} |\phi_\lambda^{(n)}\rangle - m\hbar\omega |\phi_\lambda^{(m)}\rangle = E_\lambda |\phi_\lambda^{(m)}\rangle \quad (6.21)$$

In the typical case where $\hat{H}(t) = \hat{H}_0 - 2\varepsilon\mu \cos(\omega t)$, we have that $\hat{H}^{(0)} = \hat{H}_0$ (the molecule-only Hamiltonian) and $\hat{H}^{(1)} = \hat{H}^{(-1)} = -\varepsilon\mu$ (interaction with the light field) whereas all other $\hat{H}^{(n)}$ terms are 0. This gives rise to the following effective *time-independent* Hamiltonian

$$\begin{pmatrix} \hat{H}_0 + 2\hbar\omega & -\varepsilon\mu & 0 & 0 & 0 \\ -\varepsilon\mu & \hat{H}_0 + \hbar\omega & -\varepsilon\mu & 0 & 0 \\ 0 & -\varepsilon\mu & \hat{H}_0 & -\varepsilon\mu & 0 \\ 0 & 0 & -\varepsilon\mu & \hat{H}_0 - \hbar\omega & -\varepsilon\mu \\ 0 & 0 & 0 & -\varepsilon\mu & \hat{H}_0 - 2\hbar\omega \end{pmatrix}$$

of which only a small section is shown as it is infinite in size.

Without actually going into quantum-field theory,* we have effectively included quantized photons of energy $\hbar\omega$. In this way we can understand absorption/emission processes as nonadiabatic coupling between molecular/light states and Fermi's golden-rule can be used to compute the probability of transition from one state to another. For instance absorption is described by the dynamical transformation from $|\phi^{(n)}\rangle$ to $|\phi^{(n-1)}\rangle$ via the coupling $-\varepsilon\mu$. This process starts in a state where the molecule and n photons are present (total Hamiltonian $\hat{H}_0 + n\hbar\omega$) and ends in a state where the molecule and $n-1$ photons are present (total Hamiltonian $\hat{H}_0 + (n-1)\hbar\omega$). Therefore, to conserve energy we require that the internal energy of the molecule increases by $\hbar\omega$.

6.3 Preparation of an excited electronic state

Now instead of monochromatic light, we will use a pulsed field (laser pulse) with carrier frequency ω

$$\varepsilon(t) = \varepsilon f(t) \cos(\omega t) \quad (6.22)$$

where the normalized function $f(t)$ describes the pulse shape and ε is the amplitude. The simplest pulse shape is the Gaussian:

$$f(t) = \sqrt{\frac{1}{2\pi\tau_p^2}} \exp\left\{-\frac{(t-t_p)^2}{2\tau_p^2}\right\}. \quad (6.23)$$

which is centred at t_p and has width (pulse duration) τ_p .

The Hamiltonian is no longer periodic, so we return to the more general time-dependent picture.

Generalizing equation (5.49) we get

$$|\chi_e^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} \int_{t_0}^t e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle f(t') \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt'. \quad (6.24)$$

We analyse this in the basis of vibrational eigenstates of the excited electronic state, $|\chi_{en}\rangle$, with eigenvalues $\hbar\omega_{en}$:

$$\langle\chi_{en}|\chi_e^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} \int_{t_0}^t \langle\chi_{en}|e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle f(t') \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt' \quad (6.25)$$

$$= \frac{-\varepsilon}{i\hbar} \int_{t_0}^t e^{-i\omega_{en}(t-t')} \langle\chi_{en}|\hat{\mu}_{eg}|\chi_g(t_0)\rangle f(t') \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt'. \quad (6.26)$$

Therefore the amplitude for populating a given vibrational eigenstate n is determined by two things, the Frank-Condon overlap, $\langle\chi_{en}|\hat{\mu}_{eg}|\chi_g(t_0)\rangle$, and the Fourier transform of the pulse:

$$\tilde{f}(\Omega_n) = \int f(t') e^{i\Omega_n t'} dt' = e^{-\Omega_n^2 \tau_p^2 / 2 + i\Omega_n \tau_p} \quad (6.27)$$

where $\Omega_n = \omega_{en} - \omega_g - \omega$. Therefore for long pulses, only states corresponding to $\Omega_n \approx 0$ can be occupied, but for short pulses, many states can be occupied. This is another example of the uncertainty principle: the pulse is either localized in time or energy.

*If you do want to find out more about this fascinating topic, I recommend the book Tom Lancaster and Stephen J. Blundell *Quantum field theory for the gifted amateur* (Oxford).

In previous lectures we have treated the limit of long pulse duration $\tau_p \rightarrow \infty$, from which one recovers the continuous wave limit and absorption spectrum. Here we take the limit of short pulse duration, $\tau_p \rightarrow 0$, which is equivalent to $f(t) \rightarrow \delta(t - t_p)$.

$$|\chi_e^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} e^{-i\hat{H}_e(t-t_p)/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle \cos(\omega t_p) e^{-i\omega_g(t_p-t_0)} \quad (6.28)$$

In this limit, the wavepacket is effectively just placed on the excited state at $t = t_p$.

6.4 Photochemistry

Consider a system of three electronic states. For simplicity, we employ the diabatic representation.

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) \quad (6.29)$$

$$= \frac{\hat{p}^2}{2m} + \begin{pmatrix} \hat{V}_0 & 0 & \hat{V}_{02} \\ 0 & \hat{V}_{11} & \hat{V}_{12} \\ \hat{V}_{20} & \hat{V}_{21} & \hat{V}_{22} \end{pmatrix} + \begin{pmatrix} 0 & -\mu_{10}\varepsilon(t) & 0 \\ -\mu_{01}\varepsilon(t) & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (6.30)$$

The kinetic energy term is often written like this, but a (3×3) -identity matrix is implied such that it only adds to the diagonal elements of $\hat{H}_0$. Here we have assumed that the light field couples states 0 to 1, but that state 2 is “dark” and has no transition dipole moment coupling to the ground state, perhaps because of symmetry reasons. We have also assumed that there is no diabatic coupling between states 0 and 1, but allowed there to be some between 2 and 0. This may simply be because there is no crossing between V_0 and V_1 .

A laser pulse may prepare a wavepacket in state 1 which will evolve in the coupled set of diabatic states. It may cross over to state 2 and from there find its way back to state 0. Let’s see how we can describe this using time-dependent perturbation theory and propagators.

The initial state is $|\psi(t_0)\rangle = |\phi_0\rangle |\chi_0(t_0)\rangle$ and the dipole operator is $\hat{\mu}_{10} |\phi_1\rangle \langle \phi_0|$. Taking the limit of a short pulse,

$$|\psi^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} e^{-i\hat{H}_0(t-t_p)/\hbar} \hat{\mu}_{10} |\phi_1\rangle \langle \phi_0 | \psi(t_0) \rangle \cos(\omega t_p) e^{-i\omega_g(t_p-t_0)} \quad (6.31)$$

The interesting dynamics is thus described by

$$e^{-i\hat{H}_0(t-t_p)/\hbar} |\psi^{(1)}(t_p)\rangle \quad (6.32)$$

where

$$|\psi^{(1)}(t_p)\rangle = |\phi_1\rangle |\chi_1^{(1)}(t_p)\rangle = \frac{-\varepsilon}{i\hbar} \hat{\mu}_{10} |\phi_1\rangle |\chi_0(t_0)\rangle \cos(\omega t_p) e^{-i\omega_g(t_p-t_0)} \quad (6.33)$$

We write this as a vector of nuclear wavefunctions in the basis of electronic states:

$$|\psi^{(1)}(t_0)\rangle = \begin{pmatrix} 0 \\ |\chi_1^{(1)}(t_p)\rangle \\ 0 \end{pmatrix} \quad (6.34)$$

Now we use the Trotter splitting on the propagator

$$\hat{U}(\epsilon) = e^{-i\hat{H}_0\epsilon/\hbar} = e^{-i\hat{V}\epsilon/2\hbar} e^{-i\hat{T}\epsilon/\hbar} e^{-i\hat{V}\epsilon/2\hbar} + \mathcal{O}(\epsilon^3) \quad (6.35)$$

The only difference now is that each of these operators is a (3×3) -matrix in the basis of electronic states.

For instance,

$$\hat{A} = e^{-i\hat{V}\epsilon/\hbar} = \begin{pmatrix} \hat{A}_{00} & \hat{A}_{01} & \hat{A}_{02} \\ \hat{A}_{10} & \hat{A}_{11} & \hat{A}_{12} \\ \hat{A}_{20} & \hat{A}_{21} & \hat{A}_{22} \end{pmatrix} \quad (6.36)$$

and

$$\hat{B} = e^{-i\hat{T}\epsilon/\hbar} = \begin{pmatrix} e^{-i\hat{T}\epsilon/\hbar} & 0 & 0 \\ 0 & e^{-i\hat{T}\epsilon/\hbar} & 0 \\ 0 & 0 & e^{-i\hat{T}\epsilon/\hbar} \end{pmatrix} \quad (6.37)$$

such that $\hat{U}(\epsilon) = \hat{A}\hat{B}\hat{A}$ to give

$$\begin{pmatrix} |\chi_0^{(1)}(t+\epsilon)\rangle \\ |\chi_1^{(1)}(t+\epsilon)\rangle \\ |\chi_2^{(1)}(t+\epsilon)\rangle \end{pmatrix} = \begin{pmatrix} \hat{U}_{00}(\epsilon) & \hat{U}_{01}(\epsilon) & \hat{U}_{02}(\epsilon) \\ \hat{U}_{10}(\epsilon) & \hat{U}_{11}(\epsilon) & \hat{U}_{12}(\epsilon) \\ \hat{U}_{20}(\epsilon) & \hat{U}_{21}(\epsilon) & \hat{U}_{22}(\epsilon) \end{pmatrix} \begin{pmatrix} |\chi_0^{(1)}(t)\rangle \\ |\chi_1^{(1)}(t)\rangle \\ |\chi_2^{(1)}(t)\rangle \end{pmatrix} \quad (6.38)$$

6.5 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Fermi's golden rule

- Zwanzig, Nonequilibrium statistical mechanics (Oxford), section 3.1

Bixon-Jortner Model

- Heller, The Semiclassical Way to Dynamics and Spectroscopy, chapter 14

Floquet theory

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, section 9.5
- Grossmann, Theoretical Femtosecond Physics (Springer)

Preparation of an excited electronic state

- May and Kühn, Charge and Energy Transfer Dynamics in Molecular Systems (Wiley, 2011), section 6.5
<https://onlinelibrary.wiley.com/doi/book/10.1002/9783527633791>
 (full access through ETH library – use a VPN)

Hopefully you may be able to read them on the internet.

7 Scattering theory

Quantum scattering theory plays an essential role in describing chemical reactions and photoionization. Although all of these phenomena are time dependent, scattering theory is most accessible from a time-independent perspective. Here we keep things simple by treating one-dimensional problems.

7.1 Scattering states

In a previous lecture we studied the free-particle states $|p\rangle$ which are eigenfunctions of the free-particle Hamiltonian. A more general Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad (7.1)$$

where $V(x) \rightarrow 0$ in the limit $x \rightarrow \pm\infty$.^{*} Such a barrier is represented in Figure 6. A particle incoming from the left with momentum p will have energy $E = p^2/2m$ and will either be reflected or transmitted over/through the barrier. At large distances from the barrier, the potential is zero and thus we know that the wave function will take the form of the free-particle states discussed above.

The scattering states are time-independent and thus one must consider the incoming, reflected and transmitted parts all together as well as the more complex wave function which exists in the barrier region. These states solve

$$\hat{H} |\psi_p\rangle = E |\psi_p\rangle, \quad (7.2)$$

where the energy is still $E = p^2/2m$, where p refers to the initial momentum. Although the energy E is a conserved quantity, the momentum operator no longer commutes with the Hamiltonian and thus we can only specify the momentum at the asymptotes where the potential is flat.

^{*}Actually all this theory can easily be generalized to treat exo- or endothermic reactions too. The potential simply needs to become flat at the asymptotes.

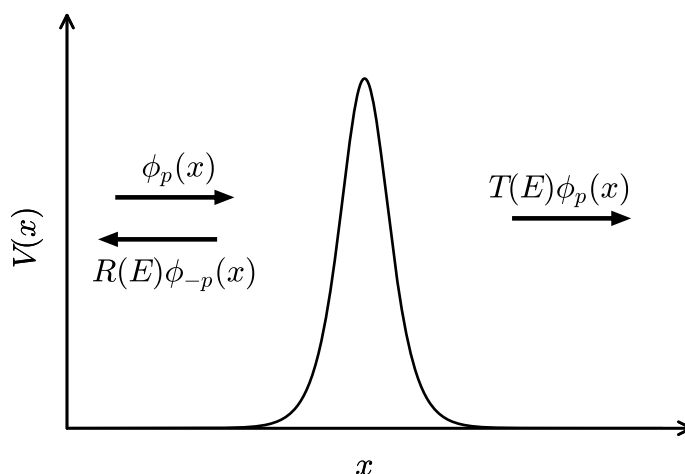


Figure 6: Schematic for scattering state for particle incoming from the left

At these asymptotes we know the values of the momentum and can thus write

$$\lim_{x \rightarrow -\infty} \psi_p(x) = \phi_p(x) + R(E)\phi_{-p}(x) \quad (7.3)$$

$$\lim_{x \rightarrow \infty} \psi_p(x) = T(E)\phi_p(x). \quad (7.4)$$

To find the values of the complex numbers $R(E)$ and $T(E)$, we must solve the time-independent Schrödinger equation which must usually be performed numerically. However, for a couple of cases it can be solved analytically such as for the square barrier, which you have probably solved in previous courses.*

7.2 Flux and rate

The probability of the particle reflecting is given by $|R(E)|^2$ and the probability of it being transmitted is $|T(E)|^2$. Of course, probabilities must sum to 1 so $|R(E)|^2 + |T(E)|^2 = 1$.

Flux through point $x = s$ is Heisenberg time derivative of Heaviside step function, $\theta(x - s)$,

$$\hat{F}(s) = \frac{i}{\hbar} [\hat{H}, \theta(\hat{x} - s)]. \quad (7.5)$$

Exercise: using this definition, show that

$$\hat{F}(s) = \frac{1}{2m} [\delta(\hat{x} - s)\hat{p} + \hat{p}\delta(\hat{x} - s)]. \quad (7.6)$$

We can calculate the flux through a point, s , far to the right of the barrier,

$$\langle \psi_p | \hat{F}(s) | \psi_p \rangle = |T(E)|^2 \langle p | \hat{F}(s) | p \rangle \quad (7.7)$$

but the flux is everywhere the same and thus also holds if s is at the barrier or even on its left.

The flux of a free particle through a point, $x = s$, is

$$\langle p | \hat{F}(s) | p \rangle = \frac{1}{2m} [\langle p | \delta(\hat{x} - s) \hat{p} | p \rangle + \langle p | \hat{p} \delta(\hat{x} - s) | p \rangle] \quad (7.8a)$$

$$= \frac{1}{2m} \int_{-\infty}^{\infty} [\langle p | x \rangle \langle x | \delta(\hat{x} - s) \hat{p} | p \rangle + \langle p | \hat{p} \delta(\hat{x} - s) | x \rangle \langle x | p \rangle] dx \quad (7.8b)$$

$$= \frac{1}{2m} \int_{-\infty}^{\infty} [\langle p | x \rangle \delta(x - s) \langle x | p \rangle p + p \langle p | x \rangle \delta(x - s) \langle x | p \rangle] dx \quad (7.8c)$$

$$= \frac{1}{2m} [\langle p | s \rangle \langle s | p \rangle p + p \langle p | s \rangle \langle s | p \rangle] \quad (7.8d)$$

$$= \frac{p}{m} (2\pi\hbar)^{-1} e^{+ips/\hbar} e^{-ips/\hbar} \quad (7.8e)$$

$$= \frac{1}{2\pi\hbar} \frac{p}{m} \quad (7.8f)$$

The probability of reaction is given by the ratio of transmitted flux to incoming flux

$$P(E) = \frac{\langle \psi_p | \hat{F}(s) | \psi_p \rangle}{\langle p | \hat{F}(s) | p \rangle} \quad (7.9)$$

$$= |T(E)|^2. \quad (7.10)$$

*If not, look at wikipedia!

Therefore, to obtain $P(E)$, we just need to know the flux of the wavefunction, $|\psi_p\rangle$ at a point, $x = s$.

The thermal rate constant for the reaction is found by a Boltzmann average over the probability of reaction,

$$k(T) = \frac{1}{2\pi\hbar Q_r(T)} \int_0^\infty P(E) e^{-\beta E} dE, \quad (7.11)$$

where for a 1D system, the reactant partition per unit length is

$$Q_r(T) = \sqrt{\frac{m}{2\pi\beta\hbar^2}}. \quad (7.12)$$

7.3 Tunnelling effect

In classical mechanics $P(E) = \theta(E - V^\ddagger)$ as only particles above the barrier can be transmitted and they all pass over. However, due to tunnelling effects, this is not true in quantum mechanics. The thermal rate constant for a classical particle in one-dimension is given by

$$k(T) = \frac{1}{2\pi\hbar Q_r(T)} \int_{V^\ddagger}^\infty e^{-\beta E} dE \quad (7.13)$$

$$= \frac{1}{2\pi\beta\hbar Q_r(T)} e^{-\beta V^\ddagger}. \quad (7.14)$$

We have therefore derived the one-dimensional version of the TST formula, equation (1.33), presented in the first lecture.

By horrendous mathematical analysis (which these days would be done by Mathematica) Bell was able to obtain the analytical expression for $P(E)$ for a parabolic barrier $V(x) = V^\ddagger - m\omega^2 x^2/2$ which is

$$P(E) = [1 + \exp(2\pi(V^\ddagger - E)/\hbar\omega)]^{-1} \quad (7.15)$$

From this we note that at an energy close to the barrier top, only half of the flux is transmitted.

For a parabolic barrier, the rate is

$$k(T) = \frac{1}{2\pi\hbar Q_r(T)} \int_{-\infty}^\infty P(E) e^{-\beta E} dE \quad (7.16)$$

$$= \frac{1}{2\pi\beta\hbar Q_r(T)} \frac{\frac{1}{2}\beta\hbar\omega}{\sin \frac{1}{2}\beta\hbar\omega} e^{-\beta V^\ddagger} \quad (7.17)$$

where we have used Mathematica to perform the integral.

This gives us a simple method for estimating the shallow tunnelling effects on any barrier which has an approximately parabolic shape near its top. However, note that the expression blows up when $\beta > 2\pi/\hbar\omega$ and thus cannot describe deep tunnelling occurring at low temperatures where the real barrier will deviate from the parabolic approximation.

A better approximation can be obtained from the Eckart barrier,

$$V(x) = \frac{V^\ddagger}{\cosh^2(x/a)}$$

for which a semiclassical approximation to the reaction probability is

$$P(E) \sim \frac{1}{1 + \exp W(E)/\hbar} \quad (7.18a)$$

$$\sim e^{-W(E)/\hbar} \quad (7.18b)$$

where $W(E) = 2\hbar\alpha(1 - \sqrt{E/V^\ddagger})$ and $\alpha = \pi\sqrt{2ma^2V^\ddagger}/\hbar$ and the second approximation is valid only for $E \ll V^\ddagger$. Note that Eckart gives the complicated exact expression in his 1930 paper.

An analytic formula for the thermal rate of the Eckart barrier in the deep-tunnelling regime can be obtained using another approximation: the steepest-descent integration.

$$k(T) \sim \frac{1}{2\pi\hbar Q_r(T)} \int e^{-W(E)/\hbar - \beta E} dE. \quad (7.19)$$

Expanding the exponent as a Taylor series around its minimum $E = E^*$, defined such that $W'(E^*) = -\beta\hbar$, gives

$$k(T) \sim \frac{1}{2\pi\hbar Q_r(T)} \int e^{-W(E^*)/\hbar - W'(E^*)(E-E^*)/\hbar - W''(E^*)(E-E^*)^2/2\hbar - \beta E} dE \quad (7.20a)$$

$$= \frac{1}{2\pi\hbar Q_r(T)} e^{-S/\hbar} \int e^{-W''(E^*)(E-E^*)^2/2\hbar} dE \quad (7.20b)$$

$$= \frac{1}{2\pi\hbar Q_r(T)} \sqrt{\frac{2\pi\hbar}{W''(E^*)}} e^{-S/\hbar}. \quad (7.20c)$$

where $S = W(E^*) + \hbar\beta E^*$.

From this you can derive

$$E^* = \frac{\alpha^2}{\beta^2 V^\ddagger} = \frac{2\pi^2 m a^2}{\beta^2 \hbar^2},$$

which means that reactions will tunnel at low energies when the temperature is low and the barrier is thin.

Exercise: make Arrhenius plots for the Eckart barrier. Note that these expressions are only valid below a certain temperature, $\beta > \alpha/V^\ddagger$, when the tunnelling mechanism dominates, i.e. when $E^* < V^\ddagger$. Using the more accurate expression, equation (7.18a), you can obtain the rate by numerical integration. This is valid at all temperatures and should be in agreement with the classical rate at high temperatures.

7.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

More on momentum states

- Feynman Lectures Volume III chapter 16, <https://www.feynmanlectures.caltech.edu>

Hard-core theory of scattering (for future reference only)

- Mark Child, Molecular collision theory

- John Taylor, Scattering theory: The Quantum Theory of Nonrelativistic Collisions (1972)

To learn more about tunnelling effects in chemical reactions, see my articles:

- “Understanding Chemical Reactions beyond Transition-state Theory”, Chimia **72**, 309-312 (2018)
- “Perspective: Ring-polymer Instanton Theory”, J. Chem. Phys. **148**, 200901 (2018)

8 Classical rate theory

For a large and complex reaction we cannot compute all the state-to-state reaction probabilities required to obtain the rate constant. However, one might guess that there is a more direct way to obtain the thermally-averaged rate constant. We shall discuss this approach within the context of classical mechanics but a similar argument also hold for the quantum case.

8.1 Linear response theory

In order to obtain a definition of the reaction rate, we perturb the system away from equilibrium and measure how it relaxes back to equilibrium when the perturbation is removed.

If our perturbed Hamiltonian is

$$H_\lambda = H - \lambda A \quad (8.1)$$

the initial thermal distribution will now be

$$\rho_\lambda = \frac{1}{Q_\lambda} e^{-\beta H_\lambda} \quad (8.2)$$

where the perturbed partition function Q_λ normalizes ρ_λ .

If at $t = 0$ we turn off the perturbation we can observe the resulting time-evolution of property B using

$$\langle B(t) \rangle_\lambda \equiv \int B(\vec{p}_t, \vec{q}_t) \rho_\lambda(\vec{p}, \vec{q}) d\vec{p} d\vec{q} \quad (8.3)$$

Unlike $\langle B(t) \rangle = \langle B \rangle$ which is time-independent (see lecture 2), $\langle B(t) \rangle_\lambda$ does depend on time as the dynamics are performed with the Hamiltonian, H , which is not the same Hamiltonian as the perturbed one used to define ρ_λ . The behaviour of $\langle B(t) \rangle_\lambda$ is rather complex in general but easier to study in the limit that $\lambda \rightarrow 0$, i.e. with a weak perturbation. Expanding to first order in λ gives

$$\langle B(t) \rangle_\lambda = \langle B \rangle + \beta \lambda [\langle A(0)B(t) \rangle - \langle A \rangle \langle B \rangle] + O(\lambda^2) \quad (8.4)$$

or using $\Delta A(t) = A(t) - \langle A \rangle$ and $\Delta B(t) = B(t) - \langle B \rangle$,

$$\langle \Delta B(t) \rangle_\lambda = \beta \lambda \langle \Delta A(0) \Delta B(t) \rangle + O(\lambda^2) \quad (8.5)$$

This powerful equation, known as the ‘Onsager regression hypothesis’ or ‘fluctuation-dissipation theorem’ allows us to express macroscopic relaxation of non-equilibrium systems in terms of spontaneous microscopic fluctuations in an equilibrium system.

8.2 Connection with macroscopic rate theory

Consider the following chemical reaction described by first-order rate constants



where k_+ is the forward rate constant and k_- is that for the reverse reaction.

Let $c_A(t)$ and $c_B(t)$ denote the concentrations then the phenomenological rate equations are

$$\frac{\partial c_A}{\partial t} = -k_+ c_A(t) + k_- c_B(t) \quad (8.7a)$$

$$\frac{\partial c_B}{\partial t} = k_+ c_A(t) - k_- c_B(t). \quad (8.7b)$$

Note that the total concentration of species, $c = c_A(t) + c_B(t)$, is constant.

At equilibrium, the concentrations $\langle c_A \rangle$ and $\langle c_B \rangle$ are stationary and thus $0 = -k_+ \langle c_A \rangle + k_- \langle c_B \rangle$ and the equilibrium constant is

$$K_{\text{eq}} \equiv \frac{\langle c_B \rangle}{\langle c_A \rangle} = \frac{k_+}{k_-}. \quad (8.8)$$

This relation between the forward and backward rate constant is known as “detailed balance”.

Defining $\Delta c_A(t) = c_A(t) - \langle c_A \rangle$ and $\Delta c_B(t) = c_B(t) - \langle c_B \rangle$, and hence $\Delta c_B(t) = -\Delta c_A(t)$, we find

$$\frac{\partial \Delta c_A}{\partial t} = -(k_+ + k_-) \Delta c_A(t) \quad (8.9a)$$

$$\frac{\partial \Delta c_B}{\partial t} = -(k_+ + k_-) \Delta c_B(t). \quad (8.9b)$$

These equations are solved by

$$\Delta c_A(t) = \Delta c_A(0) e^{-(k_+ + k_-)t} \quad (8.10a)$$

$$\Delta c_B(t) = \Delta c_B(0) e^{-(k_+ + k_-)t}. \quad (8.10b)$$

We therefore define the timescale for the reaction as $\tau = (k_+ + k_-)^{-1}$.

In this way we can relate the rate constants k_+ and k_- to the macroscopic relaxation of the concentration of reactants or products.

Multiplying and dividing equation (8.9b) by $\langle c_B \rangle$ and using the detailed-balance condition gives

$$\frac{\partial \Delta c_B}{\partial t} = -\frac{k_+ \langle c_B \rangle + k_+ \langle c_A \rangle}{\langle c_B \rangle} \Delta c_B(t) \quad (8.11)$$

$$= -k_+ \frac{c}{\langle c_B \rangle} \Delta c_B(t). \quad (8.12)$$

We define the properties $A(\vec{p}, \vec{q})$ and $B(\vec{p}, \vec{q})$ from the previous section in such a way so as to determine whether the system is in a reactant or product state. They are projection operators returning a value between 0 and 1 and $A + B = 1$, although their exact form is not important as long as they successfully identify the reactant and product basins of attraction. Again we use A to permute the equilibrium and then allow the system to relax. This permutation will create more reactants and fewer products than would be present at equilibrium and we expect that the reactant concentration will decay back to its equilibrium value.

We can therefore identify $\langle A(t) \rangle_\lambda = c_A(t)/c$ and $\langle B(t) \rangle_\lambda = c_B(t)/c$. Using $\Delta B(t) = B(t) - \langle B \rangle$, $\dot{B}(t) \equiv \frac{\partial B}{\partial t} = \frac{\partial \Delta B}{\partial t} \equiv \Delta \dot{B}(t)$ and $\Delta A(t) = -\Delta B(t)$, we find

$$\langle \Delta \dot{B}(t) \rangle_\lambda = -k_+ \langle \Delta B(t) \rangle_\lambda / \langle B \rangle. \quad (8.13)$$

Rearranging gives

$$k_+ = -\frac{\langle \Delta \dot{B}(t) \rangle_\lambda}{\langle \Delta B(t) \rangle_\lambda} \langle B \rangle \quad (8.14)$$

$$= -\frac{\langle \Delta A(0) \Delta \dot{B}(t) \rangle}{\langle \Delta A(0) \Delta B(t) \rangle} \langle B \rangle \quad (8.15)$$

$$= \frac{\langle \Delta B(0) \Delta \dot{B}(t) \rangle}{\langle \Delta A(0) \Delta B(t) \rangle} \langle B \rangle \quad (8.16)$$

For a slow reaction, very few reactants starting in A will arrive in B and therefore $\langle \Delta A(0) \Delta B(t) \rangle = \langle A(0) B(t) \rangle - \langle A \rangle \langle B \rangle \approx -\langle A \rangle \langle B \rangle$ such that

$$k_+ = -\frac{\langle B(0) \dot{B}(t) \rangle}{\langle A \rangle} \quad (8.17)$$

$$= \frac{\langle \dot{B}(0) B(t) \rangle}{\langle A \rangle} \quad (8.18)$$

In reality the dynamical equations governing the molecular motion is more complicated than suggested by equation (8.7) and at short times there may be considerable deviation. The rate constant defined by equation (8.18) would then be time-dependent. However, at longer times it seems that many reactions do obey these rate laws (otherwise we would not have developed the concept of a rate constant at all). Therefore, we implicitly only apply this equation for values of t within a plateau region where k_+ is constant.

As we shall show in the next section, this equation is the basis of the flux correlation function formalism for computing rate constants.

8.3 Flux correlation functions

Let us introduce the reaction coordinate $s(\vec{q})$ and the momentum in this direction, $p_s(\vec{q})$, which describes the progress of the reaction. For example it could report on the bond distance during a chemical reaction in which the bond is broken. Our projection operators will be defined as $A = \Theta(s^\ddagger - s)$ and $B = \Theta(s - s^\ddagger)$ where Θ is the Heaviside step function and $s^\ddagger$ is a constant. When $s < s^\ddagger$, the configuration is a “reactant” and when $s > s^\ddagger$, the configuration describes a “product.”

The time-derivative of B is easily obtained using the chain rule

$$\dot{B} \equiv \frac{\partial B}{\partial t} = \frac{\partial B}{\partial \vec{q}} \cdot \frac{\partial \vec{q}}{\partial t} = \delta(s - s^\ddagger) \frac{\partial s}{\partial \vec{q}} \cdot \frac{\vec{p}}{m} \quad (8.19)$$

This is the flux through the dividing surface $s = s^\ddagger$ and can also be written as

$$\dot{B} \equiv F(p_s, s) = \delta[s - s^\ddagger] \frac{p_s}{m_s} \quad (8.20)$$

where p_s is the momentum in the s -direction with associated mass m_s .

The classical rate constant is therefore given by

$$k_{\text{cl}}(T) = Q_{\text{r}}(T)^{-1} C_{\text{fs}}^{\text{cl}}(t) \quad \text{for } t \text{ in plateau} \quad (8.21)$$

where $Q_r(T)$ is the reactant partition function

$$Q_r(T) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \Theta(s^\ddagger - s) d\vec{p} d\vec{q} \quad (8.22)$$

and the classical flux-side correlation function is

$$C_{\text{fs}}^{\text{cl}}(t) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} F(p_s, s) \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p} d\vec{q} \quad (8.23)$$

where $\vec{q}_t \equiv \vec{q}_t(\vec{p}, \vec{q})$ is found by solving for the classical trajectory at time t with initial conditions $(\vec{p}, \vec{q})$.

The flux-side correlation function is the time-derivative of the side-side correlation function

$$C_{\text{ss}}^{\text{cl}}(t) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \Theta[s^\ddagger - s(\vec{q})] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p} d\vec{q} \quad (8.24)$$

as can be shown using Liouville's theorem and the conservation of the Boltzmann operator. Therefore

$$k_{\text{cl}}(T) = Q_r(T)^{-1} \frac{d}{dt} C_{\text{ss}}^{\text{cl}}(t) \quad \text{for } t \text{ in plateau} \quad (8.25)$$

A very important point to note is that the rate constant is independent of the choice of the dividing surface. As long as the dividing surface separates reactants from products, it can have any shape or form. The dividing surface should always be placed in a transition region between reactants and products where the thermal distribution is negligible such that $Q_r(T)$ is independent. We can then prove

$$\frac{\partial}{\partial s^\ddagger} C_{\text{ss}}^{\text{cl}}(t) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \{ \delta[s^\ddagger - s(\vec{q})] \Theta[s(\vec{q}_t) - s^\ddagger] - \Theta[s^\ddagger - s(\vec{q})] \delta[s(\vec{q}_t) - s^\ddagger] \} d\vec{p} d\vec{q} \quad (8.26)$$

$$= (2\pi\hbar)^{-3N} \left\{ \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 - \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \Theta[s^\ddagger - s(\vec{q}_0)] \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \right\} \quad (8.27)$$

$$= (2\pi\hbar)^{-3N} \left\{ \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 - \iint e^{-\beta H(\vec{p}_t, \vec{q}_t)} \Theta[s^\ddagger - s(\vec{q}_0)] \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_t d\vec{q}_t \right\} \quad (8.28)$$

$$= (2\pi\hbar)^{-3N} \left\{ \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 - \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \Theta[s^\ddagger - s(\vec{q}_{-t})] \delta[s(\vec{q}_0) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \right\} \quad (8.29)$$

$$= (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \{ 2\Theta[s(\vec{q}_t) - s^\ddagger] - 1 \} d\vec{p}_0 d\vec{q}_0 \quad (8.30)$$

where we have explicitly shown the time of the initial conditions using $\vec{q}_0 \equiv \vec{q}$ and $\vec{p}_0 \equiv \vec{p}$. Therefore

$$\frac{\partial}{\partial s^\ddagger} C_{\text{fs}}^{\text{cl}}(t) = \frac{\partial}{\partial t} \frac{\partial}{\partial s^\ddagger} C_{\text{ss}}^{\text{cl}}(t) \quad (8.31)$$

$$= 2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \frac{p_s(\vec{p}_t, \vec{q}_t)}{m_s} \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \quad (8.32)$$

$$= 2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_t, \vec{q}_t)} \delta[s^\ddagger - s(\vec{q}_0)] \frac{p_s(\vec{p}_t, \vec{q}_t)}{m_s} \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_t d\vec{q}_t \quad (8.33)$$

$$= 2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_{-t})] \frac{p_s(\vec{p}_0, \vec{q}_0)}{m_s} \delta[s(\vec{q}_0) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \quad (8.34)$$

$$= -2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \delta[s^\ddagger - s(\vec{q}_t)] \frac{p_s(\vec{p}, \vec{q})}{m_s} \delta[s(\vec{q}) - s^\ddagger] d\vec{p} d\vec{q} \quad (8.35)$$

The derivative of $C_{\text{fs}}^{\text{cl}}(t)$ with respect to $s^\ddagger$ obviously tends to 0 for large times as it would be extremely unlikely for a trajectory to return to the dividing surface after time t . The rate is therefore independent of the dividing surface.

8.4 Classical transition-state theory

Transition-state theory is one of the most important concepts in chemistry and can be rigorously understood in terms of the classical flux-side correlation function.

We know that the exact classical rate is obtained from the plateau region of $C_{\text{fs}}^{\text{cl}}(t)$ but to compute that would involve propagating chaotic trajectories on the PES which is difficult to study analytically. Instead TST is defined from the short-time limit and is much easier to study although one must bear in mind that the short-time limit may or may not be in the plateau region which greatly affects the accuracy of the approximation.

$$k_{\text{TST}}(T) = Q_{\text{r}}(T)^{-1} \lim_{t \rightarrow 0_+} C_{\text{fs}}^{\text{cl}}(t) \quad (8.36)$$

This limit can be obtained by linearizing the dynamics, $\vec{q}_t \approx \vec{q}_0 + \frac{\vec{p}_0}{m} t$, which is exact in the short-time limit:

$$\lim_{t \rightarrow 0_+} F(p_s, s) \Theta[s(\vec{q}_t) - s^\ddagger] = \lim_{t \rightarrow 0_+} F(p_s, s) \Theta \left[s^\ddagger + \frac{p_s}{m_s} t - s^\ddagger \right] \quad (8.37)$$

$$= \lim_{t \rightarrow 0_+} F(p_s, s) \Theta[p_s] \quad (8.38)$$

Note that the classical flux-side correlation function is discontinuous at $t = 0$ and gives the negative result for $t \rightarrow 0_-$.

This gives the classical TST rate

$$k_{\text{TST}}(T) = (2\pi\hbar)^{-3N} Q_{\text{r}}(T)^{-1} \iint e^{-\beta H(\vec{p}, \vec{q})} \delta(s - s^\ddagger) \frac{p_s}{m_s} \Theta(p_s) d\vec{p} d\vec{q} \quad (8.39)$$

$$= (2\pi\hbar)^{-3N} Q_{\text{r}}(T)^{-1} \int e^{-\beta \vec{p}^2 / 2m} \frac{p_s}{m_s} \Theta(p_s) d\vec{p} \int e^{-\beta V(\vec{q})} \delta(s - s^\ddagger) d\vec{q}. \quad (8.40)$$

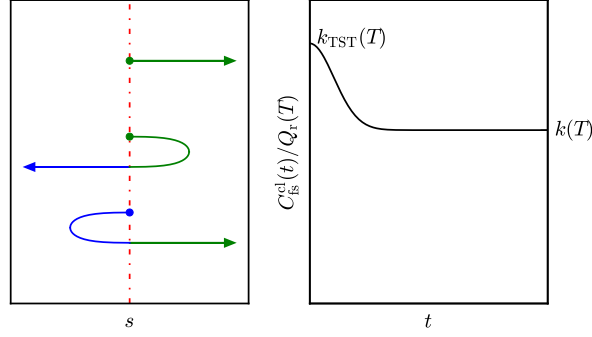


Figure 7: Trajectories starting at the dividing surface contribute (green) to the TST rate at short times and to the full rate at long times. The two lower trajectories recross and reduce the full rate.

Using $Q_r = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \Theta(s^\ddagger - s) d\vec{p} d\vec{q}$, and $\langle O \rangle = \frac{\iint e^{-\beta H(\vec{p}, \vec{q})} O(\vec{p}, \vec{q}) d\vec{p} d\vec{q}}{\iint e^{-\beta H(\vec{p}, \vec{q})} d\vec{p} d\vec{q}}$, we find

$$k_{\text{TST}}(T) = \frac{1}{2} \langle |\dot{s}| \rangle \frac{\langle \delta(s - s^\ddagger) \rangle}{\langle \Theta(s^\ddagger - s) \rangle} \quad (8.41)$$

$$= \frac{1}{2} \langle |\dot{s}| \rangle e^{-\beta \Delta A} \quad (8.42)$$

where

$$\langle |\dot{s}| \rangle = \frac{\int e^{-\beta \vec{p}^2/2m} |p_s/m_s| d\vec{p}}{\int e^{-\beta \vec{p}^2/2m} d\vec{p}} \quad (8.43)$$

and

$$\Delta A = -\frac{1}{\beta} [\ln \langle \delta(s - s^\ddagger) \rangle - \ln \langle \Theta(s^\ddagger - s) \rangle] \quad (8.44)$$

is the free-energy difference between the dividing surface and the reactants.

This gives us the simple Arrhenius picture for chemical reaction rates, in that they depend exponentially on free-energy barriers. The prefactor is an easily-obtained flux term which depends only on the dividing-surface shape and requires no evaluations of the PES.

However, TST does not always give the same result as the full classical rate as recrossing is neglected. A consideration of the possible recrossing trajectories (Figure 7) shows that recrossing will always lower the rate such that we can say that TST is an upper bound to the full classical rate.

Note that although the full rate is not dependent on the dividing surface, the TST rate is strongly dependent. The variational principle allows us to define an optimal TST with the dividing surface chosen so as to minimize the TST rate. This optimal TST is the best estimate for the full rate.

It is hard to estimate the effect that recrossing will have on the rate without computing many trajectories and observing their long-time behaviour. However, if we did know it, we could write,

$$k_{\text{cl}}(T) = \nu k_{\text{TST}}(T) \quad (8.45)$$

where $0 < \nu < 1$ is the recrossing factor.

8.5 Quantum flux correlation functions

Much of the theory derived above can be rigorously extended to describe quantum rate theory in terms of the quantum correlation functions

$$k(T) = Q_r(T)^{-1} \frac{d}{dt} C_{ss}(t) = Q_r(T)^{-1} C_{fs}(t) = Q_r(T)^{-1} \int_0^t C_{ff}(t') dt' \quad (\text{for } t \text{ in plateau}) \quad (8.46)$$

where

$$Q_r = \text{Tr} \left[e^{-\beta \hat{H}} \Theta(\hat{s} - s^\dagger) \right] \quad (8.47)$$

$$C_{ss}(t) = \text{Tr} \left[e^{-\beta \hat{H}/2} \Theta(s^\dagger - \hat{s}) e^{-\beta \hat{H}/2} e^{i\hat{H}t/\hbar} \Theta(\hat{s} - s^\dagger) e^{-i\hat{H}t/\hbar} \right] \quad (8.48)$$

$$C_{fs}(t) = \text{Tr} \left[e^{-\beta \hat{H}/2} \hat{F} e^{-\beta \hat{H}/2} e^{i\hat{H}t/\hbar} \Theta(\hat{s} - s^\dagger) e^{-i\hat{H}t/\hbar} \right] \quad (8.49)$$

$$C_{ff}(t) = \text{Tr} \left[e^{-\beta \hat{H}/2} \hat{F} e^{-\beta \hat{H}/2} e^{i\hat{H}t/\hbar} \hat{F} e^{-i\hat{H}t/\hbar} \right] \quad (8.50)$$

where the flux operator, $\hat{F}$, was defined in the previous chapter, equation (7.6). Here we symmetrized the thermal density around one of the operators, which makes the functions real, but isn't actually necessary in order to obtain the correct rate constant.

For a free particle, $H = \hat{p}^2/2m$, it is possible to find an analytic expression for the quantum flux-flux correlation function, equation (8.50). By expanding the trace in momentum states, show that

$$C_{ff}(t) = \frac{1}{2\pi\beta\hbar} \frac{(\hbar\beta/2)^2}{[t^2 + (\hbar\beta/2)^2]^{3/2}}.$$

Hence plot $C_{ss}(t)$, $C_{fs}(t)$ and $C_{ff}(t)$. Sketch the equivalent functions from classical mechanics. Compute the quantum-mechanical rate constant and show that classical rate theory gives the same result for this particular case.

Unlike the classical flux-side correlation function, $C_{fs}(t)$ is continuous at $t = 0$ and thus one cannot simply define a quantum TST by taking the short-time limit. Finding a quantum TST is an ongoing active area of research in our group and elsewhere.

8.6 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Linear response theory and classical rate

- Chandler, Introduction to Modern Statistical Mechanics (Oxford), chapter 8

Linear response theory and diffusion constants

- Tuckerman, Statistical Mechanics (Oxford), chapter 13

Hopefully you may be able to read them on the internet.

9 More on TST

9.1 Eyring transition-state theory

The form of transition-state theory which we have derived above looks quite different from Eyring TST which we used in the first lecture. Unlike our new version, Eyring TST is not rigorously derived but instead obtained using thermodynamic arguments. Here we will show how it is related to the classical TST above.

First transform from the old coordinates $\vec{q} = (q_1, \dots, q_{3N})$ to new coordinates $(s, u_1, \dots, u_{3N-1})$ and from old momenta $\vec{p} = (p_1, \dots, p_{3N})$ to $(p_s, w_1, \dots, w_{3N-1})$. The coordinate transformation should be orthogonal such that $d\vec{p}d\vec{q} = dsd\vec{u}dp_sdw$. They are mass-weighted such that the associated mass for the $(\vec{u}, \vec{w})$ degrees of freedom are all the same, m .

$$k_{\text{TST}}(T) = (2\pi\hbar)^{-3N} Q_{\text{r}}(T)^{-1} \int e^{-\beta p_s^2/2m_s - \beta \vec{w}^2/2m} \frac{p_s}{m_s} \Theta(p_s) dp_s d\vec{w} \int e^{-\beta V(s, \vec{u})} \delta(s - s^\ddagger) ds d\vec{u} \quad (9.1)$$

$$= \frac{1}{2\pi\beta\hbar} \frac{Q_{s^\ddagger}(T)}{Q_{\text{r}}(T)} \quad (9.2)$$

where we have integrated over s and p_s and

$$Q_{s^\ddagger}(T) = (2\pi\hbar)^{-(3N-1)} \int e^{-\beta \vec{w}^2/2m} d\vec{w} \int e^{-\beta V(s^\ddagger, \vec{u})} d\vec{u} \quad (9.3)$$

is the classical partition function constrained to the dividing surface.

Optimizing the dividing-surface to maximize the rate forces it to pass through regions of high potential energy and the optimum dividing surface will thus often pass through (or at least close to) a saddle point on the potential-energy surface, where $\nabla V(\vec{q}) = 0$. This saddle point is often known as the transition-state and the barrier height, $V^\ddagger$ is the difference in energy between this point and the potential of the reactants in their minimum-energy configuration. In Eyring's TST, we choose the dividing surface to pass through this saddle point such that the surface is perpendicular to the unstable normal mode corresponding to the imaginary frequency.

Shifting the energy of this constrained partition function by the barrier height, we define the transition-state partition function relative to the barrier height energy,

$$Q^\ddagger(T) = Q_{s^\ddagger}(T) e^{\beta V^\ddagger} \quad (9.4)$$

and therefore

$$k_{\text{TST}}(T) = \frac{k_{\text{B}}T}{h} \frac{Q^\ddagger(T)}{Q_{\text{r}}(T)} e^{-\beta V^\ddagger} \quad (9.5)$$

This looks like (but isn't) the traditional Eyring TST formula which you already know. The difference is that here the partition functions are evaluated by a classical phase-space average whereas Eyring's formula uses quantum partition functions within the rigid-rotor and harmonic-oscillator approximations. Note that this doesn't make Eyring's TST "quantum" because the treatment along the s coordinate is still completely classical.

The underlying assumptions of Eyring's TST are that

- the Born-Oppenheimer approximation is valid
- there is little or no recrossing of the dividing surface
- the reaction coordinate can be separated from all other coordinates
- the dynamics is classical along the reaction coordinate
- the rigid-rotor and harmonic oscillator approximations hold for the other degrees of freedom

9.2 Langevin dynamics

For a one-dimensional system (phase space p, x , mass m , potential $V(x)$) interacting with a harmonic bath, we define a system-bath model:

$$H = \frac{p^2}{2m} + \sum_j \frac{p_j^2}{2} + V(x, q)$$

with the total potential-energy surface

$$V(x, q) = V(x) + \sum_j \frac{1}{2} \omega_j^2 \left(q_j - \frac{c_j}{\omega_j^2} x \right)^2 \quad (9.6)$$

By expanding the square, it can be written in the alternative form:

$$V(x, q) = V(x) + \frac{1}{2} \lambda x^2 - \sum_j c_j q_j x + \sum_j \frac{1}{2} \omega_j^2 q_j^2 \quad \text{where } \lambda = \sum_j \frac{c_j^2}{\omega_j^2} \quad (9.7)$$

Applying Hamilton's equations of motion gives

$$\begin{aligned} \dot{x} &= \frac{p}{m} & \dot{p} &= -\nabla V(x) + \sum_j c_j \left(q_j - \frac{c_j}{\omega_j^2} x \right) \\ \dot{q}_j &= p_j & \dot{p}_j &= -\omega_j^2 q_j + c_j x. \end{aligned}$$

Unless $V(x)$ is a very simple function, these cannot be solved explicitly. However, if we imagine that $x(t)$ were known, then we can solve for the dynamics of the bath. These are just inhomogeneous differential equations which have the solution:

$$q_j(t) = q_j(0) \cos \omega_j t + p_j(0) \frac{\sin \omega_j t}{\omega_j} + c_j \int_0^t x(t') \frac{\sin \omega_j(t-t')}{\omega_j} dt'$$

Show that this is a solution and then integrate by parts to show that it can be written equivalently as

$$q_j(t) - \frac{c_j}{\omega_j^2} x(t) = \left(q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right) \cos \omega_j t + p_j(0) \frac{\sin \omega_j t}{\omega_j} - c_j \int_0^t \frac{p(t')}{m} \frac{\cos \omega_j(t-t')}{\omega_j^2} dt'.$$

Substituting into the formulas above, we get:

$$\dot{p} = -\nabla V(x) - \int_0^t \frac{p(t')}{m} K(t-t') dt' + R(t), \quad (9.8)$$

where we have defined the friction kernel and random force:

$$K(t-t') = \sum_j \frac{c_j^2}{\omega_j^2} \cos \omega_j(t-t')$$

$$R(t) = \sum_j c_j \left(q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right) \cos \omega_j t + \sum_j c_j p_j(0) \frac{\sin \omega_j t}{\omega_j}$$

Of course the force is not really random, but if there are many bath modes and we do not know their initial conditions exactly then it appears random. You could see this for yourself by selecting $q_j(0)$ and $p_j(0)$ from a random distribution and plotting the function against t .

To analyse the friction kernel further, we go to the continuum limit (i.e. assume the bath is made up of an infinite number of modes) and replace the sum by an integral weighted by the density of states

$$K(t-t') = \sum_j \frac{c_j^2}{\omega_j^2} \cos \omega_j(t-t')$$

$$\rightarrow \int_0^\infty g(\omega) \frac{c(\omega)^2}{\omega^2} \cos \omega(t-t') d\omega$$

In fact if $g(\omega) = \sum_j \delta(\omega - \omega_j)$ then these expressions are clearly identical. However, we can now also choose a smooth function for $g(\omega)$ which is easier to deal with analytically.

In fact we don't even have to specify $g(\omega)$ and $c(\omega)$ independently, but can write everything in terms of the spectral density $J(\omega) = \frac{\pi}{2} g(\omega) \frac{c(\omega)^2}{\omega}$. We will take the special case that this follows the so-called Ohmic form $J(\omega) = \gamma \omega$. Then

$$\begin{aligned} K(t-t') &= \frac{2}{\pi} \int_0^\infty \frac{J(\omega)}{\omega} \cos \omega(t-t') d\omega \\ &= \frac{2}{\pi} \int_0^\infty \gamma \cos \omega(t-t') d\omega \\ &= \frac{1}{\pi} \int_{-\infty}^\infty \gamma \cos \omega(t-t') d\omega \quad \text{because it is an even function} \\ &= \frac{1}{\pi} \int_{-\infty}^\infty \gamma e^{i\omega(t-t')} d\omega \quad \text{because the imaginary part integrates to 0} \\ &= 2\gamma \delta(t-t') \end{aligned}$$

Therefore,

$$\dot{p} = -\nabla V(x) - \gamma \frac{p}{m} + R(t)$$

which is the Langevin dynamics equation. You should recognize the term $-\gamma \frac{p}{m} = -\gamma \dot{x}$ as being the simplest type of friction force and which points opposite to the velocity. The random force describes Brownian motion, which have probably also seen before.

Using the equipartition theorem,* it is easy to show that

$$\begin{aligned} \left\langle q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right\rangle &= 0 & \langle p_j(0) \rangle &= 0 \\ \left\langle \left(q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right)^2 \right\rangle &= \frac{kT}{\omega_j^2} & \langle p_j(0)^2 \rangle &= kT \end{aligned}$$

where we have assumed that the bath is in thermal equilibrium for a given value of $x(0)$. Then we can show that

$$\begin{aligned} \langle R(t)R(t') \rangle &= \sum_j \frac{kT c_j^2}{\omega_j^2} \cos \omega_j t \cos \omega_j t' + \sum_j kT c_j^2 \frac{\sin \omega_j t}{\omega_j} \frac{\sin \omega_j t'}{\omega_j} \\ &= kT \sum_j \frac{c_j^2}{\omega_j^2} \cos \omega_j (t - t') \\ &= kT K(t - t') = 2kT \gamma \delta(t - t'). \end{aligned}$$

This means that the random force at one time t is completely independent of the random force at another time $t' \neq t$. The strength of the random force is proportional to $\sqrt{2kT\gamma}$ and is thus dependent on temperature. It is in some ways similar to a standard gaussian random variable ξ_i , which has the property $\langle \xi_i \xi_j \rangle = \delta_{ij}$, which is known as white noise. However, there are also some subtleties due to the fact that $R(t)$ is not differentiable with respect to time. For a numerical implementation of the Langevin equation, see the homework exercise and Tuckerman's textbook.

9.3 Kramers theory

Transition state theory can be easily applied to a one-dimensional double well problem and gives

$$k_{\text{TST}} = (2\pi\beta\hbar)^{-1} Q_r(T)^{-1} e^{-\beta V^\ddagger}$$

where $Q_r(T)$ is the reactant partition function and $V^\ddagger$ is the barrier height. Using classical mechanics and a harmonic approximation we find $Q_r(T) = 1/(\beta\hbar\omega_0)$ where ω_0 is the frequency of the reactant well.

Kramers argued that for Langevin dynamics in a one-dimensional double well problem, recrossing would cause the classical rate with strong friction to be lower than TST and that it would decrease as friction got stronger. However, he also predicted that for very weak friction, there would also be recrossing such that the classical rate would decrease as friction got weaker. In between there must be a “turnover” regime where the rate is neither decreasing nor increasing. In the homework exercise you can simulate this behaviour by running Langevin trajectories to compute $C_{\text{fs}}(t)$.

*This says that the thermal average of the kinetic-energy or potential-energy terms of a harmonic oscillator are always equal to $kT/2$.

Here we can study the strong friction regime using a multidimensional TST approach on the full system-bath model. It is unfortunately not so easy to study the weak friction regime in this way as TST just simply cannot describe this type of recrossing.

For simplicity, we only take one bath mode, but the analysis can be extended fairly straightforwardly to a large bath. We are interested in the transition state, which we can define as the saddle point. This is seen to be located at $x = 0$ and $q_1 = 0$. A second-order Taylor expansion around this point gives the system-bath potential:

$$V(x, q_1) = V^\ddagger - \frac{1}{2}\bar{\omega}^2 x^2 + \frac{1}{2} \frac{c_1^2}{\omega_1^2} x^2 - c_1 q_1 x + \frac{1}{2} \omega_1^2 q_1^2$$

The Hessian is

$$\mathbf{H} = \begin{pmatrix} -\bar{\omega}^2 + \frac{c_1^2}{\omega_1^2} & -c_1 \\ -c_1 & \omega_1^2 \end{pmatrix}$$

and has eigenvalues

$$\begin{aligned} \lambda_0 &= \alpha/2 - \sqrt{\bar{\omega}^2 \omega_1^2 + \alpha^2/4} \\ \lambda_1 &= \alpha/2 + \sqrt{\bar{\omega}^2 \omega_1^2 + \alpha^2/4} \end{aligned} \quad \alpha = \frac{c_1^2}{\omega_1^2} + \omega_1^2 - \bar{\omega}^2$$

It is clear that $\lambda_0 < 0$ and $\lambda_1 > 0$. Therefore the first corresponds to our reaction coordinate and the second to the perpendicular mode. It is also easy to show that the determinant of $\mathbf{H} = -\bar{\omega}^2 \omega_1^2$, and we know that this must be equal to $\lambda_0 \lambda_1$ as the determinant is always equal to the product of the eigenvalues.

The situation in the reactant well is similar but has $+\frac{1}{2}\omega_0^2 x^2$ instead of $-\frac{1}{2}\bar{\omega}^2 x^2$. Thus $Q_r(T) = (\beta\hbar\omega_0)^{-1} \times (\beta\hbar\omega_1)^{-1}$.

If we were to use the dividing surface along the x -coordinate, we would have the one-dimensional TST rate

$$\begin{aligned} k_{\text{TST,1D}} &= Q_r^{-1} \frac{1}{2\pi\beta\hbar} \frac{1}{\beta\hbar\omega_1} e^{-\beta V^\ddagger} \\ &= \frac{\omega_0}{2\pi} e^{-\beta V^\ddagger} \end{aligned}$$

as before.

However, by using the dividing surface perpendicular to the imaginary normal mode, and then can show that

$$k_{\text{hTST}} = Q_r^{-1} \frac{1}{2\pi\beta\hbar} \frac{1}{\beta\hbar\nu_1} e^{-\beta V^\ddagger}$$

where we have defined the normal-mode frequencies by $\lambda_0 = -\nu_0^2$ and $\lambda_1 = \nu_1^2$. We call this hTST as it employs a harmonic approximation to the multidimensional system and uses the optimal dividing surface for this problem. We will assume that this is a good approximation to the full classical rate.

Finally, we compute the transmission coefficient:

$$\kappa \equiv \frac{k_{\text{hTST}}}{k_{\text{TST,1D}}} = \sqrt{\frac{\omega_1^2}{\lambda_1}} = \sqrt{\frac{\omega_1^2}{\lambda_1} \frac{\lambda_0 \lambda_1}{-\omega_1^2 \bar{\omega}^2}} = \sqrt{\frac{-\lambda_0}{\bar{\omega}^2}} = \frac{\nu_0}{\bar{\omega}}$$

which is known as the Grote–Hynes transmission coefficient.

One can easily plot this function and will find that for weak coupling $c_1 \rightarrow 0$ that $\kappa \rightarrow 1$, whereas for strong coupling $c_1 \rightarrow \infty$ that $\kappa \rightarrow \omega_1^2/c_1$. This is in agreement with Kramers theory in the strong coupling regime (i.e. that the rate decreases as the coupling strength increases) and shows that when used properly on the full system TST can actually describe the rate rather well, even though TST applied only to the x -coordinate is wrong because it can't describe the recrossing in this mode.

9.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Microscopic derivation of the Langevin equation

- Zwanzig, Nonequilibrium statistical mechanics (Oxford), chapter 1
- Tuckerman, Statistical Mechanics (Oxford), chapter 15 [includes information on how to implement Langevin dynamics]
- Nitzan, Chemical Dynamics in Condensed Phases (Oxford), section 8.2.5

Kramers theory

- Baron Peters, Reaction Rate Theory and Rare Events (Elsevier), chapters 16 and 17 <https://www.sciencedirect.com/book/9780444563491/reaction-rate-theory-and-rare-events-simulations> (available through ETH library using VPN)

Hopefully you may be able to read them on the internet.

10 Electron-transfer reactions

10.1 Electron-transfer theory

The Hamiltonian written in the diabatic representation is

$$\hat{H} = |0\rangle H_0 \langle 0| + |1\rangle H_1 \langle 1| + |0\rangle \Delta \langle 1| + |1\rangle \Delta \langle 0| \quad (10.1)$$

$$= \sum_j \frac{p_j^2}{2m} + |0\rangle V_0 \langle 0| + |1\rangle V_1 \langle 1| + |0\rangle \Delta \langle 1| + |1\rangle \Delta \langle 0| \quad (10.2)$$

where $|0\rangle$ and $|1\rangle$ are the reactant and product electronic states.

We identify the important operators:

- projection onto reactants is $|0\rangle\langle 0|$
- projection onto products is $|1\rangle\langle 1|$
- flux is $\hat{F} = \frac{i}{\hbar} [\hat{H}, |1\rangle\langle 1|] = \frac{i}{\hbar} (\hat{H} |1\rangle\langle 1| - |1\rangle\langle 1| \hat{H}) = \frac{i}{\hbar} \Delta (|0\rangle\langle 1| - |1\rangle\langle 0|)$

which you check are all Hermitian.

To lowest order in Δ , the reactant partition function is

$$Z_0 = \text{Tr} [e^{-\beta \hat{H}} |0\rangle\langle 0|] \sim \text{Tr} [e^{-\beta \hat{H}_0}]$$

and the flux-flux correlation function is

$$\begin{aligned} C_{\text{ff}} &= \text{Tr} [e^{-\beta \hat{H}} \hat{F} e^{i\hat{H}t/\hbar} \hat{F} e^{-i\hat{H}t/\hbar}] \\ &= \left(\frac{i}{\hbar} \Delta\right)^2 \text{Tr} [e^{-\beta \hat{H}} |0\rangle\langle 1| e^{i\hat{H}t/\hbar} |0\rangle\langle 1| e^{-i\hat{H}t/\hbar}] - \left(\frac{i}{\hbar} \Delta\right)^2 \text{Tr} [e^{-\beta \hat{H}} |0\rangle\langle 1| e^{i\hat{H}t/\hbar} |1\rangle\langle 0| e^{-i\hat{H}t/\hbar}] \\ &\quad + \text{“other way around”} \\ &\sim \frac{\Delta^2}{\hbar^2} \text{Tr} [e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} e^{-i\hat{H}_0 t/\hbar}] + \text{“other way around”} \end{aligned}$$

We can thus compute the rate using

$$\begin{aligned} kZ_0 &= \int_{-\infty}^{\infty} \frac{1}{2} C_{\text{ff}}(t) dt \\ &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \text{Tr} [e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} e^{-i\hat{H}_0 t/\hbar}] dt \end{aligned}$$

where we have used the fact that both terms in the correlation function integrate to the same number, so we only need one (multiplied by 2).

It is possible to show that this formulation for the rate is equivalent to Fermi's golden rule given earlier in the course. We expand the trace using the eigenstate basis, defined such that

$\hat{H}_0 |\psi_0^\mu\rangle = E_0^\mu |\psi_0^\mu\rangle$ and $\hat{H}_1 |\psi_1^\nu\rangle = E_1^\nu |\psi_1^\nu\rangle$ and insert complete sets of states $1 = \sum_\mu |\psi_0^\mu\rangle\langle\psi_0^\mu|$ and $1 = \sum_\nu |\psi_1^\nu\rangle\langle\psi_1^\nu|$. This gives

$$\begin{aligned}
 kZ_0 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \text{Tr} \left[e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} e^{-i\hat{H}_0 t/\hbar} \right] dt \\
 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \sum_\mu \sum_\nu \langle \psi_0^\mu | e^{-\beta \hat{H}_0} e^{i\hat{H}_1 t/\hbar} | \psi_1^\nu \rangle \langle \psi_1^\nu | e^{-i\hat{H}_0 t/\hbar} | \psi_0^\mu \rangle dt \\
 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \sum_\mu \sum_\nu e^{-\beta E_0^\mu} \langle \psi_0^\mu | \psi_1^\nu \rangle e^{iE_1^\nu t/\hbar} \langle \psi_1^\nu | \psi_0^\mu \rangle e^{-iE_0^\mu t/\hbar} dt \\
 &= \frac{\Delta^2}{\hbar^2} \sum_\mu e^{-\beta E_0^\mu} \sum_\nu |\langle \psi_0^\mu | \psi_1^\nu \rangle|^2 \int_{-\infty}^{\infty} e^{i(E_1^\nu - E_0^\mu)t/\hbar} dt \\
 &= \frac{2\pi\Delta^2}{\hbar} \sum_\mu e^{-\beta E_0^\mu} \sum_\nu \underbrace{|\langle \psi_0^\mu | \psi_1^\nu \rangle|^2}_{\text{Franck-Condon factor}} \delta(E_0^\mu - E_1^\nu)
 \end{aligned}$$

which is identical to the previous formula if we identify the couplings $\Delta_{\mu\nu} = \Delta \langle \psi_0^\mu | \psi_1^\nu \rangle$.

Now we take the classical limit by converting quantum trace into phase-space averages.

$$Z_0 = \text{Tr}[e^{-\beta \hat{H}_0}] \rightarrow (2\pi\hbar)^{-f} \iint e^{-\beta H_0} dp dq$$

$$\begin{aligned}
 kZ_0 &= \frac{\Delta^2}{\hbar^2} \int_{-\infty}^{\infty} \text{Tr} \left[e^{-\beta \hat{H}_0} e^{i\hat{H}_0 t/\hbar} e^{-i\hat{H}_1 t/\hbar} \right] dt \\
 &\rightarrow (2\pi\hbar)^{-f} \frac{\Delta^2}{\hbar^2} \iiint e^{-\beta H_0} e^{iH_0 t/\hbar} e^{-iH_1 t/\hbar} dp dq dt \\
 &= (2\pi\hbar)^{-f} \frac{\Delta^2}{\hbar^2} \iint e^{-\beta H_0} \int e^{i(H_0 - H_1)t/\hbar} dt dp dq \\
 &= (2\pi\hbar)^{-f} \frac{2\pi\Delta^2}{\hbar} \iint e^{-\beta H_0} \delta(H_0 - H_1) dp dq \\
 &= \frac{2\pi\Delta^2}{\hbar} \langle \delta(H_0 - H_1) \rangle_0 \\
 &= \frac{2\pi\Delta^2}{\hbar} \langle \delta(V_0 - V_1) \rangle_0
 \end{aligned}$$

where the final expressions are classical averages over the Boltzmann distribution of the reactant. These can be computed using molecular dynamics simulations.

10.2 Marcus theory

In certain cases, it is appropriate to assume that the Hamiltonians can be represented in harmonic system-bath form.

$$H_0 = \frac{p^2}{2M} + \sum_j \frac{p_j^2}{2m} + \frac{1}{2}M\Omega^2(x + \xi)^2 + \sum_j \frac{1}{2}m\omega_j^2 \left(q_j - \frac{c_j}{m\omega_j^2}x \right)^2$$

$$H_1 = \frac{p^2}{2M} + \sum_j \frac{p_j^2}{2m} + \frac{1}{2}M\Omega^2(x - \xi)^2 + \sum_j \frac{1}{2}m\omega_j^2 \left(q_j - \frac{c_j}{m\omega_j^2}x \right)^2 - \varepsilon$$

We define the reorganization energies of reactant and product:

$$\Lambda_0 = V_0(x_{\min}^{(1)}) - V_0(x_{\min}^{(0)}) = 2M\Omega^2\xi^2$$

$$\Lambda_1 = V_1(x_{\min}^{(0)}) - V_1(x_{\min}^{(1)}) = 2M\Omega^2\xi^2$$

which are seen to be equal to each other.

Using this harmonic assumption, we can compute the rate (which will turn out to be Marcus theory)

$$\begin{aligned} k_{\text{MT}} &= \frac{2\pi\Delta^2}{\hbar} \frac{\int e^{-\beta V_0} \delta(V_0 - V_1) dx d\mathbf{q}}{\int e^{-\beta V_0} dx d\mathbf{q}} \\ &= \frac{2\pi\Delta^2}{\hbar} \frac{\int e^{-\beta V_0} \delta(2M\Omega^2\xi x + \varepsilon) dx d\mathbf{q}}{\int e^{-\beta V_0} dx d\mathbf{q}} \\ &= \frac{2\pi\Delta^2}{\hbar} \frac{\int e^{-\beta V_0(x)} \delta(2M\Omega^2\xi x + \varepsilon) dx}{\int e^{-\beta M\Omega^2(x+\xi)^2/2} dx} \\ &= \frac{2\pi\Delta^2}{\hbar} \frac{e^{-\beta V_0(x^\ddagger)} \frac{1}{2M\Omega^2\xi}}{\sqrt{2\pi/\beta M\Omega^2}} \\ &= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{2M\Omega^2\xi^2}} e^{-\beta V^\ddagger} \\ &= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{\Lambda}} e^{-\beta V^\ddagger} \end{aligned}$$

where $x^\ddagger = -\varepsilon/2M\Omega^2\xi$ and the activation energy is

$$\begin{aligned} V^\ddagger &= \frac{1}{2}M\Omega^2(x^\ddagger + \xi)^2 \\ &= \frac{1}{2}M\Omega^2 \left(-\frac{\varepsilon}{2M\Omega^2\xi} + \xi \right)^2 \\ &= (2M\Omega^2\xi^2 - \varepsilon)^2 / 8M\Omega^2\xi^2 \\ &= (\Lambda - \varepsilon)^2 / 4\Lambda \end{aligned}$$

such that the Marcus theory rate can be written in its usual form

$$k_{\text{MT}} = \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{\Lambda}} e^{-\beta(\Lambda - \varepsilon)^2 / 4\Lambda}$$

This equation predicts that the rate is slow when ε is small, that the rate increases as ε approaches Λ at which it reaches a maximum where the reaction is activationless. Then as ε increases past Λ , the reaction is in the inverted regime and the rate decreases again. In reality (as shown in experiment and simulations taking account of quantum effects), tunnelling causes the rate to increase substantially in the inverted regime, skewing the Marcus turnover curve.

10.3 Landau–Zener theory

According to Landau–Zener theory, the transition probability between two diabatic states is

$$P_{1 \leftarrow 0} = 1 - \exp \left[-\frac{2\pi\Delta^2}{\hbar \left| \frac{\partial}{\partial t} (E_0(x) - E_1(x)) \right|} \right]$$

This formula is an exact solution to the time-dependent dynamics of a two-level system, in which the energies change linearly with time. It is however used as an approximation to the coupled problem of a nonadiabatic transition. In this case, we use $E_n(x(t)) = V^\ddagger + vF_nt$, where v is the velocity of the particle and F_n the force in state n .

$$P_{1 \leftarrow 0} = 1 - \exp \left[-\frac{2\pi\Delta^2}{\hbar v |F_0 - F_1|} \right]$$

In the weak-coupling limit (or equivalently for fast moving particles or very sharp barriers)

$$P_{1 \leftarrow 0} \rightarrow \frac{2\pi\Delta^2}{\hbar v |F_0 - F_1|}$$

Whereas in the strong-coupling limit:

$$P_{1 \leftarrow 0} \rightarrow 1$$

This strong-coupling limit implies that a diabatic transition is inevitable, or in other words that the adiabatic transition is very unlikely to happen. This means that the adiabatic (Born–Oppenheimer) approximation is valid and is in agreement with what we noticed in previous lectures: that the adiabatic approximation is good for slow moving particles and also for large energy separation between the adiabatic states, which is a consequence of large Δ .

Compare these findings with those of the dynamics of a time-independent two-level system studied earlier in the course. There, we found that the population transfer was only large if the initial and final state were degenerate. In Landau–Zener theory, the transfer occurs mostly near the point when the two states are close together, as you will show in the homework exercise.

10.4 Nonadiabatic transition-state theory

Here we will take a one-dimensional system. Coupling to other modes can be included using the techniques of the previous chapter. First we will consider the TST rate in the adiabatic limit, which

can be written

$$\begin{aligned}
 k_{\text{TST}} &= \frac{(2\pi\hbar)^{-1} \iint e^{-\beta H_0} \delta(x - x^\ddagger) \frac{p}{m} \Theta(p) \, dx \, dp}{(2\pi\hbar)^{-1} \iint e^{-\beta H_0} \, dx \, dp} \\
 &= \frac{\beta\Omega}{2\pi} \iint e^{-\beta H_0} \delta(x - x^\ddagger) \frac{p}{m} \Theta(p) \, dx \, dp \\
 &= \frac{\Omega}{2\pi} \int e^{-\beta V_0} \delta(x - x^\ddagger) \, dx \\
 &= \frac{\Omega}{2\pi} e^{-\beta V^\ddagger}
 \end{aligned}$$

where we used the classical vibrational partition function $Z_0 = \frac{1}{\beta\hbar\Omega}$.

Let's propose a nonadiabatic transition-state theory as

$$k_{\text{NTST}} = \kappa k_{\text{TST}} = \kappa \frac{\Omega}{2\pi} e^{-\beta V^\ddagger}$$

where κ is a transmission coefficient which includes all nonadiabatic effects.

To derive a formula for κ , we start again from the flux-side correlation function formalism but also include an extra weight $T(v)$, which is the transmission probability at a given velocity.

$$\begin{aligned}
 k_{\text{NTST}} &= \frac{\beta\Omega}{2\pi} \iint e^{-\beta H_0} \delta(x - x^\ddagger) \frac{p}{M} T(p/M) \Theta(p) \, dx \, dp \\
 &= \frac{\beta\Omega}{2\pi} \iint e^{-\beta p^2/2M - \beta V_0} \delta(x - x^\ddagger) \frac{p}{M} T(p/M) \Theta(p) \, dx \, dp \\
 &= \frac{\beta M \Omega}{2\pi} \iint e^{-\beta M v^2/2 - \beta V_0} \delta(x - x^\ddagger) v T(v) \Theta(v) \, dx \, dv \\
 &= \frac{\beta M \Omega}{2\pi} \left[\int_0^\infty v e^{-\beta M v^2/2} T(v) \, dv \right] e^{-\beta V^\ddagger} \\
 &= \kappa \frac{\Omega}{2\pi} e^{-\beta V^\ddagger}
 \end{aligned}$$

so

$$\kappa = \beta M \int_0^\infty v e^{-\beta M v^2/2} T(v) \, dv$$

The transmission probability can be obtained using an argument based on Landau-Zener (LZ) theory. LZ theory tells us that a diabatic-state transition will occur with probability P . Thus we take $T(v) = P + \dots$. However, if the reaction remains on the same diabatic state (which has probability $1 - P$), it will eventually reach a region with strong forces pushing it backwards so it returns to pass the crossing point again. Here it again has a probability of P for changing diabatic state and a probability of $1 - P$ of returning to the reactant well. By following the motion of the

particle in this way, we can deduce that

$$\begin{aligned}
T(v) &= P + (1 - P)P(1 - P) + (1 - P)P^3(1 - P) + \dots \\
&= P + (1 - P)P \sum_n (P^2)^n (1 - P) \\
&= P + (1 - P)P \frac{1 - P}{1 - P^2} \\
&= P + (1 - P)P \frac{1}{1 + P} \\
&= \frac{P(1 + P) + (1 - P)P}{1 + P} \\
&= \frac{2P}{1 + P}
\end{aligned}$$

We consider the limiting case of strong coupling, where $P \rightarrow 1$ and therefore $T(v) \rightarrow 1$ such that

$$\kappa \rightarrow \beta M \int_0^\infty v e^{-\beta M v^2/2} dv = 1$$

As expected in this strong-coupling limit, we recover the adiabatic TST result.

$$k_{\text{NTST}} \rightarrow k_{\text{TST}}$$

In the weak-coupling limit we have small $P \rightarrow \frac{2\pi\Delta^2}{\hbar v|F_0 - F_1|} \ll 1$ and thus $T(v) \rightarrow 2P$

$$\begin{aligned}
\kappa &\rightarrow \beta M \int_0^\infty v e^{-\beta M v^2/2} \frac{4\pi\Delta^2}{\hbar v|F_0 - F_1|} dv \\
&= \frac{1}{2}\beta M \frac{4\pi\Delta^2}{\hbar|F_0 - F_1|} \sqrt{\frac{2\pi}{\beta M}} = \frac{2\pi\Delta^2}{\hbar|F_0 - F_1|} \sqrt{2\pi\beta M}
\end{aligned}$$

so

$$k_{\text{NTST}} = \frac{\Omega}{2\pi} \frac{2\pi\Delta^2}{\hbar|F_0 - F_1|} \sqrt{2\pi\beta M} e^{-\beta V^\ddagger} = \frac{\Delta^2}{\hbar|F_0 - F_1|} \Omega \sqrt{2\pi\beta M} e^{-\beta V^\ddagger}$$

which using $|F_0 - F_1| = |M\Omega^2(x + \xi) - M\Omega^2(x - \xi)| = 2M\Omega^2\xi$ can be shown to be equal to Marcus theory:

$$\begin{aligned}
k_{\text{NTST}} &= \frac{\Delta^2}{\hbar} \frac{\sqrt{2\pi\beta M}}{2M\Omega\xi} e^{-\beta V^\ddagger} \\
&= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{2M\Omega^2\xi^2}} e^{-\beta V^\ddagger} \\
&= \frac{\Delta^2}{\hbar} \sqrt{\frac{\pi\beta}{\Lambda}} e^{-\beta V^\ddagger}
\end{aligned}$$

Thus in the weak-coupling limit, we recover the golden-rule result and in the strong-coupling limit the adiabatic result. This shows that our NTST approach has the correct behaviour.

10.5 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Electron-transfer theory

- Chandler, Electron transfer in water and other polar environments, how it happens *from: Classical and Quantum Dynamics in Condensed Phase Simulations*, edited by B. J. Berne, G. Ciccotti and D. F. Coker (World Scientific, Singapore, 1998), pgs. 25-49.
<http://stage.cchem.berkeley.edu/~dcgrp/Pubs/DC154.pdf>

Landau-Zener theory

- C. Zener, “Non-adiabatic crossing of energy levels”, *Proc. R. Soc. Lond. A*, **137**, 696 (1932).
<http://dx.doi.org/10.1098/rspa.1932.0165>

11 Quantum states, level densities and level numbers of composite quantum systems

11.1 Harmonic oscillator

$$E_v = h\nu(v + \frac{1}{2}), \quad v = 0, 1, 2, 3, \dots, \infty \quad (11.1)$$

Number of states with energy $E_v < E$ given by

$$W(E) = \sum_{v=0}^{\infty} h(E - E_v), \quad (11.2)$$

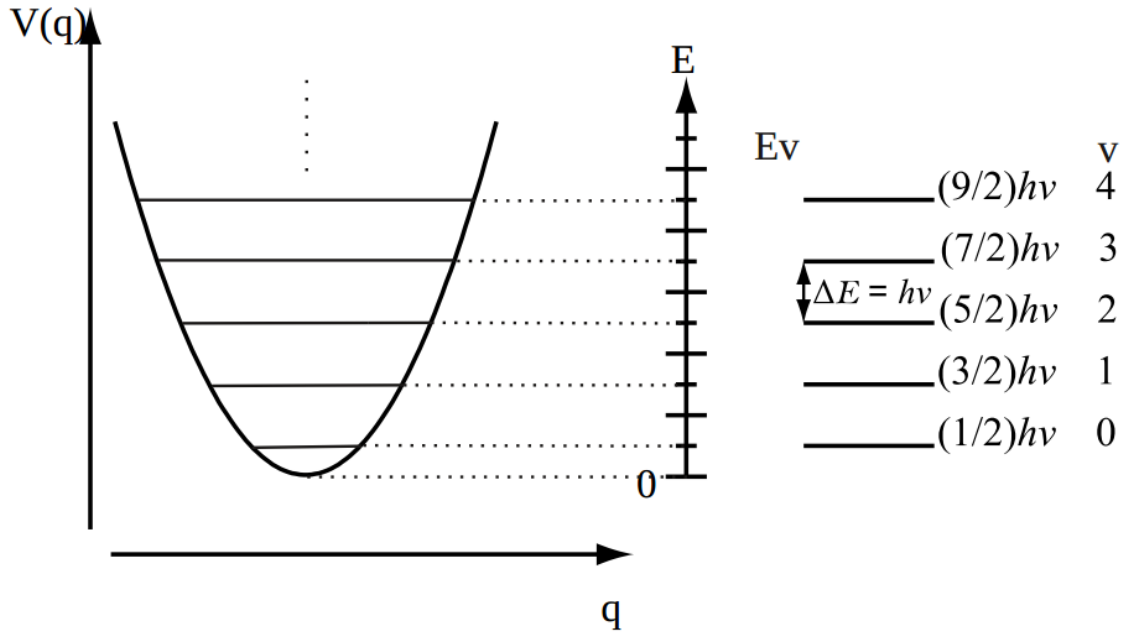


Figure 8: Potential energy $V(q)$ and energy eigenvalues E_v of the harmonic oscillator.

where $h(x)$ is the Heaviside step function defined as

$$\begin{aligned} h(x) &= 0 & \text{for } x < 0 \\ h(x) &= 1 & \text{for } x > 0 \end{aligned} \quad (11.3)$$

Density of states is defined as

$$\rho(E) = \frac{dW(E)}{dE} = \sum_{v=0}^{\infty} \delta(E - E_v) \quad (11.4)$$

Instead of $W(E)$, one often works with the smoothed version $\langle W(E) \rangle$ which describes the averaged behaviour

$$\langle W(E) \rangle = v + 1 = \frac{E}{h\nu} + \frac{1}{2} = \frac{E + E_z}{h\nu} \quad (11.5)$$

The deviation of $\langle W(E) \rangle$ with respect to energy provides the averaged level density:

$$\langle \rho(E) \rangle = \frac{d \langle W(E) \rangle}{dE} = \frac{1}{h\nu} \simeq \frac{\Delta W(E)}{\Delta E} \quad (11.6)$$

Both averaged quantities are only meaningful in the limit of many quantum states per energy interval.

11.2 Many harmonic oscillators

The levels of a quantum system composed of 2 subsystems are given by

$$E_m = E_{n_1, n_2} = E_{n_1} + E_{n_2} \quad (11.7)$$

For example, we consider 2 harmonic oscillators

$$E_k = E_{v_1, v_2} = (v_1 + \frac{1}{2})h\nu_1 + (v_2 + \frac{1}{2})h\nu_2 \quad (11.8)$$

Expressed in the level densities, we have

$$\begin{aligned} N_1(E_1) &= \rho_1(E_1) |dx| \equiv \rho_1(E') |dE'| \\ N_2(E_2) &= \rho_2(E_2) |dx| \equiv \rho_2(E'') |dE''| \end{aligned} \quad (11.9)$$

The total number of levels that fulfill $E_1 + E_2 = E$ ($|dx| \ll E$) is given by

$$N_{1,2} = N_1 N_2 = \rho_1(E_1) \rho_2(E_2) (dx)^2 \quad (11.10)$$

The corresponding level density for $E_1 = x$ and $E_2 = E - x$ is

$$\begin{aligned} \rho_{1,2}(E_1 = x, E_2 = E - x, E) &= \frac{dN_{1,2}}{dx} \\ &= \rho_1(x) \rho_2(E - x) dx \end{aligned} \quad (11.11)$$

The total level density at energy E is then obtained by integrating over the variable x

$$\begin{aligned} \rho_{1,2}(E_2) &= \int_{x=0}^{x=E} \rho_1(x) \rho_2(E - x) dx \\ &= \int_{E'=0}^{E'=E} \rho_1(E') \rho_2(E'') dE'', \end{aligned} \quad (11.12)$$

which corresponds to the convolution of the functions ρ_1 and ρ_2 , which is also written

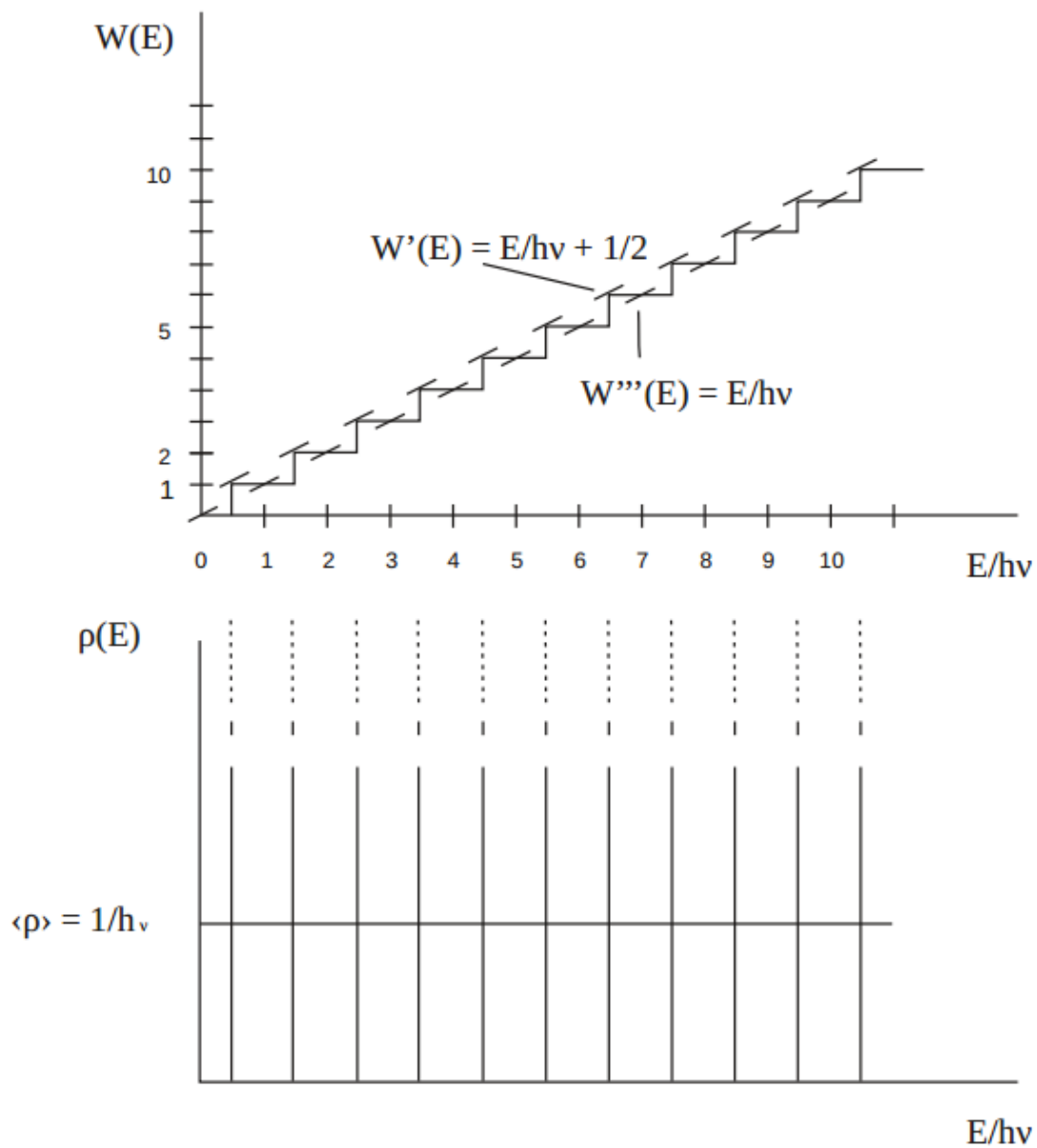


Figure 9: Number of states $W(E)$ and the density of states $\rho(E)$ for a harmonic oscillator.

$$\rho_{1,2} = \rho_1 \otimes \rho_2 = \rho_2 \otimes \rho_1 \quad (11.13)$$

Similarly, the number of levels can also be obtained by convolution

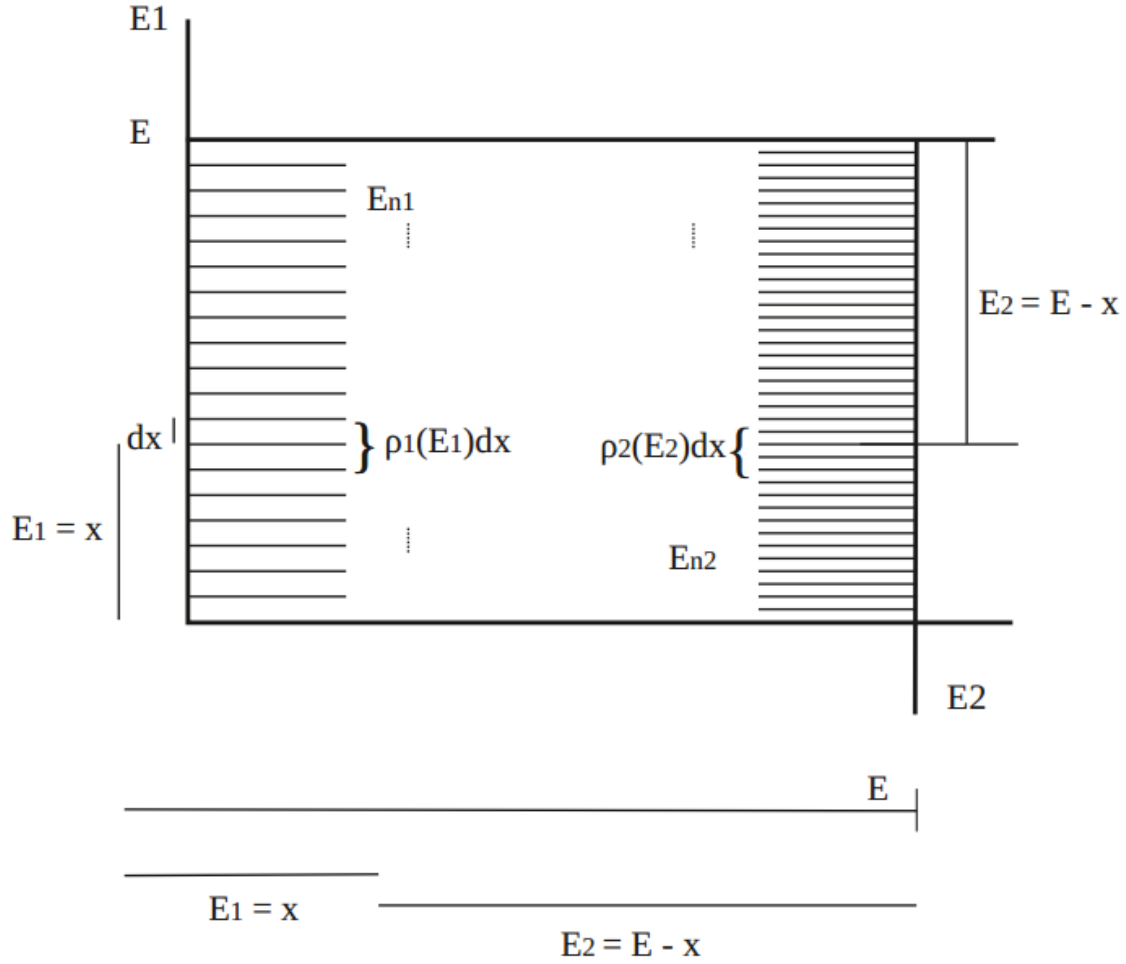


Figure 10: Convolution of the densities of states of two spectra $E_1 + E_2 = E$, see text).

$$\begin{aligned}
 W_{1,2}(E) &= \int_0^E W_1(E') \rho_2(E - E') dE' \\
 &= W_1 \otimes \rho_2 = \rho_1 \otimes W_2 \\
 &= \int_0^E \rho_{1,2}(x) dx
 \end{aligned}
 \tag{11.14}$$

The level density of a system composed of n subsystems can be obtained by a sequence of convolutions

$$\begin{aligned}
\rho_{1,2,3}(E) &= \int_0^E \rho_{1,2}(x) \rho_3(E-x) dx \\
&= \int_0^E \rho_{1,2}(E-E''') \rho_3(E''') dE'''
\end{aligned} \tag{11.15}$$

The application of these formula to the smoothed level density of harmonic oscillators is particularly simple because the level densities are not energy dependent:

$$\begin{aligned}
\rho_{1,2,3} &= (h\nu_1)^{-1} (h\nu_2)^{-1} (h\nu_3)^{-1} \int_0^E E''' dE''' \\
&= (h\nu_1 h\nu_2 h\nu_3)^{-1} \frac{1}{2} E^2
\end{aligned} \tag{11.16}$$

For s harmonic oscillators we therefore find

$$\rho_{1,2,3,\dots,s}(E) = \frac{E^{s-1}}{(s-1)! \prod_{i=1}^s h\nu_i} \tag{11.17}$$

The number of levels is obtained by integration

$$\begin{aligned}
W_{1,2,3,\dots,s} &= \frac{E^s}{s! \prod_{i=1}^s h\nu_i} \\
&= \int_0^E \rho_{1,2,3,\dots,s}(E''') dE'''
\end{aligned} \tag{11.18}$$

These simple formulae represent an approximate result following from the use of the smoothed density function $\langle \rho(E) \rangle \approx (h\nu_i)^{-1}$. A more accurate calculation provides the following better approximation

$$\begin{aligned}
\rho_{1,2,3,\dots,s} &= \frac{(E + aE_z)^{s-1}}{(s-1)! \prod_{i=1}^s (h\nu_i)} \\
W_{1,2,3,\dots,s} &= \frac{(E + aE_z)^s}{s! \prod_{i=1}^s (h\nu_i)}
\end{aligned} \tag{11.19}$$

where E_z is the total zero-point energy of the s harmonic oscillators

$$E_z = \sum_{i=1}^s \frac{1}{2} (h\nu_i) \tag{11.20}$$

and a is an energy-dependent parameter which can be chosen as $a = 1$ for the purpose of a simple approximation. This yields a upper boundary for $\rho(E)$ whereas $a = 0$ represent a lower boundary. $a = 1$ is often called the "semi-classical approximation", whereas $a = 0$ is the "classical-approximation".

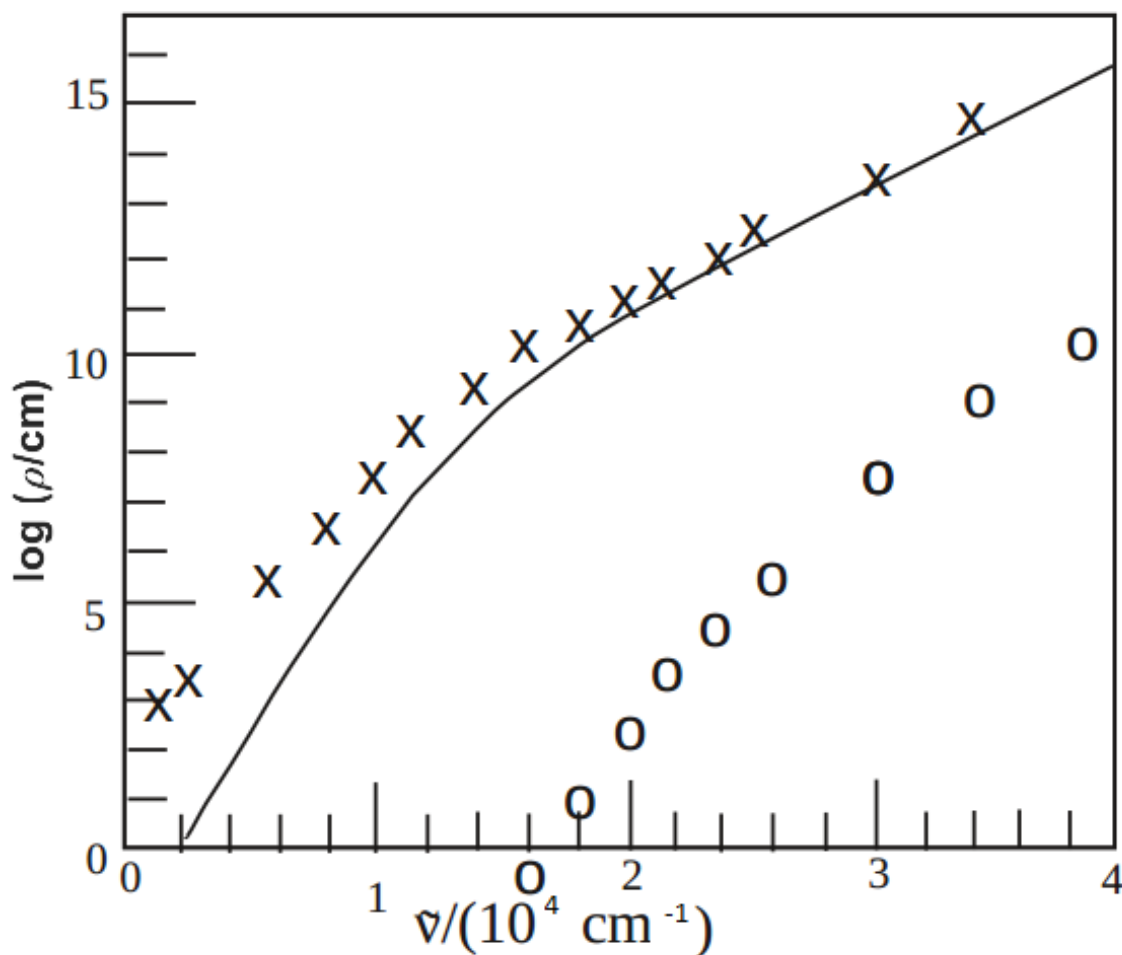


Figure 11: Logarithm (base 10) of the density of states for 30 vibrational degrees of freedom in $\text{C}_6\text{H}_5\text{F}$, Symbol x="semiclassical approximation" with $a = 1$ in Equation.(11.19); Symbol o="classical approximation". The solid line results from a step function in quantum mechanics "exact" step function $W(E)$, (from Chem.Phys. 51,353 (1980)). Wave number units have been used, which are related to energy by $E = hc\tilde{\nu}$ for the density of states $\rho(E) = \frac{\tilde{\rho}}{hc}$.

11.3 Many particles in a box (model for ideal gas)

Energy levels of a particle of mass m in one-dimensional box of length a are given by

$$E_n = \frac{n^2 h^2}{8ma^2}, \quad n = 1, 2, 3, \dots \quad (11.21)$$

The quantum number n counts the number of levels, hence

$$W(E) = n = \frac{2a}{h} \sqrt{2mE} \quad (11.22)$$

The level density is obtained through derivation

$$\rho(E) = \frac{dW(E)}{dE} = \frac{2a}{h} \sqrt{2m} \frac{1}{2} E^{-\frac{1}{2}} = \frac{a}{h} \sqrt{2m} E^{-\frac{1}{2}} \quad (11.23)$$

The level density for a particle in a three-dimensional box is obtained by convolution

$$\rho_d = \rho_{x,y,z} = \rho_x \otimes \rho_y \otimes \rho_z = m^{\frac{3}{2}} 2^{\frac{5}{2}} \pi a^3 h^{-3} E^{\frac{1}{2}} \quad (11.24)$$

In contrast to the spectrum of the harmonic oscillator, the level densities for a particle in a box (1D or 3D) depend on the energy. Explicit integration yields,

$$W_d(E) = \int \rho_d(E) dE = \pi V h^{-3} m^{\frac{3}{2}} \frac{2^{\frac{7}{2}}}{3} E^{\frac{3}{2}} \quad (11.25)$$

One can again show that a slightly better approximation of the step-like level-number function is possible:

$$\begin{aligned} \ln[W_d(x)] &= \frac{3}{2}x - ce^{-bx} + h\left(\frac{\pi}{6}\right) \\ x &= \ln(8ma^2 \frac{E}{h^2}) \end{aligned} \quad (11.26)$$

For N particles of the same mass m in a box we find

$$\rho_N^{(E)} = \rho_{d_1} \otimes \rho_{d_2} \otimes \rho_{d_3} \otimes \dots \otimes \rho_{d_N} \quad (11.27)$$

The corresponding number of states is

$$\begin{aligned} W_N(E) &= \int_0^{(E)} \rho_N(x) dx = \alpha E^{\frac{3N}{2}} \\ \text{where } \alpha &= [2\pi V \frac{(2m)^{\frac{3}{2}}}{h^3}]^N \frac{[\Gamma(\frac{3}{2})]^N}{\Gamma[\frac{3}{2}N + 1]} \end{aligned} \quad (11.28)$$

where a^3 is the volume V of the box and we call the properties of the gamma function

$$\begin{aligned} \Gamma\left(\frac{3}{2}\right) &= \frac{\sqrt{\pi}}{2} \\ \Gamma(n) &= (n-1)! \quad \text{for } n = 1, 2, 3, \dots \end{aligned} \quad (11.29)$$

In the case of N identical, non-distinguishable particle, such as gas atoms of the same isotope, we need to take into account the Pauli principle, i.e. the fact that only a function $(N!)^{-1}$ of these states is allowed. Therefore

$$W_{N,\text{Pauli}}(E) = \alpha (N!)^{-1} E^{\frac{3N}{2}} \quad (11.30)$$

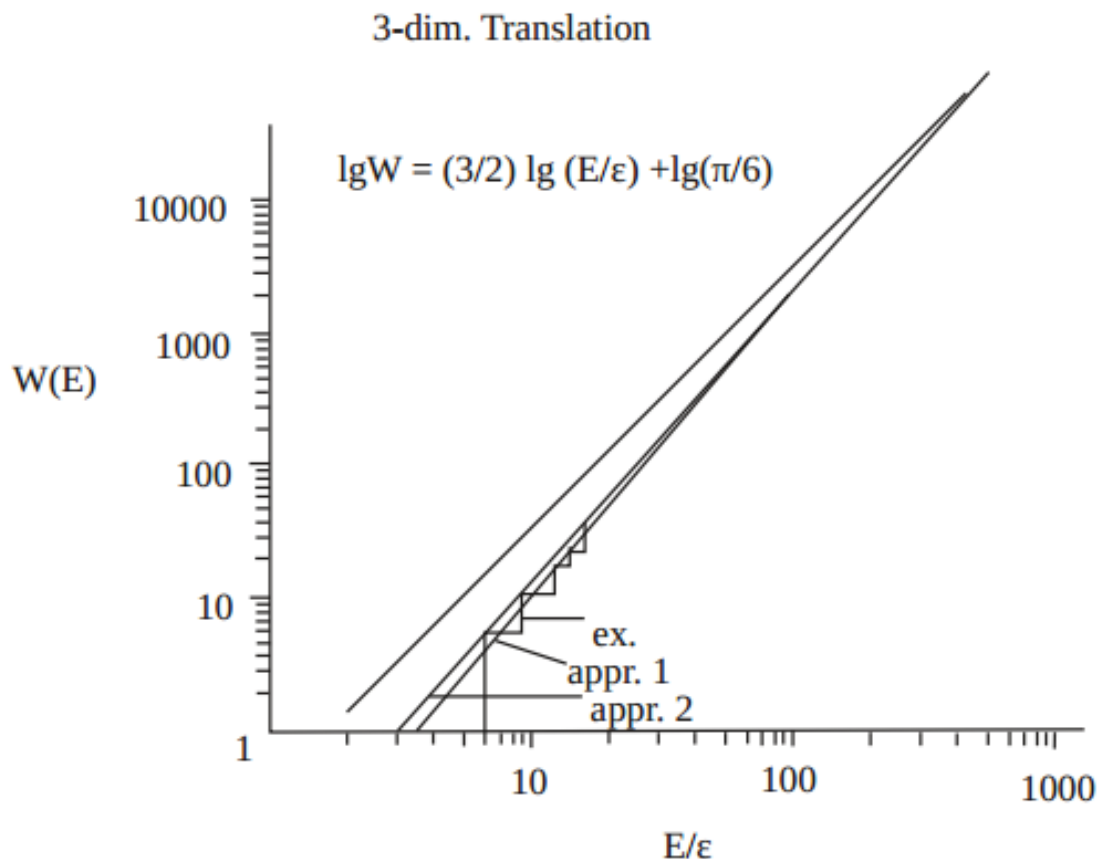


Figure 12: Partition function $W(E)$ for the three-dimensional translational motion of a particle in the box, exact step function and continuous approximations (appr. 1, appr.2 (logarithmic scale)). The approximation 2 (appr.2) has the particular advantage that $W(E) = 1$ for $E = E_z(3\epsilon, \text{Zero-point energy})$ and is of the form in Eq. (11.26) with $c = [1.5 \ln 3 + \ln(\frac{\pi}{6})]e^{b \ln 3}$, where $b = 0.413$ is a good approximation (variable). The approximation 1 corresponds to Eq. (11.26) with $c = 2.35, b = 0.48, \epsilon = \frac{h^2}{8ma^2}$.

This simple correction by division through $N!$ is only valid for high energies E and very large values of $W(E)$.

These results obtained so far allowed us to draw the following conclusions about quantum levels and properties of macroscopic systems:

1. The average energy-level separation of neighbouring quantum states is given by the inverse average level density and becomes unmeasurably small for all known forms of spectroscopy.
2. As a consequence of 1, it is experimentally impossible to prepare single quantum levels as stationary energy eigenstates. The **concept** of stationary states therefore loses its usefulness

when mesoscopic or macroscopic systems are considered.

3. Since the experimentally observable quantum states of macroscopic systems are therefore always superposition of many energy levels, they have to always be considered as **time-dependent** states.
4. The vary large numbers involved suggest or require the application of statistical methods.

11.4 Laplace transformation for derivation of level densities and level numbers via partition function

The derivation of the level densities from convolutions can be more accurately obtained via the Laplace transformation. In general, the Laplace transform $g(y)$ of a function $f(x)$ is defined as

$$g(y) = \int_0^\infty f(x)e^{-xy}dx \quad (11.31)$$

In operator notation: $g(y) = \hat{L}[f(x)]$,

The Laplace transformation operator $\hat{L}$ is an integral operator. The inverse transformation with the operator $\hat{L}^{-1}$ corresponds to integration in the complex plane

$$\begin{aligned} f(x) &= \hat{L}^{-1}[g(y)] \\ f(x) &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} g(y)e^{xy}dy \end{aligned} \quad (11.32)$$

The following properties are important for our purpose: Convolution theorem: With $F=f_1 \otimes f_2$ the convolution of functions $f_1(x)$ and $f_2(x)$, and $g_1(y) = \hat{L}f_1(x)$, $g_2(y) = \hat{L}f_2(x)$, the function $G = g_1 \cdot g_2 = \hat{L}F$. This means that **convolution** in the function space becomes a simple product in the space of the Laplace transforms. This theorem can now be exploited to obtain a basic access to level densities, provided that the Laplace transform of the level density of the sub-systems is known. We define the partition function (German: Zustandssumme) q as the Laplace transform of the level density

$$\begin{aligned} q(\beta) &= \int_0^\infty \rho(E)e^{-\beta E}dE = \hat{L}[\rho(E)] \\ &= \sum_{j=0}^\infty e^{-\beta E_j} \\ &= \sum_{j=0}^\infty g_j e^{-\beta E_j} \end{aligned} \quad (11.33)$$

where we have used a property of the Dirac δ -function

$$\begin{aligned} \int_{-\infty}^\infty \delta(x - x_0)f(x)dx &= f(x_0) \\ \int_{-\infty}^\infty \delta(E - E_j)e^{-\beta E} &= e^{-\beta E_j} \end{aligned} \quad (11.34)$$

and $\beta = (kT)^{-1}$ with the Boltzmann constant k and the temperature T .

We again consider two subsystems with energy levels E_{n_1}, E_{n_2} . The partition functions are

$$\begin{aligned} q_1(\beta) &= \sum_{n_1=0}^{\infty} e^{-\beta E_{n_1}} \\ q_2(\beta) &= \sum_{n_2=0}^{\infty} e^{-\beta E_{n_2}} \end{aligned} \tag{11.35}$$

The partition function of the total system is

$$\begin{aligned} q_{1,2} &= q_1 \cdot q_2 = \left(\sum_{n_1=0}^{\infty} e^{-\beta E_{n_1}} \right) \left(\sum_{n_2=0}^{\infty} e^{-\beta E_{n_2}} \right) \\ &= \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} e^{-\beta(E_{n_1} + E_{n_2})} \\ &= \sum_{m=0}^{\infty} e^{-\beta E_m} \end{aligned} \tag{11.36}$$

Without the need of approximations, the properties of the Laplace transformation thus lead to the result

$$\begin{aligned} \rho_{1,2} &= \hat{L}^{-1}(q_1 \cdot q_2) \\ &= \hat{L}^{-1} \hat{L}(\rho_1 \otimes \rho_2) \\ &= \rho_1 \otimes \rho_2 \quad q.e.d \end{aligned} \tag{11.37}$$

We now show that the results of section 6.1.3 can be obtained in a more straight-forward way by using the Laplace transform. We start from the translational partition function q_x using a smoothed level density ρ_x ;

$$\begin{aligned} q_x &= \int_0^{\infty} \rho_x e^{-\beta E} dE \\ &= \frac{a}{h} (2m)^{\frac{1}{2}} \int_0^{\infty} E^{-\frac{1}{2}} e^{-\beta E} dE \end{aligned} \tag{11.38}$$

Using

$$\int_0^{\infty} x^n e^{-bx} dx = \frac{\Gamma(n+1)}{b^{n+1}} \tag{11.39}$$

for all real $n > -1$ and all $b > 0$. We further recall that $\Gamma(n+1) = n!$ for all integer n . Here we need $n = -\frac{1}{2}$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, therefore

$$q_x = \frac{a}{h} (2m)^{\frac{1}{2}} \frac{\sqrt{\pi}}{\beta^{\frac{1}{2}}} = \left(\frac{2\pi m a^2}{\beta h^2} \right)^{\frac{1}{2}} \tag{11.40}$$

The product is therefore given by

$$q_d = q_x q_y q_z = \left(\frac{2\pi m a^2}{\beta h^2}\right)^{\frac{3}{2}} = V \left(\frac{2\pi m a^2}{\beta h^2}\right)^{\frac{3}{2}} \quad (11.41)$$

Using the following properties of the Laplace transform:

$$\begin{aligned} \hat{L}^{-1}\left(\frac{1}{\beta^n}\right) &= \left(\frac{E^{n-1}}{\Gamma(n)}\right) \\ \text{and } \hat{L}\left(\frac{E^{n-1}}{\Gamma(n)}\right) &= \frac{1}{\beta^n} \end{aligned} \quad (11.42)$$

We obtain

$$\hat{L}^{-1}(q_d) = \rho_d = V \left(\frac{2\pi m}{h^2}\right)^{\frac{3}{2}} \frac{2E^{\frac{1}{2}}}{\sqrt{\pi}}, \quad (11.43)$$

which is identical with Equation (11.24) The partition function for N particles is then a product

$$Q = q_d^N = V^N \left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3N}{2}} \quad (11.44)$$

The inverse of the Laplace transform with $n = \frac{3N}{2}$ provides

$$\begin{aligned} \hat{L}^{-1}(Q) &= \rho_N = V^N \left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3N}{2}} E^{(3N/2-1)} / \Gamma(3N/2) \\ W_N(E) &= \int_0^E \rho_N(x) dx = V^N \left(\frac{2\pi m}{h^2}\right)^{\frac{3N}{2}} \frac{E^{(\frac{3N}{2})}}{\frac{3N}{2} \Gamma(\frac{3N}{2})} \end{aligned} \quad (11.45)$$

Using the properties $\Gamma(x+1) = x\Gamma(x)$ and $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ and leads back to equation(11.28).

11.5 Recurrence times in dense spectra

The high level densities of "dense" spectra typical of large molecules excited to high energies or mesoscopic/macrosopic systems lead to very long recurrence times

$$\tau(E) \geq h \langle \rho(E) \rangle, \quad (11.46)$$

where $\langle \rho(E) \rangle$ is the average level density. The equality holds for an equidistant spectrum ($\langle \rho(E) \rangle = \frac{1}{\delta} = \text{constant}$).

Example:

1. $\text{C}_6\text{H}_5\text{F}$ excited to 18000 cm^{-1} : $\tau_R \geq 0.3 \text{ s}$,
2. $(\text{CF}_3)_3\text{CH}$ excited to 15000 cm^{-1} : $\tau_R \geq 0.3 \times 10^5 \text{ s}$ ($\approx 4 \text{ days}$)
3. 1 mole of Hg(s) with internal energy of 4 kJ: $\tau_R \geq 10^{10^{24}} \text{ s}$. (Longer than the age of the universe: $\tau \approx 1.5 \cdot 10^9 \text{ years} < 10^{18} \text{ s}$).

These results show that the **periodicity** of quantum dynamics is irrelevant for macroscopic systems. This however does not exclude the existence of (seemingly) periodic motion on much shorter time scales (e.g. motion of a pendulum, motion of planets). However, these motions are never really periodic but always damped on longer time scales.

11.6 Unimolecular rate theory

Unimolecular rate theory is best understood using microcanonical rate theory. A molecule must be excited to an internal energy E by collisions or by photoabsorption before reacting, which can either be a rearrangement or breaking into two pieces.

In one dimension, the microcanonical rate is easy to calculate using the flux-side correlation function with a microcanonical distribution:

$$\begin{aligned}\lim_{t \rightarrow 0+} C_{\text{fs}}(t) &= \frac{1}{2\pi\hbar} \iint \delta(x - x^\ddagger) \frac{p}{m} \theta(p) \delta(H - E) dx dp \\ &= \frac{1}{2\pi\hbar} \int_0^\infty \frac{p}{m} \delta\left(\frac{p^2}{2m} + V^\ddagger - E\right) dp \\ &= \frac{1}{2\pi\hbar} \theta(E - V^\ddagger)\end{aligned}$$

We are allowed to take the short-time limit, because in 1D, there is no recrossing as long as the dividing surface is chosen correctly. The microcanonical rate is therefore:

$$k(E) = \lim_{t \rightarrow 0+} \frac{C_{\text{fs}}(t)}{\rho_{\text{r}}(E)} = \frac{\theta(E - V^\ddagger)}{2\pi\hbar \rho_{\text{r}}(E)} = \frac{P_{\text{transmission}}(E)}{2\pi\hbar \rho_{\text{r}}(E)}$$

where we have noted that the Heaviside step function is the classical transmission function discussed in the lecture on scattering theory.

In Rice–Ramsperger–Kassels–Marcus (RRKM) theory, one deals with a multidimensional system in which it is assumed that all reactant states with energy E are equally likely to be populated. If the molecule was excited to a particular mode, this means that we assume that intramolecular vibrational redistribution occurs on a much faster timescale than the reaction. Then we say that there is a reaction mode with energy E_{rxn} , whereas all other modes in states $\mathbf{n} = (n_1, n_2, \dots, n_{f-1})$ have vibrational energy $E_{\mathbf{n}}$, such that the total energy is $E = E_{\text{rxn}} + E_{\mathbf{n}}$. Dynamics along the reaction coordinate is assumed to be classical (so there is no tunnelling) and the reaction can only proceed if $E_{\text{rxn}} > V^\ddagger$. Under these assumptions, the total reaction rate is given by

$$\begin{aligned}\lim_{t \rightarrow 0+} C_{\text{fs}}(t) &= \sum_{\mathbf{n}} \frac{1}{2\pi\hbar} \iint \delta(x - x^\ddagger) \frac{p}{m} \theta(p) \delta(H_{\text{rxn}} + E_{\mathbf{n}}^\ddagger - E) dx dp \\ &= \sum_{\mathbf{n}} \frac{1}{2\pi\hbar} \theta(E_{\text{rxn}} - V^\ddagger) \quad \text{where} \quad E_{\text{rxn}} = E - E_{\mathbf{n}}^\ddagger \\ k(E) &= \lim_{t \rightarrow 0+} \frac{C_{\text{fs}}(t)}{\rho_{\text{r}}(E)} = \frac{N^\ddagger(E)}{2\pi\hbar \rho_{\text{r}}(E)} \quad \text{where} \quad N^\ddagger(E) = \sum_{\mathbf{n}} \theta(E - E_{\mathbf{n}}^\ddagger - V^\ddagger)\end{aligned}$$

It is common to assume that the transition state can be described by a set of harmonic oscillators perpendicular to the reaction coordinate. For small molecules, one can then compute $k(E)$ by summing over $\mathbf{n}$. However, for larger molecules the number of terms in the sum soon becomes so great that one resorts to a classical approximation. This is valid for energies much higher than $V^\ddagger$, where the density of states is large.

$$k_{\text{RRKM}}(E) = \frac{N^\ddagger(E)}{2\pi\hbar \rho_{\text{r}}(E)} \quad \rho_{\text{r}}(E) = \frac{E^{f-1}}{(f-1)! \prod_{j=1}^f \hbar \omega_j}$$

where

$$\begin{aligned}
N^\ddagger(E) &= \sum_{\mathbf{n}} \theta(E - E_{\mathbf{n}}^\ddagger - V^\ddagger) = \int \sum_{\mathbf{n}} \delta(E' - E_{\mathbf{n}}^\ddagger) \theta(E - E' - V^\ddagger) dE' \\
&= \int \rho^\ddagger(E') \theta(E - E' - V^\ddagger) dE' \quad \text{where} \quad \rho^\ddagger(E') = \sum_{\mathbf{n}} \delta(E' - E_{\mathbf{n}}^\ddagger) \\
&= \int_0^\infty \frac{(E')^{f-2}}{(f-2)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \theta(E - E' - V^\ddagger) dE' \\
&= \int_0^{E-V^\ddagger} \frac{(E')^{f-2}}{(f-2)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} dE' = \frac{(E - V^\ddagger)^{f-1}}{(f-1)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \quad E > V^\ddagger
\end{aligned}$$

so the RRKM rate in this approximation is

$$\begin{aligned}
k_{\text{RRKM}}(E) &= \frac{N^\ddagger(E)}{2\pi\hbar \rho_r(E)} \\
&= (2\pi\hbar)^{-1} \frac{(E - V^\ddagger)^{f-1}}{(f-1)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \frac{(f-1)! \prod_{j=1}^f \hbar \omega_j}{E^{f-1}} \\
&= (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar \omega_j}{\prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \left(\frac{E - V^\ddagger}{E} \right)^{f-1}
\end{aligned}$$

It is clear that the rate increases with E and decreases for larger molecules (as it is unlikely to find the energy distributed such that there is enough in the reaction coordinate).

From the microcanonical rate, it is easy to get the thermal rate by averging over energies:

$$\text{thermal rate:} \quad k(\beta) = \int_0^\infty k(E) P_{\text{eq}}(E) dE$$

where for a set of classical harmonic oscillators,

$$\begin{aligned}
P_{\text{eq}}(E) &= \rho_r(E) \frac{e^{-\beta E}}{Q_r(\beta)} \\
&= \frac{E^{f-1}}{(f-1)! \prod_{j=1}^f \hbar \omega_j} e^{-\beta E} \prod_{j=1}^f \beta \hbar \omega_j \\
&= \frac{\beta^f E^{f-1}}{(f-1)!} e^{-\beta E}
\end{aligned}$$

The thermal rate is therefore

$$\begin{aligned}
 k(\beta) &= \int k_{\text{RRKM}}(E) P_{\text{eq}}(E) \, dE \\
 &= \int_{V^\ddagger}^{\infty} (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar\omega_j}{\prod_{j=1}^{f-1} \hbar\omega_j^\ddagger} \left(\frac{E - V^\ddagger}{E} \right)^{f-1} \frac{\beta^f E^{f-1}}{(f-1)!} e^{-\beta E} \, dE \\
 &= \int_{V^\ddagger}^{\infty} (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar\omega_j}{\prod_{j=1}^{f-1} \hbar\omega_j^\ddagger} (E - V^\ddagger)^{f-1} \frac{\beta^f}{(f-1)!} e^{-\beta E} \, dE \\
 &= (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar\omega_j}{\prod_{j=1}^{f-1} \hbar\omega_j^\ddagger} e^{-\beta V^\ddagger} \\
 &= (2\pi\beta\hbar)^{-1} \frac{\prod_{j=1}^f \beta\hbar\omega_j}{\prod_{j=1}^{f-1} \beta\hbar\omega_j^\ddagger} e^{-\beta V^\ddagger} = (2\pi\beta\hbar)^{-1} \frac{Q^\ddagger(\beta)}{Q_{\text{r}}(\beta)} e^{-\beta V^\ddagger}
 \end{aligned}$$

which is equivalent to classical TST, and shows that the assumptions of RRKM must be strongly related to those of TST.

11.7 Further Reading

The following textbooks and articles will give you further information on the topics of this lecture:

Unimolecular rate theory

- Baron Peters, Reaction Rate Theory and Rare Events (Elsevier), chapter 9 <https://www.sciencedirect.com/book/9780444563491/reaction-rate-theory-and-rare-events-simulations> (available through ETH library using VPN)
- Henriksen and Hansen, Theories of Molecular Reaction Dynamics (Oxford), chapter 7
- W. H. Miller, “Tunneling corrections to unimolecular rate constants, with application to formaldehyde”, *J. Am. Chem. Soc.* **101**, 6810-6814 (1979).

12 Path integrals

12.1 Path-integral description of quantum mechanics

In previous lectures, we discussed how to propagate a wavefunction using the propagator operator $\hat{U}(t)$, which can be broken up into N steps of length $\epsilon = t/N$.

$$\begin{aligned}\langle x|\Psi(t)\rangle &= \langle x|\hat{U}(t)|\Psi(0)\rangle = \langle x|\hat{U}(\epsilon) \cdots \hat{U}(\epsilon)|\Psi(0)\rangle \\ &= \int \langle x|\hat{U}(t)|x_0\rangle \langle x_0|\Psi(0)\rangle dx_0 \\ &= \int K(x_0, x, t) \langle x_0|\Psi(0)\rangle dx_0\end{aligned}$$

Here we inserted an identity (integral over a complete set of position states) and defined the position-state matrix elements of the propagator $K(x_0, x, t) = \langle x|\hat{U}(t)|x_0\rangle = \langle x|\hat{U}(\epsilon) \cdots \hat{U}(\epsilon)|x_0\rangle$.

Next we inserting complete sets of states between all short-time propagator operators (and rename the final position x to x_N for later convenience):

$$\begin{aligned}K(x_0, x_N, t) &= \langle x_N|\hat{U}(\epsilon) \cdots \hat{U}(\epsilon)|x_0\rangle \\ &= \int \cdots \int \langle x_N|\hat{U}(\epsilon)|x_{N-1}\rangle \langle x_{N-1}|\cdots|x_1\rangle \langle x_1|\hat{U}(\epsilon)|x_0\rangle dx_1 \cdots dx_{N-1}\end{aligned}$$

Therefore we need to find the representation of the short-time propagator in the position basis and can use the Trotter approximation introduced earlier in the course:

$$\begin{aligned}\langle x_{n+1}|\hat{U}(\epsilon)|x_n\rangle &= \langle x_{n+1}|e^{-i\hat{H}\epsilon/\hbar}|x_n\rangle \approx \langle x_{n+1}|e^{-i\hat{V}\epsilon/2\hbar}e^{-i\hat{T}\epsilon/\hbar}e^{-i\hat{V}\epsilon/2\hbar}|x_n\rangle \\ &= e^{-iV(x_{n+1})\epsilon/2\hbar} \langle x_{n+1}|e^{-i\hat{T}\epsilon/\hbar}|x_n\rangle e^{-iV(x_n)\epsilon/2\hbar}\end{aligned}$$

where we have used the fact that $\hat{V}|x\rangle = V(x)|x\rangle$, i.e. the position state is an eigenstate of the potential operator (like all operators of positions). Finally, we can treat the kinetic-energy part using the free particle result:

$$\begin{aligned}\langle x_{n+1}|e^{-i\hat{T}\epsilon/\hbar}|x_n\rangle &= \int \langle x_{n+1}|e^{-i\hat{T}\epsilon/\hbar}|p\rangle \langle p|x_n\rangle dp \\ &= \int \langle x_{n+1}|p\rangle e^{-ip^2\epsilon/2m\hbar} \langle p|x_n\rangle dp \\ &= \frac{1}{2\pi\hbar} \int e^{ipx_{n+1}/\hbar} e^{-ip^2\epsilon/2m\hbar} e^{-ipx_n/\hbar} dp \\ &= \frac{1}{2\pi\hbar} \int e^{-ip^2\epsilon/2m\hbar + ip(x_{n+1}-x_n)/\hbar} dp \\ &= \frac{1}{2\pi\hbar} \sqrt{\frac{2\pi m\hbar}{i\epsilon}} e^{im(x_{n+1}-x_n)^2/2\epsilon\hbar} \\ &= \sqrt{\frac{m}{2\pi i\epsilon\hbar}} e^{im(x_{n+1}-x_n)^2/2\epsilon\hbar}\end{aligned}$$

So, the total propagator is

$$\begin{aligned}
K(x_0, x_N, t) &= \int \cdots \int \langle x_N | \hat{U}(\epsilon) | x_{N-1} \rangle \langle x_{N-1} | \cdots | x_1 \rangle \langle x_1 | \hat{U}(\epsilon) | x_0 \rangle dx_1 \cdots dx_{N-1} \\
&= \int \cdots \int e^{-iV(x_N)\epsilon/2\hbar} \langle x_N | e^{-i\hat{T}\epsilon/\hbar} | x_{N-1} \rangle e^{-iV(x_{N-1})\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_{N-1})\epsilon/2\hbar} \langle x_{N-1} | \cdots | x_1 \rangle e^{-iV(x_1)\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_1)\epsilon/2\hbar} \langle x_1 | e^{-i\hat{T}\epsilon/\hbar} | x_0 \rangle e^{-iV(x_0)\epsilon/2\hbar} dx_1 \cdots dx_{N-1} \\
&= \left(\frac{m}{2\pi i \epsilon \hbar} \right)^{N/2} \int \cdots \int e^{-iV(x_N)\epsilon/2\hbar} e^{im(x_N - x_{N-1})^2/2\epsilon\hbar} e^{-iV(x_{N-1})\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_{N-1})\epsilon/2\hbar} e^{im(x_{N-1} - x_{N-2})^2/2\epsilon\hbar} \cdots e^{im(x_2 - x_1)^2/2\epsilon\hbar} e^{-iV(x_1)\epsilon/2\hbar} \\
&\quad \times e^{-iV(x_1)\epsilon/2\hbar} e^{im(x_1 - x_0)^2/2\epsilon\hbar} e^{-iV(x_0)\epsilon/2\hbar} dx_1 \cdots dx_{N-1} \\
&= \left(\frac{m}{2\pi i \epsilon \hbar} \right)^{N/2} \int \cdots \int e^{iS_N(x_0, x_1, \dots, x_{N-1}, x_N)/\hbar} dx_1 \cdots dx_{N-1}
\end{aligned}$$

where the discretized action is

$$S_N(x_0, x_1, \dots, x_{N-1}, x_N) = \sum_{n=0}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2\epsilon} - \frac{\epsilon}{2} V(x_0) - \sum_{n=1}^{N-1} \epsilon V(x_n) - \frac{\epsilon}{2} V(x_N)$$

In the infinite N limit, we have a path integral which is written

$$= \int_{x(0)=x_0}^{x(t)=x_N} e^{iS/\hbar} \mathcal{D}x$$

where the action along a given path is

$$S = \int_0^t \left[\frac{m\dot{x}(t')^2}{2} - V(x(t')) \right] dt' = \int_0^t \mathcal{L}(x(t'), \dot{x}(t')) dt'.$$

The probability for a particle to move from x_0 to x_N in time t can thus be written as $|K(x_0, x_N, t)|^2$, which is computed by considering all possible paths and summing over the phase associated with them. If two paths have completely different values of the action, the phases might cancel out, so the dominant contribution to the probability will come from paths which have similar values of the action to their neighbours. We are therefore interested in paths for which a small change to the path results in no change to the action. These are called paths of stationary action. To compute them, we use the mathematical technique called “calculus of variations”.

Consider a path $x(t')$ and a neighbouring path $x(t') + \delta x(t')$. Note that in order to obey the boundary conditions (start at x_0 and end at x_N), we require that $\delta x(0) = \delta x(t) = 0$. The actions

of these two paths are

$$\begin{aligned}
 S[x] &= \int_0^t \mathcal{L}(x, \dot{x}) dt' \\
 S[x + \delta x] &= \int_0^t \mathcal{L}(x + \delta x, \dot{x} + \delta \dot{x}) dt' \\
 &= \int_0^t \left[\mathcal{L}(x, \dot{x}) + \frac{\partial \mathcal{L}}{\partial x} \delta x + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} \right] dt' \\
 &= S[x] + \int_0^t \left[\frac{\partial \mathcal{L}}{\partial x} \delta x + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} \right] dt'
 \end{aligned}$$

such that the change in action is

$$\begin{aligned}
 \delta S &= S[x + \delta x] - S[x] \\
 &= \int_0^t \left[\frac{\partial \mathcal{L}}{\partial x} \delta x + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} \right] dt' \\
 &= \left[\frac{\partial \mathcal{L}}{\partial \dot{x}} \delta x \right]_0^t + \int_0^t \left[\frac{\partial \mathcal{L}}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) \delta x \right] dt'
 \end{aligned}$$

The change in the action is 0 if

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{\partial \mathcal{L}}{\partial x}$$

which is recognized to be the same as the Euler–Lagrange equation, from which classical mechanics can be derived.

In the limit that $\hbar \rightarrow 0$ (relative to other system parameters) the paths of stationary action dominate the path integral more and more, and thus we have found a simple link between quantum mechanics and classical mechanics. However, for finite values of $\hbar$, the other paths also contribute and can be used to explain quantum phenomena.

12.2 Path-integral quantum statistics

It is difficult to use path integrals to simulate quantum dynamics on a computer because of the enormous problems related to averaging over terms with different signs (or phases). However, the path-integral approach has been very useful in theoretical chemistry as a tool for computing quantum statistical properties. For instance, the partition function

$$Z = \text{Tr} \left[e^{-\beta \hat{H}} \right] = \int \rho(x, x) dx$$

where the density matrix is defined as

$$\rho(x', x'') = \langle x'' | e^{-\beta \hat{H}} | x' \rangle$$

Note the similarity between the density matrix and the propagator. If we choose $t = -i\beta\hbar$, they are identical, and we can thus think of the density matrix as an imaginary-time propagator.

It is not hard to convert all of the path-integral formulas to imaginary time.

$$\begin{aligned}\rho(x_0, x_N) &= K(x_0, x_N, -i\beta\hbar) \\ &= \left(\frac{m}{2\pi\eta\hbar}\right)^{N/2} \int \dots \int e^{-\bar{S}_N(x_0, x_1, \dots, x_{N-1}, x_N)/\hbar} dx_1 \dots dx_{N-1}\end{aligned}$$

using $t \rightarrow -i\beta\hbar$ and therefore $\epsilon \rightarrow -i\eta$, where $\eta = \beta\hbar/N$. The action becomes

$$\begin{aligned}S_N(x_0, x_1, \dots, x_{N-1}, x_N) &= -\sum_{n=0}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2i\eta} + \frac{i\eta}{2} V(x_0) + \sum_{n=1}^{N-1} i\eta V(x_n) + \frac{i\eta}{2} V(x_N) \\ \bar{S}_N(x_0, x_1, \dots, x_{N-1}, x_N) &= -iS_N(x_0, x_1, \dots, x_{N-1}, x_N) \\ &= \sum_{n=0}^{N-1} \frac{m(x_{n+1} - x_n)^2}{2\eta} + \frac{\eta}{2} V(x_0) + \sum_{n=1}^{N-1} \eta V(x_n) + \frac{\eta}{2} V(x_N)\end{aligned}$$

Thus

$$\rho(x', x'') = \int_{x(0)=x'}^{x(\beta\hbar)=x''} e^{-\bar{S}/\hbar} \mathcal{D}x(u) \quad \bar{S} = \int_0^{\beta\hbar} \left[\frac{1}{2} m \dot{x}(u)^2 + V(x(u)) \right] du$$

and the partition function is

$$\begin{aligned}Z &= \int \rho(x_0, x_0) dx_0 \\ &= \int \int_{x(0)=x_0}^{x(\beta\hbar)=x_0} e^{-\bar{S}/\hbar} \mathcal{D}x(u) dx_0 \\ &= \oint e^{-\bar{S}/\hbar} \mathcal{D}x(u)\end{aligned}$$

In a similar way, we can compute averages of operators which depend on position operators:

$$\begin{aligned}\langle \hat{A} \rangle &= \frac{1}{Z} \text{Tr} \left[e^{-\beta\hat{H}} \hat{A} \right] \\ &= \frac{1}{Z} \int \langle x_0 | e^{-\beta\hat{H}} \hat{A} | x_0 \rangle dx_0 \\ &= \frac{1}{Z} \int \langle x_0 | e^{-\beta\hat{H}} | x_0 \rangle A(x_0) dx_0 \\ &= \frac{1}{Z} \int \rho(x_0, x_0) A(x_0) dx_0 \\ &= \frac{1}{Z} \int \int_{x(0)=x_0}^{x(\beta\hbar)=x_0} e^{-\bar{S}/\hbar} \mathcal{D}x(u) A(x_0) dx_0 \\ &= \frac{1}{Z} \oint e^{-\bar{S}/\hbar} A(x_0) \mathcal{D}x(u)\end{aligned}$$

An alternative way of writing the path-integral formulas shows why the method is so useful.

$$\begin{aligned} Z &= \left(\frac{m}{2\pi\eta\hbar} \right)^{N/2} \int \cdots \int e^{-S_N(x_N, x_1, \dots, x_N)/\hbar} dx_1 \cdots dx_N \\ &= \left(\frac{m}{2\pi\beta_N\hbar^2} \right)^{N/2} \int e^{-\beta_N U_N(\mathbf{x})} d\mathbf{x} \end{aligned}$$

where $\mathbf{x} = \{x_1, \dots, x_N\}$ is a *ring polymer* (so called because each bead is connected to its neighbour in a ring by springs). This expression looks just like a multidimensional classical ensemble average in which the effective inverse temperature is $\beta_N = \beta/N$ and the effective potential is

$$U_N(\mathbf{x}) = \frac{S_N}{\beta_N\hbar} = \sum_{n=1}^N \frac{1}{2} m \frac{1}{\beta_N^2 \hbar^2} (x_n - x_{n-1})^2 + \sum_{n=1}^N V(x_n).$$

We can also choose to write

$$Z = (2\pi\hbar)^{-N} \iint e^{-\beta_N H_N} d\mathbf{x} d\mathbf{p}$$

where the effective Hamiltonian is

$$H_N = \sum_{n=1}^N \frac{p_n^2}{2m} + U_N(\mathbf{x})$$

and we have introduced effective momenta for each bead $\mathbf{p} = \{p_1, \dots, p_N\}$.

It is easy to show that by taking $N = 1$ in all these formulas, one recovers classical statistics (as in the first lecture). In the $N \rightarrow \infty$ limit, the path-integral approach gives the exact quantum statistical results, and for many chemical systems a good approximation is provided by a small but finite value for N . By consideration of the harmonic oscillator, one sees that the result will only be converged if $\beta_N \hbar \omega \ll 1$, such that for typical simulations such as liquid water at room temperature, $N = 32$ is sufficient.

12.3 Path-integral rate theory

Now we have an effective ring-polymer Hamiltonian, we can define ring-polymer dynamics using Hamilton's equations of motion.

$$\dot{x}_n = \frac{p_n}{m} \qquad \dot{p}_n = -\frac{m}{\beta_N^2 \hbar^2} (2x_n - x_{n-1} - x_{n+1}) - \nabla V(x_n)$$

Note that the Hamiltonian has been derived by analogy between quantum and classical *statistics*, and we should not assume that these dynamics also describe quantum *dynamics* correctly. However, in some situations, it turns out that it does quite a good job. We can for instance look at the motion of the centre of mass of the ring-polymer, called the “centroid”:

$$\begin{aligned} \bar{x} &= \frac{1}{N} \sum_{n=1}^N x_n & \bar{p} &= \frac{1}{N} \sum_{n=1}^N p_n \\ \dot{\bar{x}} &= \frac{\bar{p}}{m} & \dot{\bar{p}} &= \frac{1}{N} \sum_{n=1}^N \nabla V(x_n) \end{aligned}$$

Note that in the special case of a harmonic oscillator or free particle the force on the centroid is equal to $\nabla V(\bar{x})$ (because the gradient is linear or non-existent) and will thus move as if it were a classical particle. This is actually good news, as we know that classical mechanics is basically correct for these systems. In more complicated problems, $\nabla V(x_n) \neq \nabla V(\bar{x})$, so the dynamics will be different from classical (good because classical mechanics is wrong), but unfortunately not necessarily the same as quantum.

Something that it does do well is calculating rates. We simply use the ring-polymer Hamiltonian in the flux-side correlation function

$$C_{\text{fs}}(t) = (2\pi\hbar)^{-N} \iint \delta[\bar{x} - x^\ddagger] \frac{\bar{p}}{m} \theta(\bar{x}_t - x^\ddagger) e^{-\beta_N H_N} d\mathbf{x} d\mathbf{p}$$

and then can define the ring-polymer molecular dynamics (RPMD) rate:

$$k_{\text{RPMD}} = \lim_{t \rightarrow \infty} \frac{C_{\text{fs}}(t)}{Z_r}$$

and by analogy to the classical case, we define an ring-polymer TST:

$$k_{\text{RPTST}} = \lim_{t \rightarrow 0_+} \frac{C_{\text{fs}}(t)}{Z_r}$$

Note that RPTST can be (in some cases greatly) improved by using a better dividing surface than the one suggested here.

12.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Path integrals

- Feynman, QED: the strange case of light and matter [*easy reading which emphasizes concepts rather than maths*]
- Feynman, Path integrals and quantum mechanics [*classic*]

12.5 What are you expected to be able to do in the oral exam?

Explain how to calculate the rate of a chemical reaction from first principles, describing and assessing the approximations used. In particular:

- Compare classical vs quantum mechanics (correlation functions, equilibrium averages, partition functions, etc.);
- Describe Born-Oppenheimer approximation and how to go beyond it;
- Describe how molecules interact with light in terms of wavefunction propagation;
- Explain how the quantum dynamics of a two-level system differs from that of a system with a continuum of states;

- Define Eyring TST and describe how to evaluate it;
- Describe approximations of TST and how to go beyond them (friction, nonadiabatic effects, microcanonical ensembles, tunnelling, etc.).